

Employers and Refugee Economic Integration: The Effect of Early Employer Quality*

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Abstract

Drawing on matched employer–employee data from Denmark, we provide the first causal evidence on role of early employers in the labor market integration of immigrants. First, using the two-way fixed effects model of Abowd, Kramarz, and Margolis (1999), we estimate firm-specific wage premia that we use as a proxy for workplace quality. Second, we leverage the role of social connections and a dispersal policy, which quasi-randomly allocated refugees across municipalities, to obtain exogenous variation in their exposure to the quality of first potential employers. Placement in municipalities where, at arrival, co-nationals are employed by high-quality employers significant increases refugees’ employment and earnings for up to ten years. Incorporating employer quality into a data-driven algorithm to optimally match refugees to municipalities leads to a 46% increase in short-run employment probability relative to the status quo policy. Our results imply that the type of employers available upon refugees’ first entry is an important determinant of their success in host countries.

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1 Introduction

Across a wide range of high-income destinations, immigrants from lower-income countries experience persistent gaps in their economic performance, even compared to otherwise similar workers (Bratsberg et al., 2017; Brell et al., 2020; Fasani et al., 2022). With the near doubling of forcibly displaced people over the past decade, these difficulties extend beyond migrants’ own well-being, placing increasing pressure on public authorities, who face short-run crowding effects as well as longer-term fiscal, social, and political costs from delayed integration (Bahar et al., 2024; UNHCR, 2025).¹ Among the determinants of immigrant–native earnings differences, the role of employers has only recently begun to receive attention, with emerging evidence showing that between-firm sorting, rather than unequal pay within the same firm, accounts for a substantial share of the earnings gap (Damas de Matos, 2017; Dostie et al., 2023; Arellano-Bover and San, 2023; Yumoto et al., 2026; Åslund et al., 2025; Hermansen et al., 2025). As early career experiences are strongly related to workers’ long-run economic success, this raises a fundamental question: whether and to what extent does an early match with a “good” firm provide immigrants with a pathway for integration?

Answering this question is econometrically challenging. While mounting evidence indicates that firms play an important role in wage determination (Card et al., 2013, 2016, 2018), endogeneity and selection are pervasive in employer–employee matching, making the causal effect of early employers on employees difficult to identify. Nevertheless, there are two reasons an answer is important. First, workplaces vary widely across many dimensions, such as productivity, on-the-job training, and management practices, and early career matches with different firms can have long-term consequences for employees (see von Wachter (2020) for a review). Second, for immigrants specifically, the first workplace represents initial exposure to an unfamiliar labor market, offering opportunities for training, country-specific human capital development, and valuable connections, making early employers arguably even more consequential than for native workers. This is consistent with a large literature on the role of initial conditions for migrants’ outcomes at destination (see Arendt et al. (2022) and Foged et al. (2024) for reviews), which has, however, scarcely considered the role of labor demand.

In this paper, we provide the first causal evidence on the impact of first accessible employers on immigrants’ labor market outcomes, developing an identification strategy that leverages a government policy experiment and the role of social connections. Utilizing multiple linked administrative datasets from Statistics Denmark, we proceed in two stages. First, following the methodology of Abowd et al. (1999) (AKM), we estimate establishment-specific pay premia and use them as proxies for establishment quality, after validating that they are strongly associated with measures of firm productivity and desirability for employees.² Second, we exploit the conditionally random initial placement of refugees across municipalities under a Danish national dispersal policy. Conditional on individual characteristics known to authorities at the time of allocation, the initial assigned location was uncorrelated with newly arrived

¹At the end of 2024, the number of internationally displaced people surpassed 45 million, with the average duration of displacement steadily increasing. Figure A1 shows the trend over the past 40 years.

²Firm pay premia capture the wage component attributable to working at a given firm and are obtained by modeling wages as a function of additive worker and firm fixed effects to account for unobserved individual heterogeneity. See Kline (2024) for a review of methodologies and interpretations of firm wage differences.

refugees’ observable skills and locational preferences. Crucially, using firm effects in a second stage with a quasi-experimental design sets our work apart from the literature on firm wage-setting policies and allows us to derive causal estimates of exposure to early employer quality. Our measures progressively refine the set of first accessible employers, from all active firms in the municipality of assignment to the narrower, more relevant set of those employing co-nationals.³

The dramatic surge in refugee migration makes this group highly relevant for studying the integration of immigrants from lower-income countries, as one in seven international migrants today is forcibly displaced. In a context where Europe has experienced the fastest growth in refugee arrivals, Denmark offers a particularly advantageous setting to address our question. First, the country has accepted refugees from a large variety of origins over the past four decades and is therefore widely considered the “ideal laboratory” to study integration ([Arendt et al., 2022](#)). Second, its unusually rich administrative employer–employee data allow us both to estimate firm quality with accuracy and to track individual refugees over a 15-year period following arrival, providing a longer horizon than most previous studies. Finally, the implementation of a quasi-random national dispersal policy between 1986 and 1998, under which allocation decisions were made strictly according to administrative rules, helps address concerns about non-random worker sorting across locations and employers.

We begin by documenting three novel stylized facts about refugees and firms that motivate our empirical approach. First, for refugees, starting one’s career at a higher-quality firm in the host country is strongly correlated with higher hourly wages, annual earnings, and employment probabilities, with effects that persist for more than 15 years after the first employment spell. Second, employer quality is “sticky” for refugees. While we show that pay premia for natives who begin in low- or high-quality firms quickly converge toward one another, the initial advantage (or disadvantage) for refugees remains much more persistent, suggesting that lower mobility across rungs of the firm ladder amplifies the importance of early matches.⁴ Third, social connections shape refugees’ opportunities. Using a dataset of worker–establishment pairs, we show that having a co-national employed at an establishment upon arrival significantly increases a refugee’s probability of later employment there, highlighting the role of co-national (or co-ethnic, terms we use interchangeably) networks in the labor market.

Results from our main analysis imply that successful integration depends not only on obtaining a job but also on the employer providing it, as the quality of employers accessible through co-national networks significantly shapes labor market outcomes. We find that placement in a municipality where, at arrival, members of the local co-national network are employed by higher-quality firms, has positive and significant effects on both employment and earnings in the short (1–5 years) and medium (6–10 years) run after arrival.⁵ By contrast,

³In practice, we construct three measures of early employer exposure, each refining the set of establishments initially accessible to refugees: (i) all establishments active in the municipality of assignment at arrival, (ii) those that had ever hired individuals from refugee-sending countries, and (iii) those employing members of the local co-national network at arrival. For each set, we compute the average establishment-level pay premium. Figure A2 outlines the construction of our main measure of employer quality.

⁴Even among refugees who change employers or municipalities, the effect of first-employer quality persists, consistent with portable human capital development rather than a mechanical effect.

⁵The effects remain positive in the long run (11–15 years after arrival) but lose statistical significance.

higher pay premia among all firms in the municipality do not improve outcomes, underscoring the role of co-national connections. Magnitudes are economically meaningful: in the first five years, the effect sizes correspond to increases of between 8 and 10% relative to sample means of both outcomes, and over the first decade our effects amount to between one-third and one-sixth of estimates for other direct, potentially more costly integration policies, such as active labor market policies and language training. These results extend the vast body of work on the determinants of immigrant integration, which has largely focused on asylum policies as well as local conditions at arrival.⁶ Since the early years are a crucial integration period, when refugees often face very low employment rates (Brell et al., 2020), our work bridges the literature on refugee integration and job ladder models, revealing employer accessibility as a largely overlooked but promising policy lever.

This paper makes several additional contributions. First, we shed new light on dynamic effects not captured by the commonly used two-way fixed effects models, which view wage determination as fundamentally static. A group of recent papers point out the effects of past experience at heterogeneous firms, providing evidence of accumulation of portable skills when starting in larger firms or in firms offering greater on-the-job learning (Abowd et al., 2018; Arellano-Bover, 2024; Arellano-Bover and Saltiel, 2024; Gendron-Carrier, 2025). By showing the persistent influence of the first employer fixed effect on immigrants' future outcomes, our case reveals an analogous but distinct source of dynamic effects that can severely undermine the conventional AKM framework when applied to subgroups such as refugees or immigrants without further scrutiny or assumptions. To our knowledge, we are the first to document a difference between natives and immigrants in the importance of early employers. This distinction seems economically reasonable, as for foreign-born workers the first workplace may represent a crucial stepping stone in an unfamiliar labor market, while for natives, local knowledge, mobility, and established networks mitigate such dependence on the first match.⁷ Although our estimates are unaffected, since refugees are excluded from the panel used in the AKM estimation, our evidence highlights a source of time-dependence violating the separability of worker and firm fixed effects that underpins the AKM model.⁸

Furthermore, to illustrate the importance of characterizing accessible employers using co-nationals, we show that co-national networks act as key information-transmission mechanisms connecting new members with employers. This evidence is consistent with two complementary channels: job referrals and broader information sharing. The role of networks for labor market outcomes is well established, especially for migrants, due to their limited knowledge of the language, cultural norms, and effective job search methods.⁹ However, while network

⁶Some policies examined include language training, active labor market policies, changes in welfare benefits, and regulations for permanent residency and employment (Arendt et al., 2024; Foged et al., 2022; Arendt and Bolvig, 2023; Abbiati et al., 2025; Dustmann et al., 2024b; Arendt et al., 2023; Marbach et al., 2018; Fasani et al., 2021; Ahrens et al., 2023); initial conditions considered include the size of co-ethnic networks, employment levels, crime rates, and urbanization, among others (Edin et al., 2003; Damm, 2009a; Åslund and Rooth, 2007; Azlor et al., 2020; Damm and Dustmann, 2014; Dustmann et al., 2023; Eckert et al., 2022).

⁷This aspect also speaks to the empirical literature on the long-term consequences of entry labor market conditions for young adults (Kahn, 2010; Oreopoulos et al., 2012; Altonji et al., 2016).

⁸Bonhomme et al. (2019) note that the static formulation of the AKM framework limits its ability to capture mechanisms emphasized in the dynamic structural literature. They propose an alternative framework to modeling unobserved heterogeneity and document state dependence in earnings.

⁹Notable works in this literature include Ioannides and Loury (2004), Bayer et al. (2008), Hellerstein et al.

size has been widely examined, studies on co-ethnic networks have rarely focused on network quality, typically measured through the outcomes of co-ethnic members, such as employment and earnings (Edin et al., 2003; Damm, 2009a, 2014; Martén et al., 2019). Instead, we define a measure of network quality based on the characteristics of employers connected to the network, isolating the effect of the firm component from other network attributes.¹⁰ Crucially, this allows us to address a more fundamental question in the network literature: what drives the “warm embrace” of network to new members (Borjas, 2000)—human capital externalities from high-achieving peers, or friction-reducing mechanisms connecting people to strong labor demand? By relying on the two-way AKM structure to estimate both firm and worker fixed effects, we separate the effect of the quality of network members from that of employers connected to the network, showing that employer quality is the main driver of integration and of the widely documented positive network effects.

The last part of the paper leverages our evidence to extend a data-driven algorithm that optimally resettles refugees to maximize their employment prospects. We combine machine learning methods that predict individual employment probabilities across municipalities with integer optimization to determine optimal refugee–municipality matches. Importantly, in addition to the individual- and municipality-level characteristics used in previous work on automated resettlement, we incorporate information on the quality of firms within networks into a LASSO model.¹¹ This alters the predicted employment probabilities across municipalities, as different origins feature different best-suited destinations. The optimized assignment prescribed by the algorithm, while respecting municipality quotas, increases refugees’ average predicted probability of employment within five years to 42.1%, compared with 28.9% under the status quo dispersal policy, with gains that are stable across the counterfactual policy scenarios we evaluate. While many countries adopted dispersal policy to distribute refugees evenly and share integration costs more equitably, our findings emphasize the importance of considering early access to high-quality employment opportunities. Because more informed assignment decisions involve no direct monetary costs, they represent an effective, low-cost policy improvement that can be tailored to policymakers’ priorities, fully automated, and implemented in real time, reducing inefficiencies associated with manual assignment. With increasing rates of migration worldwide, recognizing the role of early employers in shaping integration trajectories is essential, particularly as successful immigrant integration can yield significant economic benefits for host countries, from greater labor market flexibility, through improved fiscal sustainability, to support in counteracting population aging (Clemens, 2022; Bahar et al., 2022).

(2011), Topa (2019), Dustmann et al. (2016), Goel and Lang (2019), Wahba and Zenou (2005), Beaman (2012).

¹⁰This network dimension has not yet been explored, with the exception of Schmutte (2015), who uses 2002–2003 U.S. data to study the effect of firm-based local network quality on pay premia from job transitions. Our work differs in three important ways. First, our identification strategy is stronger, as the dispersal policy allows us to avoid assuming no endogenous sorting at the block level within neighborhoods. Second, our data allow a richer analysis, tracking multiple labor market outcomes over a much longer period at destination. Third, we build our main network measure at arrival to capture early exposure, complementing it with other AKM-based aggregations, such as averages over all local firms, to provide alternative proxies for accessible employers.

¹¹Examples from this recent literature at the intersection of economics and operations research include Bansak et al. (2018), Ahani et al. (2021), Jones and Teytelboym (2018), Nguyen et al. (2021), and Delacrétaz et al. (2023).

The remainder of the paper is organized as follows. Sections 2 and 3 describe the institutional setting in Denmark and the data used to construct the sample for our analysis, respectively. Section 4 outlines the procedure used to build a proxy for establishment quality. Section 5 presents a set of novel stylized facts on refugees and establishments. In Section 6, we discuss our empirical framework, while Section 7 presents the main results and robustness checks. Section 8 provides evidence on the mechanisms. In Section 9, we develop a data-driven approach to implement an optimal assignment of refugees, considering alternative counterfactual scenarios. Finally, Section 10 concludes.

2 Institutional Setting

Denmark represents a compelling context to study the role of employers for refugee integration due to both large inflows of displaced people and the implementation of a national dispersal policy during the period of study. The country once had some of the most liberal refugee immigration laws ([Dustmann et al., 2024b](#)), and 155,752 individuals were granted refugee status between 1984-2019 ([Arendt et al., 2022](#)). In 2000, at the end of the dispersal policy, the number of asylum seekers who were granted protection was 96 per 100,000 inhabitants, about four times the EU average at the time.

In 1956, the Danish Refugee Council (DRC) was established in response to Denmark’s 1952 ratification of the 1951 United Nations Convention on the Status of Refugees.¹² The DRC was tasked with assisting asylum seekers in applying for refugee status and residence permits in Denmark. In response to a large inflow of refugees during the early 1980s, the Danish government implemented a dispersal policy in 1986 for individuals whose asylum cases had been approved (i.e., refugees). This policy was carried out by the DRC.

The dispersal policy consisted of two stages and was in effect from 1986 to 1998 ([Damm and Dustmann, 2014](#); [Dustmann et al., 2019, 2023](#)). Its aim was to distribute refugees arriving in Denmark across counties and municipalities in proportion to the population size of each locality. Prior to 1986, refugees were primarily housed in the largest cities ([Hasager and Jørgensen, 2024](#)). In the first stage of the policy, refugees were allocated to Denmark’s 15 counties based on the population size of the county. In the second stage, they were further assigned to municipalities within those counties, again in proportion to each municipality’s population. At the time, Denmark had 275 municipalities.¹³ The goal of achieving proportional allocation at the municipal level within 3–5 years was pursued through a rotation scheme, with DRC offices rotating between towns within each county.

When asylum seekers arrived in Denmark, they were initially placed in Red Cross reception centers across the country while waiting for their case to be adjudicated. These centers, both then and now, are usually located in sparsely populated areas. Until 2013, asylum seekers were not permitted to work while residing there. Consequently, most refugees likely had very little contact with the broader Danish population before being granted asylum. In the

¹²Denmark was the first country to sign and ratify the 1951 Refugee Convention ([Gammeltoft-Hansen and Madsen, 2021](#)).

¹³Pre-2006-reform Danish municipalities are best compared in the U.S. context to incorporated towns or cities, or to small counties.

late 1990s, the average waiting time for a decision exceeded one year (Arendt et al., 2024).

Once granted asylum, refugees were relocated to temporary housing in one of Denmark's 15 counties within the first 10 days. Subsequently, the local DRC office assigned them to a municipality within the county and assisted them in securing permanent housing. During the policy period, the vast majority of refugees obtained permanent housing within 18 months. According to Damm and Dustmann (2014), only 0–4% did not secure permanent housing within this time frame. Because some refugees initially lived in temporary housing near the municipality to which they were later assigned, typically within 3 months and on average after 6–7 months, we define the municipality of assignment as the municipality in which the refugee resided in the year following the receipt of their residence permit (Damm, 2009b).

Municipality allocation decisions were made by the DRC without face-to-face meetings with the refugees. However, after receiving asylum, refugees completed a short questionnaire that collected basic personal information such as nationality and family size. These characteristics were sometimes used to guide the allocation process. To support the claim of conditional random assignment to municipalities, Table 1 reports the correlation between individual refugee characteristics and municipality characteristics. While characteristics observed by the authorities—such as nationality, family size, and other demographics (Panel A)—are not expected to be uncorrelated with municipality characteristics, given that assignment was conditional on them, we show that other individual variables are orthogonal to municipality characteristics. In particular, education measures (Panel B), which are likely correlated with unobserved skills, are orthogonal to municipality-level co-national presence, labor market outcomes, and employer quality, thereby supporting the validity of the claim of quasi-randomness in the assignment.¹⁴ Moreover, taking the validity of this claim one step further in our exercise, we also observe that even the variables in Panel A are uncorrelated with the municipality characteristics most relevant for our analysis (columns 3 and 4). This is fully consistent with the intuition that, although some demographic features were used in the allocation process, they are unlikely to be correlated with origin–municipality traits such as network employer quality.

While there were no restrictions on mobility for refugees following their initial assignment to municipalities, they were incentivized to remain in the assigned municipality for the first 18 months. This was because the introduction program, which included courses in Danish language, culture, and job training, was offered only in the assigned municipality, despite the fact that eligibility for means-tested social benefits was not conditional on remaining there (Damm, 2005). Furthermore, reassignment requests were considered by the council only after the initial municipal assignment. Figure A3 (Panel A) shows the proportion of refugees residing in their assigned municipality in the years following asylum approval. In the early years, the majority remained in their assigned municipality, and even by year 15, more than 40% still resided there.¹⁵ This persistence is important, as we emphasize the role of initial exposure to employers in the municipality of assignment.

Two additional clarifications regarding the context are in order. First, refugees were eligible

¹⁴Additional checks are provided in Damm and Rosholm (2010), Damm (2014) and Foged et al. (2024).

¹⁵In the long run, this mobility rate is comparable to that observed for the overall Danish workforce in the 20 years following the start of the dispersal policy (Panel B of Figure A3).

to work immediately upon receiving asylum, so there were no formal restrictions on the ability to seek employment for refugees in our sample. Second, public attitudes toward refugees were not particularly hostile during this period in Denmark. If anything, sentiment began to shift in the early 2000s, when the Danish People’s Party, a populist right-wing party, began supporting the governing coalition, and “Start Aid”, a reform that reduced welfare benefits for newly admitted refugees by around 40%, was implemented in 2002 ([Dustmann et al., 2024b](#)).

3 Data and Sample

Our study relies on Danish administrative data provided by Statistics Denmark. Denmark is among the countries offering the most detailed longitudinal data on refugees in recent decades, along with exceptionally rich information on multiple dimensions. The datasets we use cover the entire population of individuals and firms in the country and include detailed information on aspects such as residence location, demographic details, socioeconomic characteristics, and employer (establishment and firm) information. To analyze labor market outcomes, we use employer–employee matched data from the Integrated Database for Labor Market Research (IDA), which we link to the Income Register (IND) and the Register for Classification of Employment (AKM). Additionally, we draw demographic information from the Population Register (BEF) and data on educational attainment from the Education Register (UDDA). Finally, we use the Migration Register (VNDS) and the Country Admission Register (OPHG) to gather information on refugee admissions and the timing of their initial settlements.

As the main outcomes in the analysis, we consider employment probabilities and earnings for refugees following their arrival in Denmark. Employment is defined as a binary indicator equal to one if the individual was employed at any point during the year, and 0 otherwise. Earnings are measured as annual gross labor market income, expressed in thousands of US dollars, deflated to 2015 prices, and include zero earnings. For the analysis of short-term (1–5 years), medium-term (6–10 years), and long-term (11–15 years) outcomes, we calculate the simple average of individual yearly outcomes within each interval and use this average as the outcome variable.

Our analysis sample consists of refugees who were granted asylum between 1987 and 1998.¹⁶ Since actual refugee status is not directly observable prior to 1997, we follow other studies and impute this status for the years 1987–1996 according to the following two criteria ([Damm and Dustmann, 2014](#); [Foged et al., 2024](#)). An individual is considered as a refugee if: i) they arrived in Denmark from one of nine refugee-sending countries—Palestine, Ethiopia, Somalia, Afghanistan, Sri Lanka, Iraq, Iran, Lebanon, or Vietnam; and ii) they arrived unmarried to either someone from a non-refugee country or an immigrant from a refugee-sending country who had arrived one year or more earlier.¹⁷ The second criterion ensures that the sample consists of refugees subject to the quasi-random dispersal policy, excluding subsequently arriving spouses, who were almost always assigned to the location of their preexisting family. Similarly, migrants from Yugoslavia are excluded from the set of refugee-sending countries

¹⁶We start from the 1987 cohort because of data limitations.

¹⁷Conflicts (e.g., Iran and Iraq) and civil wars (e.g., Somalia) triggered an exodus from many of these countries.

because they were subject to non-random dispersal patterns upon arrival in Denmark as part of the Bosnian program during the 1990s. Lastly, we restrict our sample of refugees to those aged 18 to 55 at arrival and to those remaining in Denmark for at least 15 years in order to extend the analysis horizon relative to most previous studies.

Table 2 presents summary statistics for the final analysis sample of refugees who were *de facto* randomly dispersed and observed for at least 15 years in Denmark.¹⁸ The average age at arrival is approximately 30 years, the majority are male (58%), and one fourth of those reporting educational attainment have completed academic education prior to arrival. Figure 1 displays the size and origin composition of the different arriving cohorts of refugees in our sample, which consists of approximately 15,570 individuals. In the period we study, the yearly number of arrivals ranged from around 900 to roughly 1,700.

4 Establishment Quality

This section describes the high dimensional fixed effects method we use to estimate a proxy for establishment quality available to refugees in Denmark. We adopt the popular methodology originally proposed by [Abowd et al. \(1999\)](#), known as the AKM decomposition, and build on frontier refinements developed in the recent empirical literature, which we now describe. We leave supportive evidence and robustness checks for this framework to Appendix B.

To estimate establishment pay premia, we fit a linear wage model with additive person and establishment fixed effects ([Abowd et al., 1999](#)). Using Danish matched employer–employee data, we construct a job spell panel with person-year observations for N workers employed in J establishments during our period of interest (1986–1998). We assume that the log hourly wage y_{it} of worker i in year t are determined by the sum of a worker component α_i , an establishment component $\psi_{J(i,t)}$, time-varying characteristics $X'_{it}\beta$, and an error term r_{it} :

$$\ln(y_{it}) = \alpha_i + \psi_{J(i,t)} + X'_{it}\beta + r_{it} \quad (1)$$

Following the standard interpretation of the AKM decomposition ([Abowd et al., 1999](#); [Card et al., 2013](#)), the person effect α_i captures a combination of personal, permanent skills leading to different earnings capacity; the establishment effect ψ_j can be interpreted as the time-invariant pay premium offered by the employer to all employees working at j ; and X_{it} is a vector of time-varying controls that affect worker productivity, which in our case include year dummies and age terms (quadratic and cubic). Importantly, the model assumes that ψ_j remains constant throughout the sample period from 1986 to 1998.¹⁹

For the estimation of equation 1, we adopt a common restriction in the literature and consider only private sector employees to obtain a set of comparable firms with market-based

¹⁸As we impose various restrictions on the initial sample of refugees, selection may raise concerns about the external validity of our conclusions. For instance, the requirement that individuals be observed for at least 15 years at destination may result in a particularly positively selected sample. To examine this, Table A1 reports summary statistics for the refugee sample *before* excluding those observed for fewer than 15 years and later-arriving spouses.

¹⁹We consider this an innocuous assumption given that firm wage effects tend to be highly temporally persistent ([Lachowska et al., 2020](#); [Engbom et al., 2023](#)). Shortening or extending the sample period has very limited effects on our estimates.

wage-setting. We winsorize hourly wages by excluding the bottom and top 0.5% in each year, and further retain only observations indicated as high quality by Statistics Denmark to handle outliers and avoid reporting errors. Moreover, although refugees represent a small group compared to the rest of the population, we exclude them from the panel used for estimation to avoid mechanical effects.²⁰ As is well known, workplace effects in this model can only be identified within a “connected set” of establishments linked by worker transitions, as movers provide the variation needed to disentangle α_i from ψ_j in wages. Consequently, we restrict the analysis to the largest connected set of establishments, reporting various statistics in Table B1 (Panel A). Our largest connected set includes 98% of workers and 90% of establishments, with the mean log hourly wage for observations within it equal to that of the full panel.

Consistent with this methodology, we interpret the vector of establishment fixed effects ψ_j as capturing the pay premia workers receive, with one premium associated with each establishment j . As noted in previous work, establishment fixed effects reflect employer-specific, time-invariant compensation policies (Baker et al., 1994) or, more broadly, the advantages associated with being employed by a given employer (Card et al., 2013). These advantages can derive, for instance, from rent-sharing or efficiency wages (for theoretical explanations see, for instance, Burdett and Mortensen (1998), Moscarini and Postel-Vinay (2012), Shapiro and Stiglitz (1984), Akerlof and Yellen (1990)). We use the estimated establishment fixed effects as our primary measure of employer quality in the analysis, following a broad empirical literature linking firm wage effects to firm productivity, size, workplace amenities, and revealed-preference measures of desirability for employees (Kline, 2024; Sockin, 2024). Supporting this interpretation, Tables B2 and B3 show a positive and significant correlation between the estimated establishment fixed effects and other establishment characteristics, such as value added per worker, establishment size, and the share of high-skill employees in our setting.

Panel B of Table B1 presents the resulting variance decomposition from estimating equation 1. The variance of hourly wages is divided into five components: worker effects, employer effects, year effects, covariance terms, and a residual. While worker fixed effects account for a large share of the variation in our outcome (44%), employer effects are also important, explaining approximately 16% of the variation. This is comparable to 13% reported by Lachowska et al. (2020) for Washington state and 18–21% reported by Card et al. (2013) for Germany.

Finally, consistent OLS estimation of equation 1 requires the assumption of conditional exogenous mobility. In Appendix B.1, we discuss this assumption and present a variety of tests to show the consistency of our setting with it. Furthermore, because we estimate this model without imposing origin restrictions on the sample of workers, we also assess whether our estimates accurately reflect workplace qualities for migrants in Denmark. If employers engage in forms of within-firm differentiation between natives and immigrants, establishment effects estimated on the full sample of workers in Denmark may not capture the workplace quality actually experienced by foreign-born individuals, particularly refugees.²¹ While we find some evidence of within-firm differentiation, Appendix B.2 shows that it does not threaten our estimates since the workplace quality ranking is broadly similar for natives and immigrants.

²⁰In Section 5, we present another important justification for this choice.

²¹This could arise from native-immigrant differences in bargaining power, outside options, firm-specific labor supply elasticities, or reservation wages (Adda et al., 2022; Arellano-Bover and San, 2023; Dustmann et al., 2024a).

We therefore proceed to use the workplace effects estimated in equation 1, based on the full sample of natives and non-refugee immigrants, in the remainder of our analysis.

5 Stylized Facts

In this section, we document three novel stylized facts about refugees and their employers that motivate our empirical analysis. These relationships, while descriptive, reveal patterns about refugees and firms that have not previously been documented but are important to establish given the limited existing evidence on this group. To do so, we use the proxy for employer quality estimated in the previous section and apply it to our refugee sample.

Fact 1. First, we examine the relationship between establishment quality and the labor market outcomes of refugees in our sample. Table 3 presents the correlation between the first employer’s fixed effect and refugees’ individual employment probability (columns 1-3) and annual earnings (columns 4-6), including individual controls as well as the main set of fixed effects used throughout the analysis—municipality of assignment fixed effects, cohort fixed effects, and origin fixed effects.²² Panels B and C restrict the sample to refugees with at least one employer transition and at least one inter-municipality move, respectively.

Two clear results emerge. First, the quality of the initial workplace is strongly correlated with employment probability and annual earnings of refugees in the short run (1–5 years), medium run (6–10 years), and long run (11–15 years). A one standard deviation increase in the pay premium offered by the first employer is linked to a higher probability of being employed by 3.6 percentage points, and to higher yearly earnings by 3416 USD in the short run (Panel A). Second, the persistence of the effect for up to 15 years after the beginning of the first employment spell is remarkable. Varying the fixed-effects specification leaves the magnitude and significance of the estimates unchanged, as does restricting the sample to a subset of refugee movers. Importantly, in Panel B we focus on refugees who had at least one transition between employers to rule out the possibility that these persistent effects are mechanically due to refugees staying with the same employer throughout the period, finding only small decreases in the magnitudes. While there is no causality in these regressions, as sorting of more productive workers to better firms is not completely accounted for, this descriptive evidence suggests that initial workplace quality is directly connected to refugees’ outcomes.

Beyond its economic relevance, it is worth considering the methodological implications that this persistence has for the two-way fixed effects models commonly used to document firm wage differences. Table 4 confirms that this long-run correlation also holds for refugee wages, the outcome typically used in the AKM decomposition.²³ A time-dependence effect suggesting that individual productivity is shaped by the first employer fixed effect would directly violate the fundamental assumptions of the AKM model when applied to specific groups such as refugees, and potentially other immigrants, who display this pattern. Specifically, the AKM framework views wage determination as a fundamentally static process, assuming that worker and firm effects enter the equation additively and separately. Not only does this framework

²²Table A2 reports estimates from an alternative specification without fixed effects.

²³We run the same regressions on a larger sample of refugees who arrived in Denmark during this period and find that the observed patterns are not driven by our restrictions of the analysis sample (Table A3).

ignore dynamic effects, but it cannot reliably estimate person and employer effects if past match quality has a lasting impact on an individual’s future earnings capacity.²⁴

Our observation is similar in spirit to recent works documenting dynamic wage effects from starting in large firms (Arellano-Bover, 2024), from experience in firms with greater on-the-job learning opportunities (Arellano-Bover and Saltiel, 2024), and from spending the first formative years in the labor market in larger cities (De La Roca and Puga, 2017).²⁵ However, the source of dynamic effects that we describe, arising from the fixed effect of past employers, cannot be easily accounted for with an augmented static AKM specification. Relatedly, Bonhomme et al. (2019) document an effect of the previous employer on workers’ current earnings, conditional on the type of the current firm, using an alternative two-step empirical approach in which firms are grouped into classes via k -means clustering. By contrast, here we highlight a departure from AKM assumptions while remaining within the AKM framework, emphasizing the dynamics triggered by first employers rather than previous ones.²⁶

Importantly, this pattern only pertains to our foreign-born group, as Table 4 does not reveal the same dynamic pattern for a comparable sample of native labor market entrants, confirming the validity of our approach. On the one hand, this reassures us that, given our controls and fixed effects, our estimates are not solely capturing sorting on unobservable skills. If sorting was driving our results for refugees, then it would imply that natives are not sorting in the same way. This seems implausible because, if anything, we would expect natives to be able to sort more easily and effectively in the labor market.²⁷ On the other hand, this discrepancy provides further justification, beyond the rationale of avoiding mechanical effects and the lack of within-firm differentiation evidence discussed in Appendix B.2, for estimating employer fixed effects using a pooled job spell dataset that includes natives and non-refugee immigrants, as we do in Section 4. We further rule out sorting as the sole driver of these estimates by estimating, in a separate AKM decomposition on a pooled sample, individual fixed effects for refugees and then plotting them against their first employer fixed effects. Figure A4 shows no evidence of positive assortative matching.²⁸

We interpret this difference in the role of the first employer as evidence of a novel and additional aspect that differentiates refugees from other workers. For refugees, the first workplace represents a critical point of entry into a new labor market. It carries greater potential for learning, acquiring context-specific human capital, and building professional connections. A better first employer may also buffer against negative initial shocks and offer stability in early

²⁴Rather than reflecting improvements in earnings capacity, the persistent first-employer effect could arise from network channels. Refugees who start at high-quality employers may have stronger initial connections that continue steering them toward high-quality employers after switching. Panel C of Table 3 addresses this concern by re-estimating the same specification on refugees who ever moved away from their initial municipality, and thus their initial network, finding very similar results. While the precise mechanism is hard to pin down, this suggests that the persistent first-employer effect cannot be fully driven by network-induced patterns either.

²⁵To account for the latter dynamic, Card et al. (2025) estimate augmented AKM specifications that incorporate the dynamic effects of time spent in large, high-wage US commuting zones. They find that doing so has very little impact on the estimated place effects in the aggregate, and therefore proceed with the simpler model.

²⁶Our evidence points to mechanisms more consistent with human capital development than with an offer-counteroffer mechanism in the spirit of Postel-Vinay and Robin (2002).

²⁷Refugees’ skill signals are likely much noisier at arrival, and their opportunities for sorting into better jobs are more limited.

²⁸In fact, Figure A4 reveals some degree of negative assortative matching for refugees, consistent with the hypothesis that they face greater difficulty signaling their skills upon entering the host labor market.

years at destination. In contrast, natives possess local skills, face fewer barriers, can move freely, know the labor market better, including how and where to search for other jobs. As a result, while a large literature has emphasized the importance of entry labor market conditions for workers in general, the long-term consequences of an initial job match are likely to be even more consequential for refugees than for natives.

Fact 2. We then explore the stickiness of initial employer quality for refugees. Figure 2 plots the evolution of firm pay premia earned by quartile of the first employer quality distribution, separately for refugees and a comparable set of natives entering the labor market for the first time. We draw two key insights from this chart. First, refugees do not move up and down the firm ladder as easily as natives, whose average pay premium by quartile group converge rapidly over the first five years since the beginning of the first job spell. Second, this pronounced stickiness in initial quality for refugees appears to persist, as a sizeable gap remains open at years 10 and 15.

These results are consistent with the extensive evidence documenting the numerous obstacles refugees face, as well as the persistent labor market gaps between refugees and other groups of workers in host countries (Bratsberg et al., 2017; Schultz-Nielsen, 2017; Brell et al., 2020; Fasani et al., 2022). At the same time, these findings underscore the importance of the initial conditions refugees encounter upon arrival, including the quality of accessible workplaces. Our results align with Åslund et al. (2025), who show that immigrants in Sweden are less likely than natives to climb the productivity ladder and transition across firms. We confirm a similar pattern for refugees, whose mobility is particularly constrained.

Fact 3. Lastly, we assess the role of social connections in shaping job search outcomes for refugees. Specifically, we aim to determine whether the presence of a connection to a firm influences the probability that a refugee worker is hired by that firm. We begin by defining a connection as a co-national residing in the same municipality to which the refugee is initially assigned and employed at a given establishment when the refugee arrives. Following the approach of Eliason et al. (2023) and Åslund et al. (2024), we construct a dataset of refugee-establishment dyads, pairing each refugee i with every establishment j active in their municipality of assignment at the time of arrival. We then use the following specification to recover the effect of being connected to establishment j through a member of the local co-national network at arrival on the probability of being hired by j :

$$[Pr(\text{Hired by } j)]_{ij} = \alpha + \beta D_{ij} + X_i' \gamma + \lambda_j + \epsilon_{ij} \quad (2)$$

where the outcome variable is an indicator for whether refugee worker i is employed at establishment j at any point in their working career, multiplied by 100, and D_{ij} is a dummy variable equal to one if refugee i has a co-national connection at j . Importantly, we restrict our sample to dyads where establishment j has hired at least one refugee with a connection there and at least one without. This restriction eliminates pairs with no variation in the existence of a connection, making the cardinality of the sample tractable.

Table 5 reports estimates of parameter β using different sets of controls and fixed effects.²⁹

²⁹Regardless of the specification employed, our estimates likely represent a lower bound, as we do not exclude refugees who relocate to a municipality different from their initial assignment. While this relocation choice is

In column 1, we include a set of individual controls consisting of variables observed by Danish authorities when assigning refugees to municipalities as part of the dispersal policy. In column 2, we add education controls. In column 3, we also include establishment fixed effects to focus on within-establishment variation in connectedness, effectively controlling for factors such as differing hiring strategies across establishments. Column 4 presents the most restrictive specification, including establishment-by-cohort fixed effects to account for different hiring situations at the same establishment faced by different cohorts of refugees. Our preferred specification (column 3) indicates that having a local co-national connection at arrival increases the probability of being hired at a given establishment by 0.06 percentage points for newly arrived refugees.

Overall, we interpret the evidence discussed in this section as suggesting that establishment quality is strongly related to refugees' labor market outcomes, and that initial conditions influence their subsequent transitions across workplaces. While refugees move along the employer ladder, they do so sluggishly, relying on co-national networks upon arrival when searching for a job. We will explore these dynamics more formally in the next sections.

6 Empirical Approach and Identification

The main aim of this paper is to estimate the effect of employer quality on refugees' labor market outcomes. Ideally, one would randomly assign identical refugees to firms of varying quality upon arrival and track their subsequent integration. However, such an experiment is not feasible in practice. As a consequence, sorting across locations and employers complicates drawing causal conclusions, and the evidence presented in Section 5 cannot completely rule out the possibility that unobserved productivity drives matching.

In this section, we outline the empirical strategy used to circumvent this issue and estimate instead the effect of the quality of refugees' first accessible employers on their outcomes. To do so, it is important to address this question in Denmark, where we can rely on exogenous variation in refugees' exposure to employers by leveraging the implementation of the 1986–1998 dispersal policy. For this approach to credibly address sorting, avoiding that newly arrived migrants choose their residential location based on individual skills or preferences that also affect their outcomes, we require the identifying assumption that the placement policy is independent of unobserved individual characteristics. As discussed in Section 2, Table 1 presents a test of the conditional random assignment of refugees to municipalities. Family structure and country of origin are, by design, systematically correlated with initial location characteristics, since placement was conditional on them. Crucially, however, education at arrival, likely related to unobserved skills, does not appear to be correlated with municipality characteristics. This supports the assumption that individual refugee characteristics are independent of initial location characteristics, conditional on the observables available to the authorities, which we control for throughout our analysis.

This randomization allows us to obtain quasi-experimental evidence on the impact of exogenous, it mechanically reduces the likelihood that, after relocating, they will work at an establishment where they had a connection.

posure to different employers.³⁰ While individual refugee–employer matches are not strictly random, the dispersal policy ensures that refugees were quasi-randomly exposed to the set of active establishments in their assigned municipality upon arrival, as well as to those that hired members of the co-national network they initially joined. After estimating AKM-based establishment fixed effects as described in Section 4, which serve as our proxy for establishment quality, we then proceed to aggregate them at three different levels of exposure, progressively refining our definition to identify the most relevant employers for newly arrived refugees.

First, we investigate the effect of “pure” geographic exposure by averaging individual establishment effects, ψ_j , across all active establishments in the municipality of assignment at the time of arrival.³¹ Second, we assess the role of employer quality by focusing only on establishments in the municipality of assignment that, at the time of arrival, were employing or had previously employed at least one individual from a refugee-sending country. Finally, we estimate the effects of an exposure measure based on averaging individual establishment effects in the municipality of assignment for all establishments that, at the time of arrival, were employing at least one member of the local co-national network. To mitigate remaining endogeneity concerns, these measures are constructed one year prior to the admission year of each newly arrived refugee.

Depending on the level of exposure considered, we use the function f to average individual establishment effects, $\hat{\psi}_j$, across these three different sets of establishments. While we will explore all three levels, we emphasize and use the last as our preferred measure of exposure, as it captures the effects of having co-national network links to high-paying local firms at arrival. Thus, our main specification estimates the effect of exposure to employer quality at arrival on refugees’ labor market outcomes as follows:

$$y_{ioc,t} = \alpha + \beta_1 f[\hat{\psi}_j]_{om,c-1} + \beta_2 Sh.Conat.om,c + \beta_3 Emp.NW_{m,c} + \mathbf{X}_{i,t} + \gamma_m + \delta_o + \mu_c + \epsilon_{ioc,t} \quad (3)$$

where $y_{ioc,t}$ denotes either employment probability or real annual earnings for refugee i from country of origin o at time t , who arrived in Denmark in year c (cohort) and was assigned to municipality m . Here, $f[\hat{\psi}_j]_{om,c-1} \equiv [\overline{\hat{\psi}_j}]_{om,c-1}$ captures the average quality of establishments in municipality m employing other migrants from country o , one year prior to the arrival (i.e., $c - 1$) of refugee i .³² The parameter of interest is β_1 , which, given the research design, can be interpreted causally when estimated by OLS. To facilitate interpretation, we divide our main regressor by its standard deviation. To account for the fact that the average of $\hat{\psi}_j$ is a generated regressor, we will also report p-values after calculating bootstrapped standard errors, as in [Bana et al. \(2023\)](#).³³

³⁰Other studies have exploited variation at highly disaggregated geographical levels, such as residential housing blocks, under the assumption of no endogenous sorting at the block level within neighborhoods ([Boeri et al., 2015](#); [Schmutte, 2015](#)). However, the policy experiment created by spatial dispersal policies arguably represents the most methodologically rigorous approach ([Schüller and Chakraborty, 2022](#)).

³¹Importantly, [Card et al. \(2025\)](#) show that AKM-based aggregations used to estimate location effects avoid bias from selective mobility that affected earlier place-effect estimates. This bias arises in two-way fixed-effects models with person and place effects because geographic movers sort selectively across establishments at different levels of the local job ladder.

³²Had we focused on the impact of geographic quality by averaging establishment effects of all establishments active in the municipality of assignment at the time of arrival, the main regressor would have been $f[\hat{\psi}_j]_{m,c-1}$.

³³We adopt a wild cluster bootstrap procedure, which performs better than traditional simpler bootstrap pro-

Equation 3 includes controls for the population share of co-nationals in the municipality of assignment at arrival, $Sh.Conat_{om,c}$, and for the employment rate of non-Western immigrants, $Emp.NW_{m,c}$. The former controls for country o 's comparative advantage in m , which can influence geographic sorting by generating a greater inflow of co-nationals. The latter allows to control for broadly-defined local economic characteristics favoring the integration of non-Western migrants in general. Additionally, the vector $X_{i,t}$ contains individual background characteristics in the year of assignment, including those observed by the authorities during the dispersal process.³⁴

Our preferred specification includes municipality of assignment fixed effects, γ_m , which account for time-constant differences in economic advantages across municipalities, effectively capturing location m 's absolute advantage for all refugees. Municipality fixed effects are particularly important in our setting as the location of high-paying employers is not random but likely linked to other location-specific factors. We also include origin fixed effects, δ_o , to control for permanent differences across origin groups, and cohort fixed effects, μ_c , to account for aggregate differences across arrival cohorts.³⁵ Robust standard errors are clustered at the municipality of assignment level throughout the analysis.

Our estimand should be interpreted as capturing *intention-to-treat effects*. We estimate the average impact of being exposed through the network to high-quality firms, regardless of whether individuals ultimately work in those firms or relocate after the initial assignment. In this sense, refugee movers do not pose a threat to our approach. On the contrary, mobility that depends on the initial quality of employers is itself an outcome that we can and will analyze.³⁶ Still, as discussed in Section 2, the dispersal policy had lasting effects on refugees' settlement patterns: after eight years, more than 50% remained in their municipality of initial assignment, and after 15 years, more than 40%.

Importantly, a locality's firm structure likely correlates with other local characteristics. While the quasi-randomization of refugees to localities ensures the estimation of a causal parameter, such a parameter may in principle not measure the effect that firm quality itself has on labor market outcomes, but rather represent the effects of firm quality through its correlation with municipality characteristics affecting integration.³⁷ Table A5 explores the unconditional relationship between establishment-based network quality and other local characteristics. On the one hand, this assesses the extent to which our measure captures a distinct dimension not already reflected in existing observable characteristics of municipalities. Reassuringly, network employer quality, averaged over countries of origin and years within each municipality for the purpose of this table, is not perfectly correlated with any single characteristic. On the other hand, this allows us to assess whether the correlations have the expected signs—that is, whether the bundling is of the kind one would anticipate—which is indeed the case here.

cedures (Cameron and Trivedi, 2022), adopting standard Rademacher weights and bootstrapping standard errors 9999 times.

³⁴More specifically, we include age at entry (quadratic and cubic terms as well) and a dummy for gender, for being married, for having kids (in two age ranges), for having the spouse in Denmark.

³⁵The set of fixed effects we adopted is commonly used by existing studies in this literature (e.g., Edin et al. (2003); Battisti et al. (2022)). We consider alternative, more restrictive fixed-effects specifications in Section 7.

³⁶We address concerns related to multiple treatments and nonrandom refugee mobility in Sections 7.4 and 8.1.

³⁷The challenge of disentangling which specific characteristic of a location is responsible for the observed causal impact is sometimes called the “bundling problem” in the peer effects and neighborhood effects literature.

Nevertheless, the nature of our setting gives us confidence that we eliminate most correlated effects and can interpret our estimates as intended. The prolonged nature of the dispersal policy and the construction of our main regressor at the co-national network level allow us to condition on municipality fixed effects, not just on a large set of time-varying municipality controls, thereby removing also time-invariant unobserved local effects that may be correlated with both integration outcomes and firm quality. In addition, in Section 7.4.4, we conduct several robustness tests to show that any remaining influence of time-varying unobserved local characteristics that are correlated with municipality firm quality and labor market outcomes is likely to be small.

We now characterize our main regressor. Figure A5 shows the raw variation in our main treatment variable, the average quality of accessible employers in the co-national network, measured using AKM establishment effects, in the analysis sample. The standard deviation of 0.11 log points corresponds roughly to an 11% wage gain linked to better employers' pay policies when moving from a network at the 25th percentile to one at the 75th percentile of the quality distribution.³⁸ This amount of variation is slightly lower but comparable to the standard deviation of U.S. location wage premia, measured by averaging establishment effects at the commuting zone level, which is 0.15 log points (Card et al., 2025). Furthermore, Figure A7 shows the distribution of the number of connections (both employed network members and distinct establishments) that contribute to the average employer quality measure for each co-national network in the sample.

7 Results

7.1 Employment and earnings

Our main results derive from the estimation of equation 3 at the three different levels described in Section 6. Panel A of Table 6 shows the effects of geographic exposure to employers of differing quality on employment and earnings for our sample of refugees. Specifically, the table indicates that assignment to a municipality with higher-quality employers leads to a modest reduction in the probability of employment and earnings in the short run (1–5 years after arrival). *Ceteris paribus*, being surrounded by high-paying firms seems to be detrimental immediately after arrival, as these establishments might be more difficult to access. Our estimates reveal no significant effects on employment or earnings in the medium run (6–10 years) or the long run (11–15 years) after arrival.

Because, for consistency with Panels B and C, the coefficients in Panel A are estimated using the full set of fixed effects from equation 3, we re-estimate these regressions focused on geographic effects including only cohort fixed effects. This exercise allows us to avoid relying solely on within-municipality variation by effectively comparing two refugees who arrived in the same year and were assigned to different municipalities. Results in Table A6 confirm the short-run negative effect found in Panel A from the main, more saturated model.

³⁸Table A4 also reports the residual variation in the treatment variable after partialling out the main set of fixed effects used in the analysis, showing that substantial variation remains. Figure A6 further shows that all three dimensions (municipality, arrival cohort, and origin country) contribute meaningfully to the overall variation.

In Panel B of Table 6, we present estimates of the effect of exposure to higher-quality employers that had previously employed, or were employing at the time of arrival, at least one individual from a refugee-sending country. We view this measure of exposure as a refinement of the geographic measure in Panel A, as these employers have already demonstrated a willingness to hire refugees, making them a better proxy for the set of first accessible employers. The results are small in magnitude and show no significant effects on employment or earnings. Moreover, the signs of the point estimates alternate between negative and positive across the three intervals, suggesting null effects.

In Panel C of Table 6, we present our preferred estimates, which reflect the effects of exposure to higher-quality employers that employ co-nationals at the time of arrival. The results show positive, statistically significant effects on employment and earnings in the short run (1–5 years after arrival) and the medium run (6–10 years after arrival). A one standard deviation increase in the average workplace pay premium earned by the local co-national network at arrival leads to a higher probability of being employed by 0.9 percentage points in the short run and by 1.7 in the medium run, and to higher yearly earnings by 328 USD in the short run and 798 USD in the medium run. The effect remains positive in the long run (11–15 years after arrival) but without being statistically significant. These magnitudes are economically meaningful: in the first five years, the effect sizes correspond to an 8% increase in employment probability and a 10% increase in annual earnings, relative to the respective sample means.³⁹

Our main results show that having network links to high-quality firms benefits the labor market integration of refugees for up to ten years after arrival. As our measure of early employer quality is refined to incorporate co-national network links, the effects of exposure shift from being negative to positive and from detectable only in the short run to detectable in both the short and medium run. This suggests that geographic exposure to higher-quality employers alone does not drive the labor market integration of refugees; rather, it is exposure to high-quality employers that employ members of the network that provides these benefits. Crucially, the positive effects are larger when considering employers that employ co-nationals at the time of arrival compared to those that had previously employed workers from refugee-sending countries, indicating that the benefits are driven by contemporaneous co-national links to high-quality employers rather than by employers' general willingness to hire refugees. In Section 8, we will examine the specific network mechanisms that drive our main results.

Interestingly, comparing Panel C of Table 6 with the potentially endogenous estimates of Table 3, we can see that the effects are much smaller in magnitude. Two explanations are plausible. First, estimates from the initial exercise considering the actual first employer may include some upward bias, possibly through match-specific components between refugee workers and establishments that are not fully accounted for. While we did not find evidence supporting positive assortative matching between refugees and their first employers (see Figure A4), we cannot completely rule out this possibility. Second, the discrepancy may arise because of a difference in estimands: the first-employer specification captures a direct effect that accrues immediately upon employment for those hired, whereas our main estimates are intention-to-treat effects of network quality, measuring the average impact of exposure

³⁹Section 7.2 further discusses these magnitudes in relation to the existing literature.

to higher-quality employers, which are estimated on a broader sample and may or may not materialize for each individual.

It is also important to note that, while virtually all refugees in our sample are assigned a measure of geographic exposure to employer quality, this is not the case when the measure is refined to account for the average quality of firms currently employing network members. Refugees assigned to municipalities without co-nationals, or where co-nationals work for establishments outside the largest connected set, do not receive a network exposure measure. Table A7 reports differences in characteristics between refugees matched with a network measure and those unmatched. As expected, matched refugees are more likely to reside in municipalities with a higher number of co-nationals and in larger, more educated, and more urbanized municipalities. The two groups also differ in some individual characteristics, all of which are important to consider when generalizing our results.⁴⁰ In this light, the estimates from Panel C of Table 6 can be interpreted as the effect of firm quality within the co-national network on refugees' outcomes *conditional* on having a measurable network measure. In Table A8, we compare outcomes for individuals with no observable network to those with networks of varying firm quality, split into terciles. This evidence suggests that medium- and high-quality networks in terms of employers are more beneficial for integration than either a low-quality network or no network at all, particularly for employment probabilities. In line with economic intuition, having a low-quality network and having no network seem to have comparable effects on labor market outcomes, likely reflecting some form of free disposal.

Finally, we examine whether the effect found on earnings is driven by higher earnings among those employed or by higher employment probability. Specifically, we want to make sure that higher firm quality in the network boosts earnings not only by bringing more refugees into work, but also through better jobs, higher wages and productivity, and/or improved job stability. In Table A9, we estimate the effect on annual earnings conditional on refugees reporting positive earnings, expressed in thousands in columns 1 to 3 and in logs in columns 4 to 6. The positive and statistically significant pattern is confirmed in both the short and medium run. The fact that the coefficient magnitudes do not decline when conditioning on positive earnings indicates that the increase in earnings documented in the main results is driven disproportionately by the intensive margin, or more precisely by a shift in the distribution of earnings among those employed.

Influencing refugees' employment and income can generate spillover effects beyond the labor market, shaping social integration more broadly. One such dimension is crime outcomes for refugees and their families. We leave estimates for crime outcomes to Appendix C.

7.2 Comparison with prior studies

It is important to frame our estimates within the extensive work on integration. Our analysis speaks more directly to the literature on the role of initial labor market conditions for refugee integration, which has mostly focused on the strength of the local labor market and

⁴⁰In particular, refugees in the matched group are more likely to be female, less likely to come from predominantly Muslim countries, and less likely to be native speakers of Latin languages. They also tend to have higher initial labor income and are more likely to be employed in jobs with a higher complexity index.

availability of co-national support, finding significant effects for both (e.g., [Edin et al. \(2003\)](#); [Damm \(2009a\)](#); [Damm and Rosholm \(2010\)](#); [Azlor et al. \(2020\)](#)).⁴¹ We include these variables as controls but do not focus on their coefficients, and their lack of statistical significance does not contradict earlier studies. By introducing municipality fixed effects in our main specification, we absorb part of the effects estimated in previous work, which, even though best seen as municipality-by-year of arrival effects, tend to be relatively stable over time. As a result, rather than originating from general differences across municipalities, our results can be seen as being “on top of” most municipality effects documented by the literature, and thus complementing it by adding the employer dimension to the analysis.⁴²

Specifically, we present evidence that efforts to promote labor market integration should not focus only on obtaining a job, as much of the existing literature has emphasized, given refugees’ typically low labor market attachment ([Brell et al., 2020](#)). We show that the *quality of the employer* offering the job plays a crucial role in shaping refugees’ trajectories at destination over time, highlighting the importance of employer characteristics for successful integration. This idea complements the finding by [Degenhardt and Nimczik \(2025\)](#) that access to gig jobs in online food delivery initially accelerates job finding among refugees in Austria, but that refugee gig workers tend to remain in low-paid, unstable jobs with limited career prospects.

Compared to other integration policies evaluated in Denmark, the magnitude of our effects is meaningful but not huge. Relative to the 1999 language training policy, considered the most effective intervention for refugees’ employment probability and earnings in the long run ([Foged et al., 2024](#)), our estimates are about one-sixth as large in the short run and about one-third as large in the medium run.⁴³ Nevertheless, the effects we find in the first ten years after arrival are important, given that the early years are a crucial period for integration.

Our results also align with the AKM literature on wage inequality, which attributes a substantial share of earnings gaps between groups to between-firm sorting (e.g., the gender gap in [Card et al. \(2016\)](#) and the native-immigrant gap in [Dostie et al. \(2023\)](#)). As discussed, our estimates should be treated as intention-to-treat effects, capturing the average impact of being exposed, through the network, to high-quality firms. This approach is appealing because it allows us to draw causal conclusions, but it limits direct comparability with AKM studies that focus on the consequences of direct employment in different firms without quasi-experimental designs. Still, using a job ladder model based on a firm productivity grouping, [Åslund et al. \(2025\)](#) estimate that working in the fifth decile of the firm productivity distribution yields earnings returns of 8.5 log points for natives and 12.1 log points for immigrants in Sweden, with the differential returns driven primarily by non-Western immigrants. While not identical in interpretation, our earnings effects are in the same ballpark, highlighting that network-provided access to better firms can have economically important consequences, particularly

⁴¹When analyzed together in Denmark, higher employment rates in the host local labor market, but not larger networks of co-nationals, are associated with better employment and earnings outcomes for refugees, in both the short run and the long run ([Foged et al., 2024](#); [Damm, 2014](#)).

⁴²The early literature on the effects of living in co-ethnic enclaves on individual earnings ([Edin et al., 2003](#); [Damm, 2009a](#)) also included municipality fixed effects. However, unlike our approach and the more recent dispersal policy literature discussed here, it estimated a contemporaneous relationship between ethnic concentration and individual earnings, rather than focusing on conditions at arrival, even though the importance of cumulative effects was already documented in [Edin et al. \(2003\)](#).

⁴³The language training policy provided 400 extra hours of training to refugees.

for non-Western immigrants, who are precisely the focus of our study.

7.3 Decomposing network quality: employer or employee quality?

Our causal estimates indicate that refugees' labor market outcomes improve with the quality of employers accessible through their network. However, this measure may not fully isolate the role of firms if the employer quality in the network partly reflects the quality of network members employed there.⁴⁴ As a result, high firm quality in the network may also capture correlated member quality, which is itself a potential driver of individual labor market outcomes. Put differently, well-established and high-achieving ethnic peers can act as role models, potentially motivating newly arrived immigrants to invest in skill acquisition.⁴⁵

Addressing this concern allows us to answer a more fundamental question in the network literature by relying on a simple decomposition of the network quality effect using the two-way AKM structure. Previous studies on networks have documented that the returns to living in an ethnic enclave increase with enclave "aggregate" quality, most commonly proxied by average annual earnings in the group, but also by alternative measures such as the share of self-employed or the share of highly educated (e.g., [Edin et al. \(2003\)](#); [Damm \(2009a\)](#)). In line with a linear two-way fixed-effects model for individual earnings, the average earnings in network k at any given time can be expressed in terms of members' earnings capacity and the firm pay premia of their employers as follows:

$$\overline{\ln(y)}_k = \frac{1}{N_k} \sum_{i \in N_k} \ln(y_i) = \underbrace{\frac{1}{N_k} \sum_{i \in N_k} \alpha_i}_{\text{average member ability}} + \underbrace{\sum_{j \in J} \pi_{kj} \psi_j}_{\text{average firm pay premium}} + \bar{\mathbf{X}}_k' \beta + \nu_k \quad (4)$$

where i indexes individual members, N_k is the total number of members in the network, α_i is individual ability, ψ_j is establishment j 's pay premium, π_{kj} are exposure shares to establishments with $\sum_j \pi_{kj} = 1$, and ν_k captures deviations due to estimation error and model misspecification. Hence, including both the average member ability and the average firm pay premium terms in our main specification allows for a cleaner assessment of the role of firms, while also contributing to the network literature by shedding light on the mechanism behind the positive effects of living in networks.

We do so by constructing two network-level controls that capture the quality of network members. First, by including refugees into the same job spell dataset used in Section 4 and re-running the AKM decomposition over the dispersal policy period, we estimate a vector of individual fixed effects (α_i) that proxy for unobserved permanent ability related to earnings potential. We then average these coefficients within each network to create a measure of member quality. Second, we compute the share of network members with academic education prior

⁴⁴Even though our AKM-estimated measure of firm quality, by construction, purges establishment fixed effects of worker unobserved heterogeneity in the full economy, it does not rule out nonrandom sorting within the specific subset of workers in an individual's network.

⁴⁵Examples of positive human capital externalities are documented in [Åslund et al. \(2011\)](#) and [Chakraborty et al. \(2019\)](#); negative effects detrimental to success in [Borjas \(1995\)](#), [Cutler and Glaeser \(1997\)](#), and [Bertrand et al. \(2000\)](#).

to arrival, providing an alternative and more traditional measure. Each of these controls is included separately in our main specification 3, along with the average quality of employers within the network.

Table 7 presents the results of this horse race exercise. Panel A reports estimates for the effect of aggregate quality, using average real annual earnings in log in the network.⁴⁶ The coefficients are small, positive, but not statistically significant. Panels B to C include controls for average network-member quality, measured by AKM worker effects in Panel B and by the share of highly educated members in Panel C. The coefficients on average firm quality remain positive, statistically significant, and stable in magnitude, while those on average member quality are small, negative and insignificant. This suggests that our establishment-based network quality measure primarily reflects firm quality rather than the sorting of high-ability individuals, so that the positive impact of network quality previously documented in this setting is driven by connections to better-paying firms.

More broadly, while disentangling the relative importance of different channels behind the positive effects of co-ethnic enclaves is empirically challenging, these results suggest that our measure of firm quality primarily captures the opportunity-augmenting role of the network. Therefore, consistent with the idea of networks providing a “warm embrace” (Borjas, 2000), the causal effects documented previously seem to reflect social interactions as friction-reducing mechanisms in the job-search process, rather than arising from human capital externalities. Put differently, in our setting networks are helpful insofar as they connect new members to strong labor demand; while high-quality members may provide that access, they may also generate stronger competition, so that ultimately employer quality matters more than employee quality. We will expand on this point in our discussion of mechanisms in Section 8.

7.4 Robustness

We perform several tests to ensure the robustness of our main results for refugees’ employment and earnings. These checks are briefly summarized below, and the corresponding evidence is presented in Appendix D.

7.4.1 Aggregation of establishment quality

We begin by exploring alternative constructions of the network measure of firm quality, replacing the simple average used in equation 3 with different functions f to aggregate individual establishment effects $\hat{\psi}_j$ within networks. This allows us to assess whether our results depend on a particular aggregation choice. First, we consider a weighted average using full-time equivalents at each establishment as weights. The results from re-estimating equation 3 are reported in Table D1. Panel A examines geographic exposure to firms, using the weighted average of all employers in the municipality, while Panel B instead considers network-level

⁴⁶Edin et al. (2003) and Damm (2009a) examine the role of enclave quality using 2SLS specifications where the main effect of the quality measure is not separately identified from country-of-origin fixed effects. Their variables of interest are interactions between quality and ethnic concentration. By contrast, we can include and estimate the main effect directly in our design. Focusing on neighborhoods within municipalities, Damm (2014) estimates the main effect of quality of co-nationals, defined as the employment rate of men aged 18–60, their mean years of education, and log mean real annual earnings. Results for the latter, however, are not reported.

exposure, using the weighted average of local employers that employ co-nationals at the time of arrival. For this latter measure, estimates are remarkably similar in magnitude compared to those in Panel C of Table 6.

Second, instead of averaging establishment fixed effects, we assign a dummy variable equal to one for establishments in the top quartile of the Danish workplace pay premium distribution, and then calculate the average of this dummy within each network. This approach yields the proportion of top employers in the local co-national network. The estimates obtained using this alternative measure are reported in Panel C of Table D1 and confirm the sign and significance of our main results. However, their magnitudes are larger, which we attribute to the emphasis this specification places on top employers.

7.4.2 Specifications and sample restrictions

We also estimate alternative specifications of equation 3. Including both our geographic and network-level measures of exposure to firm quality jointly, the coefficients of interest for employer quality in the co-national network, reported in Table D2, remain virtually unchanged in magnitude, sign, and statistical significance compared to the separate estimation. This suggests a significant degree of orthogonality between the effects of geographic and network-based measures on refugees' outcomes. This result is consistent with Figure 3, which shows that municipalities with higher average establishment quality do not necessarily coincide with those having higher average network-based establishment quality. The stability of the coefficients on network-based employer quality, even after controlling for municipality-level employer quality, reinforces our confidence that social connections play a primary role in facilitating refugee integration by improving access to higher-quality employers.

One potential concern is that establishments employing co-nationals provide particular types of jobs that network members are more likely to obtain, independent of the role of social connections in facilitating access. To address this possibility, we use industry information for each establishment to classify jobs held by employed network members into one of nine industry groups.⁴⁷ Controlling for the industry shares of the network in Panel B of Table D2 leaves our main results very similar. Next, we perform a falsification test by restricting our sample to refugees who relocate to a municipality different from their assigned one within their first five years after arrival, whom we refer to as "early movers".⁴⁸ Consistent with the idea that networks, by construction, can only provide valuable connections to good firms in the municipality of assignment, the estimates reported in Panel C of Table D2 for this subset of refugees are substantially smaller in magnitude and statistically insignificant.

We then consider alternative, more restrictive fixed-effects specifications in Table D3. In Panel A we include origin-by-cohort fixed effects to control for selection along unobservable factors in inflows from origin o . The quality of employers remains identified because it varies

⁴⁷The nine NACE industry categories are: Agriculture, fishing, and quarrying; Manufacturing; Electricity, gas, and water supply; Construction; Wholesale and retail trade, hotels, and restaurants; Transport, storage, and communication; Financial intermediation and business activities; Public and personal services; and Activity not stated. Unfortunately, we cannot explore the role of occupations of co-national workers, as occupation codes are unavailable in our data prior to 1991.

⁴⁸The average duration of residence in the municipality of initial assignment for this group is 2.7 years.

across origin-by-cohort groups within municipalities. In Panel B we use municipality-by-cohort fixed effects to absorb all factors common to refugees assigned to the same municipality in a given arrival cohort. Our main results are confirmed.

Finally, we check the sensitivity of our results to sample restrictions. Table D4 shows that coefficients remain statistically significant and stable after trimming the top or bottom 10% of observations in our refugee sample based on size of co-ethnic network at arrival (Panel A), population in the municipality of assignment (Panel B), and municipality of assignment's share of national full-time equivalents (Panel C). This confirms that our results are robust to changes in sample selection and hold more broadly within the population of interest.⁴⁹

7.4.3 Measurement of establishment quality

The accuracy of estimates of individual establishment effects is crucial for our empirical exercise. Tables B2 and B3 confirm that these estimates are not purely driven by noise, as they display a positive and statistically significant correlation with other observable measures of establishment productivity and desirability, in line with the existing literature (Kline, 2024). Building on this, we introduce split-sample instrumental variable (IV) estimates to correct for measurement error that results from the well-known “limited mobility bias” in AKM model estimates (Kline et al., 2020; Bonhomme et al., 2019).⁵⁰ In practice, we randomly split the sample of workers used in the estimation of equation 1 into two equal sized samples (keeping person histories together), calculate two sets of AKM estimates separately in both subsamples, aggregate establishment effects at the network level, and then use the network-level quality from one set of establishment effects as instrumental variable for the other in the same specification as equation 3.⁵¹ Table D5 shows that the IV estimates are comparable in magnitude to our main results, indicating that our findings are robust to this correction. This finding is unsurprising, as the bias is known to be particularly problematic in short panels and in the presence of small employers. These concerns are mitigated in our setting: our panel spans a relatively long period (1986–1998), establishment effects are aggregated at the network level, and worker mobility rates are reasonably high (Table B1).

7.4.4 Local correlates

Another potential concern is that our network-based measure of employer quality may capture local confounding variables that make a municipality generally advantageous for refugees. If this were the case, we would expect the effects associated with the co-national network not to differ significantly from those of a network including all refugees. In principle, we can already rule out this possibility by comparing Panel B and Panel C of Table 6, but we can take this one step further. To assess whether co-national network effects drive our findings, rather than other hidden local characteristics related to the treatment variable, we perform a

⁴⁹Figure A9 further shows that our results are not driven by the largest municipalities of assignment.

⁵⁰Establishment effects are only identified through worker mobility. When the number of movers per establishment is small, the estimated effects from equation 1 may be noisy, potentially leading to attenuation bias in the estimates of equation 3.

⁵¹Similar strategies are implemented in Goldschmidt and Schmieder (2017), Schmieder et al. (2023) and Gerard et al. (2021).

permutation-based pseudo-placebo test. Specifically, we run 1,000 replications of equation 3, each time assigning to dispersed refugees of a given country the average employer quality from the network of one of the other eight origins.⁵² By permuting the treatment across countries of origin, we generate a distribution of placebo-like effects under the null hypothesis that the treatment has an attenuated effect. We repeat this procedure for each outcome for which we found an effect in our preferred specification (employment over 1–5 years after arrival, employment 6–10 years, earnings 1–5 years, and earnings 6–10 years) and plot the resulting distributions in Figures D1 and D2. In every case, the true effect estimated in Panel C of Table 6, depicted by the black solid line, is more extreme than the 95th percentile of its respective placebo-like distribution. This test confirms that the relevant connections come from the co-national network rather than from an aggregate network of all refugees.

To further address concerns about omitted local confounders beyond the two main variables relevant for refugee integration included as controls in equation 3, we re-estimate our main specification iterating over possible combinations of local controls.⁵³ Figures D3 and D4 report the distribution of the resulting coefficient estimates and show very limited dispersion. The stability of the main coefficients is reassuring and suggests that we are not omitting important local variables that also affect integration.⁵⁴

7.4.5 Local skill-complementarity

We do not require network members to be randomly assigned to firms to recover causal estimates. However, absent such randomization, it remains possible that some unintended origin-related factors contribute to the observed effects. Suppose that municipality m is particularly attractive to migrants from origin o because of skill-complementarity between o -types and the local mix of most productive firms. While the municipality fixed effect γ_m controls for any systematic differences across locations affecting all refugees, it does not account for such group-specific advantages. To threaten our main findings, one would need unobservables specific to municipality-origin pairs that simultaneously increase the likelihood that migrants from a given origin enter high-quality firms and improve the integration of newly arrived co-nationals. Our extensive battery of robustness checks has already ruled out several relevant channels, suggesting any remaining influence of such unobservables is likely small. For instance, since comparative advantages of this kind would likely result in geographic sorting, our main specification already accounts for this by controlling for network size ($Sh.Conat_{om,c}$). Moreover, these unobserved factors would need to operate independently of individual network members' unobserved abilities and the industry composition of the network, both of which we have already accounted for in prior checks.

Nevertheless, we perform three additional exercises. First, we compute our main measure

⁵²A placebo test typically repeats the same analysis in a setting where no true effect is expected. In our setting, however, alternative permutations should attenuate, rather than eliminate, the estimated effects. In practice, we perform a derangement (i.e., no element appears in its original position) with repetitions. In this case, there are 14,833 such alternative ways to reshuffle the origin-specific measures.

⁵³To provide a more complete picture of the range of possible estimates, we generate $2^k - 1$ permutations using a set of k municipality covariates.

⁵⁴Because our regressions include municipality-of-assignment fixed effects, the scope for potential “multiple treatment bias” is already greatly reduced.

of employer quality in the network using only refugees who were randomly assigned to locations through the dispersal policy and remained in their assigned municipality. While this procedure restricts the number of networks we can observe, it still allows to construct a meaningful sample and compare this more restrictive, arguably more exogenous quality measure with the baseline version. Figure D5 plots the bivariate relationship between the baseline network measure (x-axis) and the more restrictive version (y-axis), showing that two measures are strongly related and share substantial variation.

Second, we replace the employment rate of non-Western immigrants in the municipality of assignment at the time of arrival ($Emp.NW_{m,c}$) with the co-national employment rate ($Emp.NW_{o,m,c}$) as a control. Any origin-specific comparative advantage would likely manifest itself in overall employment outcomes even before employment at high-quality firms. Panel A of Table D6 confirms the robustness of our main results.

Finally, we re-estimate equation 3 including region-of-origin-by-municipality and arrival-cohort fixed effects attempting to absorb unobserved municipality-origin factors. As origin-by-municipality fixed effects would be too demanding for our data, we grouped origin countries by broad geographic region (Middle East, Africa, and Southwest Asia) and grouped cohorts into three 4-year cohort bins. This adjustment may be particularly relevant if, for instance, top firms in certain municipalities systematically discriminate against particular nationalities, based for instance on visual or cultural stereotypes.⁵⁵ Table D7 presents the results employing this alternative set of fixed effects and using both our main network quality measure (Panel A) and the alternative focused on firms in the top quality quartile (Panel B) as main regressors. While medium-run estimates are affected, short-run estimates for employment and earnings across both panels remain virtually unchanged in magnitude and statistical significance, further ruling out this type of bias.

8 Mechanisms

This section examines the mechanisms through which exposure to good employers provided by co-ethnic networks affects refugees' labor market integration in Denmark. The literature on social connections in the labor market identifies two primary ways in which connections benefit job seekers. First, network members may share information about openings they have knowledge of. Second, both workers and employers may use job referrals to reduce information asymmetry in the search process. In both cases, social networks act as information-transmission mechanisms for their members, which is particularly beneficial for immigrants, who often lack the country-specific knowledge necessary for a successful job search.⁵⁶ We evaluate these complementary channels below, while we leave an exploration of the role of co-

⁵⁵Broader native anti-migration attitudes may be captured through local voting behavior. Using data from [Dustmann et al. \(2019\)](#), we estimate a specification that includes the municipality-level vote share for parties with anti-migration platforms across six election cycles (three municipal elections: 1989, 1993, 1997; and three parliamentary elections: 1990, 1994, 1998) as an additional control. The estimates reported in Panel B of Table D6 confirm the robustness of our results.

⁵⁶While networks may also accelerate integration through non-job-search channels, such as encouragement from family and friends or information about training and language programs ([Åslund et al., 2024](#)), this support is unlikely to vary systematically with employer quality, and our controls for network size and network employment outcomes further mitigate this concern.

national managers to Appendix E, as it is not central to our results given the limited number of such managers.

8.1 Referrals

A compelling explanation is that connections affiliated with good firms can provide specific recommendations and thereby facilitate employment in those same high-quality firms as well as access to better jobs. Although our data do not include direct information on job referrals, we examine additional job search outcomes and patterns among refugees in our sample that may offer indirect evidence of their use.

First, we assess whether exposure to higher-quality employers leads newly arrived refugees to work at better firms themselves, as measured by establishment fixed effects of their employers. We do so by re-estimating equation 3, using their employers' fixed effects as the outcome variable. Results are reported in columns 1 to 3 of Table 8. Panel A uses our baseline exposure measure to firm quality in the network as the main regressor, while Panel B uses the alternative based on the share of top-quartile firms in the network. Because of the smaller sample size available, we do not include municipality fixed effects in these specifications but instead control for the municipality's share of college-educated individuals aged 18-65 and its population share of the country total aged 18-65, key characteristics related to the presence of high-quality firms.

The estimates in columns 1 to 3 of Table 8 indicate that better employers in the co-national network help new refugees access better firms themselves, with statistically significant effects in the short, medium, and also long run. Specifically, a one standard deviation increase in the average workplace pay premium in the network leads to a 1.2% increase in the wage premium earned at the firms accessed in the first ten years after arrival. A back-of-the-envelope calculation suggests that roughly 60% of the main effect on annual earnings ten years after arrival that we have previously presented stems from access to better employers.⁵⁷ However, since both the dependent and independent variables in columns 1–3 are AKM-estimated, the resulting regression estimates may, in principle, be affected by sampling error to an even greater extent than in our main specification, as previously discussed in Sections B.1 and 7.4.3. To alleviate this concern, we follow [Dostie et al. \(2023\)](#) and [Gerard et al. \(2021\)](#), and re-estimate the same regressions using the split-sample IV approach described in Section 7.4.3. The results, shown in Table D8, closely resemble the estimates in columns 1–3 of Table 8.⁵⁸

Next, we consider whether exposure to better firms improves access to better jobs, using a measure of occupational complexity to proxy for job quality. We define job quality based on task content, constructing an index equal to the ratio of communication and cognitive tasks to manual tasks, which therefore increases with job complexity. To do so, we draw information on

⁵⁷To translate this wage premium effect into annual earnings, we multiply the OLS coefficients in Table 8 by the average hourly wage and then by the total number of annual working hours, rescaled to reflect the average employment probability of refugees in this sample.

⁵⁸It is not surprising that OLS estimates for this regression were not downward biased as ex-ante we do not know the direction of the bias when regressing an individual AKM firm effect on a network-level aggregation of AKM firm effects, unlike in regressions of person effects on firm effects, where the sampling errors in the estimated worker and firm effects are negatively correlated, leading to a downward bias (see, e.g., [Dostie et al. \(2023\)](#)).

task content from O*NET data, using the ISCO classification for jobs.⁵⁹ We use this measure of job quality as the outcome variable in regressions based on the same specification as equation 3 and report results in columns 4 to 6 of Table 8. Our estimates show that refugees with better employer connections are more likely to access better jobs, with statistically significant effects persisting into the long run, at least when considering the exposure measure to employer quality based on the presence of top-quality firms in the network (Panel B).

In Table E1 we also explore possible effects on the geographic mobility of refugees, which can be another channel for labor market success by facilitating movement towards job opportunities (Foged et al., 2024). The absence of a robust, significant effect on this margin is consistent with two countervailing forces. On the one hand, having connections to high-quality local firms may provide incentives to stay; on the other hand, early access to better firms and jobs may encourage subsequent moves in search of even better matches or amenities.⁶⁰

Complementing the evidence from Table 8, we then examine which individuals benefit most from these connections. Heterogeneity by education completed prior to arrival (Figure 4) and by age at arrival (Figure A8) reveals that middle-age refugees with academic education experience the largest gains in labor market outcomes. This pattern is consistent with highly educated immigrants from disadvantaged countries facing downskilling and benefiting most when employers learn about their true abilities. In this context, referrals may help bridge credential gaps.⁶¹ Interestingly, for these groups of refugees, the employment and earnings effects are permanent, remaining positive and statistically significant also in the long run.

Last, we study the wage and turnover dynamics among refugees to assess whether they are consistent with implications of referral-based job matching. Consistent with the models of search with referrals presented in Dustmann et al. (2016) and Glitz and Vejin (2021), where workers' match-specific productivity is more uncertain in the external labor market than in the referral market, we expect workers hired through referrals to exhibit higher initial wages and lower turnover due to the higher expected match quality at the outset. We also anticipate that these initial differences will decline as tenure increases, driven by continuous learning about match-specific productivity and selective separations.⁶²

Following Glitz and Vejin (2021), we construct a proxy for referrals by using an indicator variable equal to one if a newly arrived refugee i begins their current job in an establishment where at least one member of their initial local co-ethnic network is still present at the time of

⁵⁹The US Department of Labor O*NET database (https://www.onetcenter.org/db_releases.html) measures the importance of various physical and language abilities for each occupation.

⁶⁰Since our main regressor is constructed from the municipality of initial assignment and the pre-existing conational presence, it does not change if a refugee subsequently moves to another municipality. In this sense, onward mobility does not generate a violation of SUTVA. To address any remaining concerns discussed in Schmandt et al. (2025), Table E1 further shows that neither the decision to move nor its timing is systematically related to our regressor for integration, complementing the evidence from Section 7.4.

⁶¹Dostie et al. (2023) show that employment reallocation over time toward higher-paying employers drives the reduction in immigrant-native wage gaps in Canada. This effect is particularly significant for highly educated immigrants from disadvantaged countries, who often experience substantial downskilling but benefit as employers learn about their abilities. Our findings for refugees speak directly to the importance of this mechanism.

⁶²We expect $\beta_1 > 0$ and $\beta_2 < 0$ for wages, and $\beta_1 < 0$ and $\beta_2 > 0$ for turnover in equation 5.

job start (τ).⁶³ We then estimate the following specification:

$$y_{i,t} = \alpha + \beta_1 JR_{i,\tau} + \beta_2 (JR_{i,\tau} * \text{Tenure}_{i,t}) + X_{i,t} \gamma + \phi_t + \lambda_{\bar{j}} + \epsilon_{i,t} \quad (5)$$

where $y_{i,t}$ represents either (log) wages or an indicator variable for leaving one's current employer, while JR is our proxy for referrals. The model also includes a vector of individual and establishment characteristics, $X_{i,t}$, year fixed effects (ϕ_t), and, in the preferred specification, terciles of establishment productivity fixed effects ($\lambda_{\bar{j}}$). Standard errors are clustered at the municipality of assignment level.⁶⁴

Table 9 reports estimates from equation 5, with columns 1 to 4 presenting results for the wage regressions and columns 5 to 8 for the turnover regressions. Our preferred specifications, which account for the fact that different types of firms rely on referrals to varying degrees (e.g., low- vs. high-productivity firms, as discussed in [Galenianos \(2013\)](#)) by including terciles of establishment productivity fixed effects, are shown in columns 2 and 6; in columns 3 and 7, we restrict the sample to refugees' first job spells in Denmark; and in column 4 and 8, we include establishment fixed effects. The estimate on turnover at the time of entry is consistently negative and significant across specifications, suggesting the presence of higher initial expected match quality.⁶⁵ While other coefficients are generally not statistically significant, likely due to limited power for this demanding analysis, the pattern of signs is consistent with the predictions of referral-based models.

Overall, we interpret the evidence provided so far to suggest that job referrals are an important driver of our results for refugees, although other mechanisms may also be at work.

8.2 Information sharing

Beyond referrals, co-national connections may share information about vacancies, amplifying the reach of networks in influencing the integration of newly arrived refugees. We find evidence of such broader information sharing across three complementary dimensions: industry sorting, network size heterogeneity, and residential clustering.

We begin by considering industry sorting. We assess whether the industry concentration of the co-national network at arrival, measured using a Herfindahl–Hirschman Index (HHI) based on the distribution of network members across industries, predicts industry sorting for newly arrived refugees.⁶⁶ Our outcome of interest in these regressions is defined as the share of a refugee's network employed in the same industry as the individual during the first years after arrival.⁶⁷ Whether the HHI is included as a continuous variable (columns 1 and

⁶³In other words, we assume that when a refugee follows a network member into the same establishment, the new job is obtained through a referral.

⁶⁴The controls in $X_{i,t}$ include tenure, tenure squared, age, age squared, accumulated experience in Denmark, accumulated experience squared, occupation-specific experience, occupation-specific experience squared, establishment size, industry dummies, education group dummies, and a gender dummy. All dummies are interacted with tenure and tenure squared to ensure that heterogeneous tenure profiles across subgroups that differentially rely on referrals do not bias the estimates ([Glitz and Vejlín, 2021](#)).

⁶⁵We lose significance in column 8, where very likely the inclusion of establishment fixed effects substantially reduces the variation available to identify turnover effects.

⁶⁶As in the classic antitrust applications, the HHI is computed as the sum of squared industry shares.

⁶⁷All regressions include the standard set of controls. While we do not observe refugees' occupation prior to

3) or as a binary indicator equal to one for networks with above-median HHI (columns 2 and 4), the estimates in Table E2 show that greater industry concentration within an individual's network is associated with stronger alignment between the individual's industry and that of their network in the first five years after arrival (columns 1 and 2). This pattern is consistent with a channeling mechanism driven by shared information about job opportunities or sector-specific knowledge. The magnitude of the effect declines when we consider sorting in the medium run (6–10 years after arrival).

We then expand the analysis to study the interaction between employer quality in the network and network industry concentration.⁶⁸ Columns 5 to 7 of Table E2 report the corresponding estimates using binary indicators for the HHI and quality. The positive and significant coefficient on their interaction indicates that when networks with high-quality employers are also concentrated, they become particularly powerful drivers of industry sorting, likely reflecting transmission of sector-specific job information. As before, the effect diminishes with time since arrival.

Next, we explore whether the impact of employer quality in the network depends on network size. We categorize networks into size-based terciles and interact the three dummy variables with our employer-based quality measure. Table E3 reports estimates of the effect of these interaction terms on refugees' labor market outcomes, using mid-sized networks as omitted category and including the usual set of controls and fixed effects. While small networks may not have enough information to share, the negative and statistically significant coefficients on the interaction of quality and the large network dummy points to information congestion, further supporting the idea that information flow is a relevant mechanism (Beaman, 2012).

Finally, we examine whether information flows matter most at firms on the margin of hiring refugees. If members of networks with high-quality employers work at firms with a stronger propensity to hire refugees, information flows add little beyond the access new arrivals would have obtained on their own. Conversely, information within the network should matter most when connected firms are neither systematically open nor systematically closed to hiring refugee workers. To test this, we predict the probability of hiring a refugee for each establishment using a logit model with establishment size (log of full-time equivalents), industry, municipality, share of foreign employees, presence of a foreign manager, and AKM firm effect as predictors. We do this separately for each year to account for employer learning, as firms may update their beliefs about refugees' skills based on past refugee hiring experiences (Loiacono and Silva-Vargas, 2024). We then classify networks into terciles of average predicted willingness to hire and interact these with network employer quality.

Table E4 reports estimates from regressions of refugees' labor market outcomes on dummies for terciles of network average willingness to hire refugees and their interaction with employer quality in the network, using the bottom tercile of openness to hire ("Low") as the omitted category.⁶⁹ The largest significant effect of network employer quality on refugees' outcomes arises in networks whose employers display medium levels of average predicted willingness to hire a refugee. In other words, connections at high-quality firms help refugees

migration, we add a control for their education attainment at arrival.

⁶⁸We first explore this descriptively in Figure E1, which shows the unconditional correlations.

⁶⁹The rest of the specification is the same as in equation 3.

most when these firms are moderately likely to hire them, pointing to a reversed U-shaped pattern. Consistent with information flows mattering the most for firms on the margin, connections with good firms that on average are not likely to hire refugees are not as useful, nor are those with good firms that on average are very likely to hire them and access is less of a barrier. Overall, this relationship between predicted firm openness and network employer quality, along with the evidence in this section, provides support for the important role of information flows within networks.

9 Optimal Assignment

We have estimated positive effects of employer quality within co-national networks on refugees' labor market outcomes. Since the resettlement of refugees to different locations within the host country is one of the earliest and most consequential policy decisions in the integration process, a natural question arises: *Can our findings be leveraged to provide a cost-efficient policy tool that improves refugees' outcomes?* In this section, we extend a series of data-driven methodologies for refugee resettlement (Bansak et al., 2018; Ahani et al., 2021). We contribute by explicitly including network measures among the variables used to train prediction models underlying the data-driven assignment, accounting in particular for the quality of accessible employers at arrival through co-national networks.⁷⁰ As documented, this is an important refugee–municipality match-specific characteristic that would otherwise remain unobserved without either first estimating a fixed effects wage model or accessing additional firm-specific information. Using the generated predictions, we then compare various assignment scenarios for Denmark, in the spirit of recent work for the United States and Switzerland.

9.1 Methodology and results

The integer optimization problem of matching refugees to municipalities takes the form of a multidimensional knapsack problem, and can be formulated as follows:⁷¹

$$\max \sum_{i=1}^{|I|} \sum_{m=1}^{|M|} v_m^i z_m^i \quad (6a)$$

$$\text{s.t.} \quad \sum_{m=1}^{|M|} z_m^i \leq 1, \quad \forall i, \quad (6b)$$

$$\underline{s}_k^m \leq \sum_{i=1}^{|I|} s_k^i z_m^i \leq \bar{s}_k^m, \quad \forall m, \forall k, \quad (6c)$$

$$z_m^i \leq a_m^i, \quad \forall i, \forall m, \quad (6d)$$

$$z_m^i \in \{0, 1\}, \quad \forall i, \forall m. \quad (6e)$$

⁷⁰Ahani et al. (2021) acknowledge that “additional information—such as housing information, social networks, or new job opportunities—likely exists to at least some degree at the local community level and could prove very useful in supplementing the decision process”.

⁷¹We adopt a notation similar to Ahani et al. (2021).

where the objective is to maximize the sum of individual values v , a measure of individual integration, over all refugees i assigned Danish municipalities m . The binary variable z_m^i equals one if refugee i is matched to municipality m , and zero otherwise. Two constraints are specified. Constraint 6c ensures that each municipality receives at least a minimum number and no more than a maximum number of refugees, denoted by $\underline{s}_k^m$ and $\bar{s}_k^m$ respectively. We use the actual number of refugees originally assigned to each municipality to infer our municipality quotas. In practice, we almost always impose only this upper bound, but in some cases, municipality capacities are pre-approved, requiring that a minimum number of refugees also be assigned.⁷² For binary support service $a_m^i \in \{0, 1\}$, constraint 6d ensures that matches occur only when the municipality can accommodate specific needs. In our case, we ensure that the optimal allocation matches the actual number of refugees with specific family structures assigned to each location.⁷³ The variable domain specified in constraint 6e characterizes the problem as an integer optimization, where the optimal solution is the vector z^* of refugee-municipality matches.

For the integration metric in the objective function, we focus on individual employment probability, as in [Bansak et al. \(2018\)](#) and [Ahani et al. \(2021\)](#), and consider employment in the first five years after migration to align with the short-run horizon used in our main analysis. Crucially, however, employment probabilities are not observed for incoming refugees, nor for past refugees in municipalities other than the one to which they were actually assigned. Hence, to obtain the potential outcome distribution for each refugee, we rely on machine learning methods to estimate individual probabilities of employment at each municipality m from:

$$v_m^i = \mathbb{E}[y_i | \mathbf{X}_{i,m}] \quad (7)$$

where y_i is a binary outcome indicating whether individual i was ever employed within the first five years after migration. In the baseline version, which follows other recent works, $\mathbf{X}_{i,m}$ includes observable individual characteristics and their interactions with municipalities. In our alternative version, we augment instead the set of variables for statistical learning of employment probabilities with $\tilde{\mathbf{X}}_{i,m}^{network}$, which explicitly includes employer quality within the networks, so that:

$$v_m^i = \mathbb{E}[y_i | \mathbf{X}_{i,m}, \tilde{\mathbf{X}}_{i,m}^{network}] \quad (8)$$

We proceed in the following steps. First, because the potential outcome distribution is not observed, following [Bansak et al. \(2018\)](#) we begin with a modeling stage that predicts expected individual success in terms of early employment for a random 80% of refugees from cohorts 1987–1997 across all potential resettlement locations (training sample). We employ two supervised machine learning models in this stage: we introduce a Least Absolute Shrinkage and Selection Operator (LASSO) constraint to a logit model featuring individual characteristics

⁷² $\underline{s}_k^i$ denotes the required units of service k . It is possible to define other constraints similar to constraint 6c, where instead of municipality quotas, refugee processing capacity at the municipality or slots in language classes are considered. In our case, for municipality quotas, $\underline{s}_k^i = 1 \forall i$.

⁷³We implement this by creating indicators for refugees with children in the 0–5 and 6–17 age ranges, consistent with the dummy variables used in our main analysis.

and their interactions with municipalities, and we also estimate a gradient boosted regression tree (GBRT), featuring the same set of predictors and municipality interactions as in the LASSO specification.

Next, we evaluate the performance of these models on a holdout (test) sample (the remaining random 20%), comparing them with two second-best alternatives: a naive constant estimator and a standard logit model that includes individual characteristics but does not account for municipality-specific effects. Based on predictive performance, we select LASSO and use this method to estimate the potential outcome distribution for the last cohort of refugees allocated under the dispersal policy (the 1998 cohort), for whom we implement the optimal assignment. Importantly, we estimate the potential outcome distribution with LASSO using, in turn, both the baseline in expression 7 and our refinement in expression 8. Further methodological details and performance comparisons are provided in Appendix F.

Finally, in a matching stage, we solve the maximization problem described above using our predictions for the 1,258 refugees in the 1998 cohort and evaluate the employment gains from the optimal assignment relative to the actual one. Restricting our exercise to this cohort offers three advantages. First, we rely on the integration of earlier-arriving refugees in the same municipalities to train our models. Second, we use networks formed in earlier periods to construct our measures and thus inform our models. Third, we avoid complications from multi-period assignments, which may become difficult to manage.⁷⁴ We use the CBC MILP solver within the PuLP Python modeling environment to find the optimal municipality for each of these refugees, i.e. vector z^* .

Intuitively, given the municipality quotas, the optimal allocation cannot assign all refugees to a handful of economically dynamic municipalities, no matter how beneficial these might be for integration. At the same time, when minimum capacities are pre-approved, weaker locations cannot be left without refugees. Respecting the constraints of problem 6 therefore requires balancing the advantages and disadvantages of different municipalities across different refugees. Our extension, by explicitly incorporating networks and employer quality, reinforces the idea that optimal assignments vary across immigrant types: different origins may benefit from different locations, and the best placement depends not only on individual characteristics and the overall economic activity of a municipality, but also on an origin-municipality match-specific component determined by the structure of accessible co-ethnic networks, as we show below.

Several important results emerge from solving problem 6. First, using baseline predictions from expression 7 and implementing the assignment solution raises the average predicted probability of ever being employed within five years since migration for the 1998 cohort by 16.6 p.p., from 28.9% to 45.5%. This optimized assignment increases the probability of finding employment across the entire distribution of refugees, as shown by the empirical cumulative distribution functions of refugees' predicted employment probabilities in Figure 5. The employment probability distribution after this optimization (dashed green line) first-order stochastically dominates the estimated distribution under actual assignment (solid red

⁷⁴As noted by [Ahani et al. \(2021\)](#), "experiments with $n > 1$ placement periods introduce some additional nuances that required equally detailed implementation strategies".

line). These results are in line with employment improvements obtained in other works for the United States (+25 p.p.) and Switzerland (+11 p.p.) using very similar methodologies.⁷⁵

Second, accounting for the role of co-ethnic networks, in particular the quality of employers accessible to newly arrived refugees through these connections, in predicting the potential outcome distribution as in expression 8 with the LASSO, improves predictive accuracy relative to baseline predictions from expression 7. Details are reported in Table F1. Using this refined distribution in problem 6, the optimized refugee-municipality matches are different relative to the solution obtained without accounting for network quality. This new solution to the assignment problem leads to an average predicted probability of ever being employed within five years of 42.1%, confirming the considerable improvement from algorithmic assignment over the status quo, with gains across the entire distribution (solid blue line in Figure 5).

Importantly, accounting for firms connected through the networks also improves on the baseline optimization. Had we used the more accurately predicted probabilities from expression 8, while holding fixed the matches prescribed by the first optimized assignment, we would have obtained an average predicted employment probability of 36.0%. In other words, accounting for network information in our data-driven approach updates the potential outcome distribution and leads to a slightly modified optimized assignment, which improves not only on the actual assignment under the dispersal policy but also on the optimized assignment that does not account for networks. Indeed, the new optimized assignment (solid blue line) first-order stochastically dominates both the estimated distribution under actual assignment (solid red line) and the baseline optimized one (solid green line).

Concrete examples illustrate how network quality shapes the optimized allocation, making some municipalities particularly attractive for some origins and less for others. For instance, about 17% of refugees from Vietnam in the 1998 cohort were assigned to Odense under the dispersal policy, even though the quality of employers connected to their co-ethnic network there was relatively low, despite Odense being Denmark's third-largest and one of its most economically active cities. In our optimized allocation, by contrast, no Vietnamese refugee is assigned to Odense. Another example is represented by Viborg, which in 1998 received around 7% of all randomly dispersed refugees, the second most after Copenhagen. About 15% of refugees assigned to Viborg under the dispersal policy were from Somalia, even though their network employers were not of particularly high quality. In the optimized allocation, however, two-thirds of these Somalis are reallocated elsewhere, many to Esbjerg, Denmark's fifth-largest city, where their network was substantially stronger and reported some of the highest quality values for this origin. At the same time, the number of Iranian refugees assigned to Viborg by our algorithm grows by a factor of four relative to the dispersal policy, consistent with their stronger network there. These reallocations occur while respecting municipality quotas, ensuring, for instance, that the overall number of refugees from the 1998 cohort in Viborg remains unchanged, and at the same time better aligning origins with localities to improve integration prospects.

⁷⁵Bansak et al. (2018) find a 25 p.p. gain in median employment during the first 90 days for the United States in 2016 ($n = 919$), and a 11 p.p. gain in third-year employment for Switzerland in 2013 ($n = 888$). Ahani et al. (2021) find gains between 22% and 38%, depending on the constraints, in employment outcomes within 90 days since arrival for refugees in the United States in 2017 ($n = 498$).

Third, refugees predicted to fare worst under the status quo gain the most from an optimized assignment that explicitly accounts for network-connected employers. Since incorporating networks into the algorithm changes the optimal matches for refugee–municipality pairs, we examine which refugees gain or lose when reallocated from municipalities chosen without accounting for networks to those selected optimally with them. Figure F1 plots the predicted gain or loss for individuals who would be reassigned against their predicted employment in the actual assignment, revealing heterogeneous gains across the distribution. This is an important distributional result, as improving the condition of the most vulnerable individuals is often an ancillary goal for policymakers.

Finally, compared to the status quo assignment, implementing the optimized assignment prescribed by our approach does not disproportionately benefit the largest or most economically active municipalities. Figure F2 shows gains and losses by municipality. Copenhagen and Aarhus, Denmark’s two largest cities, see gains that are not among the highest, and Odense, the fourth-largest, records small losses. This is consistent with the origin–municipality complementarities described above: once quality of employers in the network is accounted for, the optimal allocation differs across origins rather than concentrating all refugees in a few top cities.⁷⁶ This further highlights the importance of an approach that accounts for origin–municipality complementarities rather than treating municipalities in isolation, even if this complicates the design of resettlement policies. Overall, most Danish municipalities would have experienced increases in the average predicted employment of their refugee population arriving in 1998 had the authorities relied on this optimized resettlement method.

9.2 Counterfactual policy scenarios

We conclude by comparing four counterfactual policy scenarios generated with our preferred employment-optimization procedure, illustrating the sensitivity of employment gains to policy choices. The scenarios we consider are:

1. *Optimized assignment subject to family structure restrictions and inferred municipality quotas (with no lower bound).*
2. *Optimized assignment subject to family structure restrictions and allowing for 110% of inferred municipality quotas (with no lower bound).*
3. *Optimized assignment subject to family structure restrictions and allowing for 150% of inferred municipality quotas (with no lower bound).*
4. *Optimized assignment subject to family structure restrictions and allowing for 110% of inferred municipality quotas, with a lower bound of 90%.*

Counterfactual 1 represents our preferred optimization setup, accounting for networks and featuring default constraints for family structure and inferred quotas for each municipality. Relaxing municipality quotas by allowing an additional 10% (counterfactuals 2 and 4) or an

⁷⁶For instance, Esbjerg, mentioned earlier among our examples, records some of the largest gains in Figure F2.

additional 50% (counterfactual 3) capacity increases the average predicted employment probability in all three scenarios. The average predicted probability of ever being employed within five years since migration rises to 43.6% in counterfactual 2, to 50.0% in counterfactual 3, and to 43.2% in counterfactual 4. Empirical cumulative distribution functions for refugees allocated under these scenarios are reported in Figure F3.

Counterfactual 4 also imposes a 90% lower bound on municipality capacity that must be met by the optimal assignment in every municipality. This additional constraint reduces the average predicted employment probability only slightly relative to counterfactual 2. This is encouraging, as it indicates that our optimization performs well and delivers gains even under tighter distributional constraints. Ultimately, while increasing municipality quotas generates gains by allowing more refugees to benefit from favorable matches, the presence of family-structure restrictions in the form of binary service constraints and of minimum capacity requirements has only a modest impact on employment gains.

Overall, this algorithm produces substantial employment gains while respecting important distributional considerations. Resettlement authorities can modify the constraints we specified or introduce new ones (e.g., limits on the number of refugees from certain origins), as this data-driven approach is highly flexible to accommodate policymakers' priorities. While one resettlement agency in the United States (HIAS) has already adopted a software based on a similar algorithm (*Annie*TM, developed in [Ahani et al. \(2021\)](#)), refining these methodologies to consider which local features and complementarities offer the best potential to improve refugee outcomes as well as encouraging their wider adoption could enhance integration and reduce host-country costs.

Although these findings highlight the benefits of exploiting origin-municipality complementarities, we do not advocate concentrating all co-nationals in very few specific locations, which would in any case likely be constrained by existing quotas. Our results suggest that leveraging complementarities between origins and municipalities can improve integration outcomes, but identifying where the tipping point more generally lies remains an open question. Excessive co-national concentration may instead foster segregation and hinder integration, and our evidence also indicates that beyond a certain size, networks may experience congestion effects. Hence, understanding precisely when the marginal returns to co-national complementarities turn negative is a promising direction for future research on resettlement policies.

10 Conclusion

This paper presents the first causal evidence on the effect of early employer quality on immigrants' labor market integration, focusing on refugees. Using administrative data from Denmark, we first estimate establishment-specific pay premia that serve as a proxy for employer quality using a two-way fixed effects wage model. We then exploit co-national networks and Denmark's 1986–1998 dispersal policy, which quasi-randomly assigned newly arrived refugees to Danish municipalities, to obtain exogenous variation in refugees' exposure at arrival to different levels of employer quality.

The type of employers available upon refugees' first entry is an important determinant

of their labor market success. Our results show that higher-quality employers in the local co-national network at arrival improve employment and earnings for newly arrived refugees, with a clear positive impact in the short run (1–5 years after arrival) and medium run (6–10 years after arrival). The absence of similar effects from employer quality among all firms in the municipality underscores the importance of focusing on co-national networks. Job referrals and information sharing among network members emerge as key channels through which connections to good employers matter. The benefits are strongest for highly educated refugees, who may be subject to severe downskilling.

Beyond our main results, we make other important contributions. We document a lasting influence of early employers on refugees' outcomes, which we do not observe for other workers facing lower barriers and greater mobility, and we discuss the resulting concerns for the widely used AKM decomposition that may affect recent work on migrant subgroups. Our paper also offers a new perspective on networks by disentangling the effect of the quality of connected firms from that of network members on newly arrived refugees' outcomes, pointing to friction reduction over human capital externalities as mechanism explaining the positive effects of co-national networks on new members. We conclude by incorporating information on employer quality within networks into a data-driven algorithm to optimally resettle refugees. The optimized assignment raises refugees' employment within the first five years after arrival by about 46% relative to the dispersal policy, with stable gains across counterfactual policy scenarios.

Our findings have important policy implications for host countries. We show that refugee integration is sensitive to employer quality and that the connections provided by co-nationals can offer valuable access to firms. Rather than focusing solely on logistical considerations or on broad economic characteristics of the local labor market, the design of placement policies—and of integration policies at destination in general—should also take into account the role played by employers, which has so far been largely overlooked. Improving migrants' access to better employers, encouraging stronger early matches with firms, and more broadly incorporating a wider range of demand-side considerations into resettlement policies can promote a faster and more effective integration while limiting costly mismatches.

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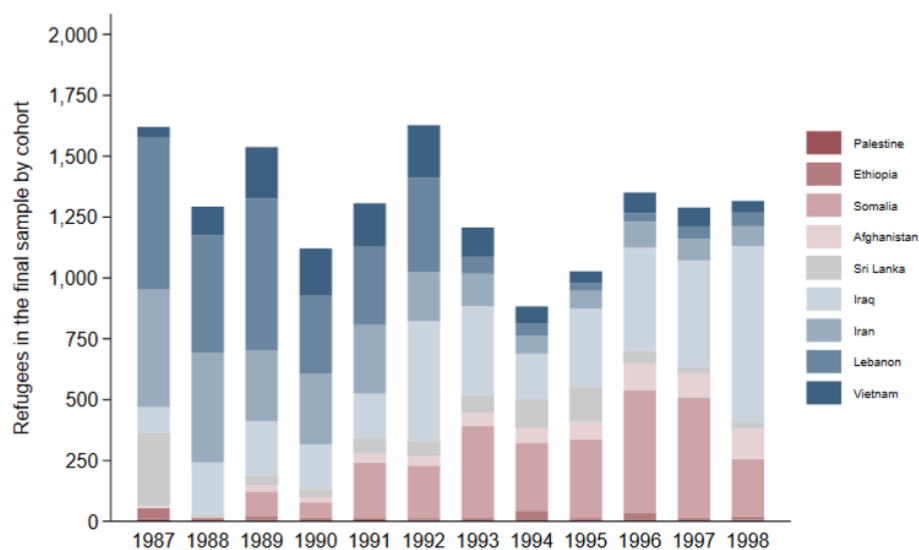
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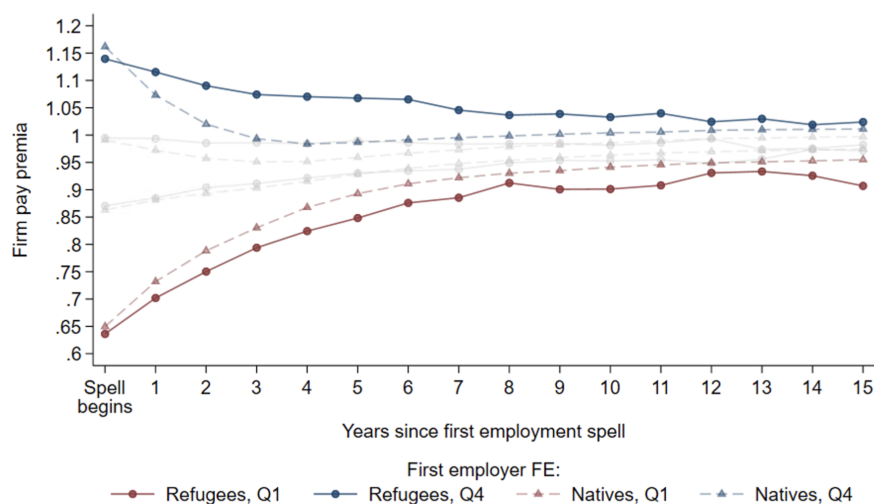
Figures and Tables

Figure 1: Cohort size and composition



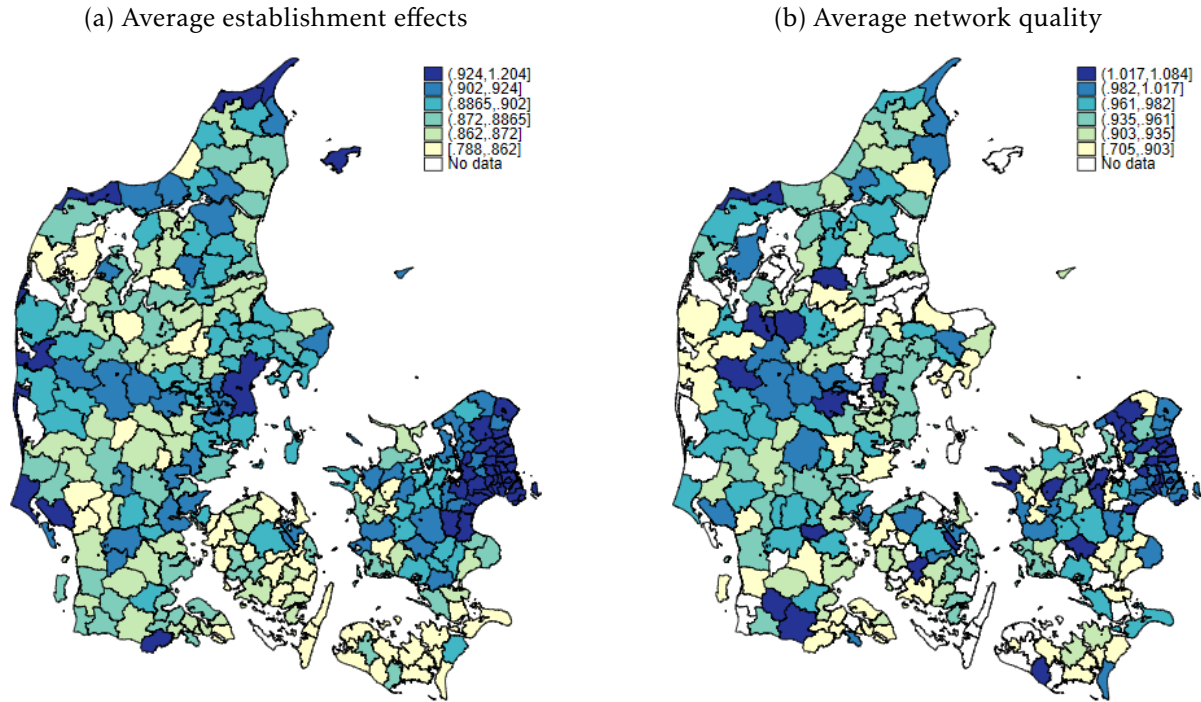
Notes: This figure plots the number of refugees subject to the 1986–1998 national dispersal policy by cohort and country of origin. We start from 1987 because of data limitations.

Figure 2: Stickiness of initial firm quality



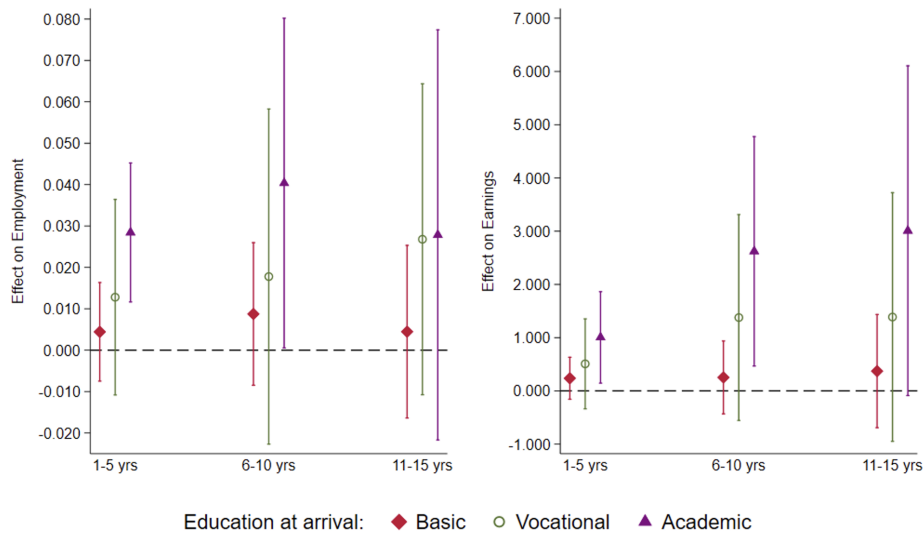
Notes: This figure plots the evolution of firm pay premia earned by natives and refugees. Both samples of native and refugee workers are categorized into quartiles based on the fixed effect of their first employer. Each point in the figure is the average firm pay premium by group (natives, denoted by triangle-shaped markers, and refugees, denoted by circle-shaped markers), by initial firm effect quartile (maroon for the bottom quartile, blue for the top quartile), and by year since first employment spell.

Figure 3: Geography of establishment and network



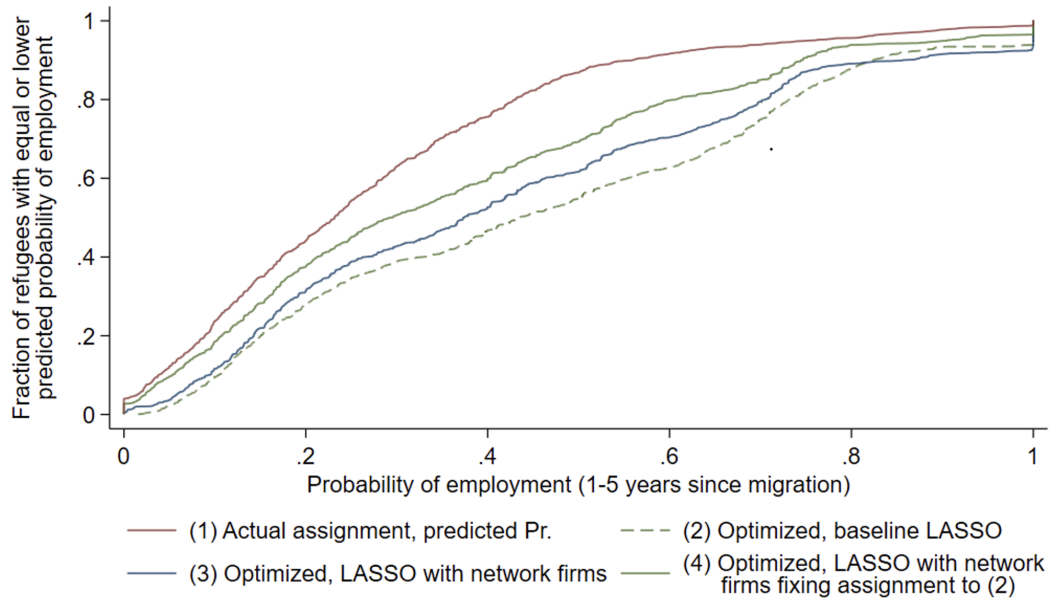
Notes: These maps display the average establishment effects (Panel A) and the average establishment effects within networks (also averaged over origin and year, Panel B) by sextiles across the 275 Danish municipalities (pre-2007 reform). In certain locations data have been suppressed to comply with Statistics Denmark's confidentiality rules.

Figure 4: Heterogeneity analysis by education



Notes: This figure plots the estimates and 95 percent confidence intervals of the differential effect of network quality (unweighted measure) on the main outcomes (employment in the left panel and earnings in the right panel) by education completed prior to arrival. Refugees are divided into three groups: basic education, vocational education and academic education. Each outcome is examined over three time intervals: short run (1 to 5 years since arrival in Denmark), medium run (6 to 10 years), and long run (11 to 15 years). The sample is restricted to refugees reporting education attainment.

Figure 5: Empirical cumulative distribution functions under different assignments



Notes: This figure displays the empirical cumulative distribution functions (ECDFs) of the refugees' predicted employment probabilities within the first five years since migration under their actual and optimized assignments. The solid red line refers to predicted probabilities in municipality of assignment under the dispersal policy. The dashed green line refers to probabilities computed with the baseline LASSO (no network measures) in the corresponding optimized assignment. The solid blue line refers to probabilities computed with the refined LASSO (including network measures) in the new optimized assignment. The solid green line refers to probabilities computed with the baseline LASSO in the new optimized assignment. All optimized cases employed family structure restrictions and inferred municipality quotas (with no lower bound) as constraints.

Table 1: Conditional random assignment of the policy

	Employment rate of non-Western immigrants (1)	Share of conationals (2)	Network employer quality (mean) (3)	Network employer quality (weighted mean) (4)
<i>Panel A: Variables observed by the authorities</i>				
Age 30-39 years	-0.042** (0.017)	-0.098** (0.040)	0.003 (0.036)	-0.041 (0.038)
Age 40-49 years	-0.018 (0.024)	-0.097* (0.057)	-0.002 (0.048)	0.019 (0.052)
Age 50-55 years	0.042 (0.038)	-0.239** (0.115)	0.190* (0.099)	0.113 (0.100)
Female	-0.002 (0.005)	0.053*** (0.011)	-0.019 (0.012)	-0.021* (0.012)
No. children, 0-2 yrs	0.003 (0.008)	-0.020 (0.015)	0.005 (0.021)	-0.006 (0.021)
No. children, 3-5 yrs	-0.001 (0.009)	-0.030** (0.015)	0.007 (0.018)	0.010 (0.018)
No. children, 6-12 yrs	0.007 (0.005)	-0.012 (0.009)	-0.001 (0.011)	-0.006 (0.011)
No. children, 13-17 yrs	-0.009 (0.009)	0.038*** (0.015)	-0.014 (0.016)	0.005 (0.016)
Single	-0.006 (0.007)	-0.020 (0.014)	-0.014 (0.016)	-0.022 (0.016)
Africa	0.294* (0.168)	0.223 (0.367)	0.566*** (0.220)	0.213*** (0.065)
Asia	0.277** (0.111)	-0.127 (0.730)	-0.690 (0.655)	-0.101 (0.068)
<i>Panel B: Variables not observed by the authorities</i>				
Basic education	0.019* (0.011)	0.004 (0.024)	0.022 (0.026)	0.007 (0.026)
Academic education	0.009 (0.013)	-0.007 (0.025)	0.011 (0.031)	-0.003 (0.031)
Unknown education	0.017 (0.010)	0.004 (0.022)	-0.011 (0.025)	-0.026 (0.025)
Obs.	15571	15571	10271	10246
Adj. R^2	0.770	0.504	0.406	0.497
Municipality FE	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes
F	1.125	0.131	1.353	1.436
Pr. > F	0.337	0.942	0.255	0.230

Notes: This table reports a balancing test for the conditional random assignment of the dispersal policy. The sample is refugees from admission cohorts 1987 to 1998 and subject to the dispersal policy. Outcomes of these regressions are the main regressors in our preferred specification. Network employer quality represents the average employer quality of establishments employing at least one co-national in the municipality of assignment at the time of arrival (unweighted mean in column 3, weighted mean using full-time equivalents in column 4). Variables reflecting family structure and country of origin in Panel A are not (nor are they expected to be) uncorrelated with initial location characteristics, as placement was conditional on these factors. F denotes the F -test statistic of joint insignificance of the dummies for educational attainment in Panel B: basic education, academic education, and unknown education (vocational education is the omitted category of reference). $Pr > F$ denotes the corresponding p-value from the F -test. Robust SE in parentheses are clustered at the family level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table 2: Sample of refugees: Summary statistics

	Observations	Mean	S.D.
<i>Panel A: Individuals</i>			
Age at Entry	15,578	29.8	8.25
Female	15,578	0.42	0.49
Married/cohabiting	15,578	0.65	0.48
Number of Children	15,578	1.06	1.62
Academic Education (prior to arrival)	6,566	0.26	0.44
Latin Alphabet of Mother Tongue	15,578	0.33	0.47
From Predominantly Muslim Country	15,578	0.14	0.35
Asia	15,578	0.15	0.36
Middle East	15,578	0.66	0.48
Africa	15,578	0.20	0.40
Employment (any)	15,578	0.22	0.27
Annual Earnings (thousand USD)	15,578	8244.6	12925.7
Avg. Hourly Wage Rate in Estab.	8,080	28.4	12.6
Complex Job (indicator)	15,578	0.12	0.22
<i>Panel B: Worker years</i>			
Agriculture	40,893	0.01	
Manufacturing	40,893	0.27	
Utilities	40,893	0.00	
Construction	40,893	0.02	
Trade and hospitality	40,893	0.18	
Transport	40,893	0.11	
Business and finance	40,893	0.08	
Public and personal services	40,893	0.33	
Not stated	40,893	0.00	

Notes: This table presents summary statistics for the sample of refugees used in the main analysis. Panel A reports individual and economic characteristics (observations are individual refugees). Panel B reports the industry distribution of the annual main jobs of employed refugees during the first fifteen years after arrival in Denmark (observations are worker years). Panel B includes only jobs with non-missing establishment and industry information. The nine NACE industry categories used are: Agriculture, fishing, and quarrying; Manufacturing; Electricity, gas, and water supply; Construction; Wholesale and retail trade, hotels, and restaurants; Transport, storage, and communication; Financial intermediation and business activities; Public and personal services; and Activity not stated.

Table 3: First employers and refugees' labor market outcomes

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Main specification</i>						
First employer quality	0.036*** (0.007)	0.036*** (0.009)	0.032*** (0.010)	3.416*** (0.376)	2.563*** (0.451)	2.626*** (0.554)
boot. p-val.	[0.000]	[0.002]	[0.010]	[0.000]	[0.000]	[0.001]
Mean of Y	0.721	0.577	0.522	23.511	24.500	25.185
Obs.	4,193	4,086	3,859	4,193	4,091	3,872
Adj. R ²	0.072	0.096	0.125	0.108	0.085	0.101
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Panel B: At least one firm transition</i>						
First employer quality	0.016* (0.008)	0.028*** (0.010)	0.029*** (0.012)	2.933*** (0.642)	2.152*** (0.607)	2.596*** (0.772)
boot. p-val.	[0.068]	[0.021]	[0.035]	[0.000]	[0.003]	[0.010]
Mean of Y	0.818	0.730	0.667	26.976	31.993	33.559
Obs.	2,108	2,099	2,031	2,108	2,099	2,036
Adj. R ²	0.027	0.032	0.084	0.103	0.049	0.065
Ever left first employer	Yes	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Panel C: Moved from initial municipality</i>						
First employer quality	0.041*** (0.013)	0.040*** (0.013)	0.039*** (0.013)	3.375*** (0.559)	2.890*** (0.668)	2.966*** (0.743)
boot. p-val.	[0.006]	[0.010]	[0.012]	[0.000]	[0.002]	[0.002]
Mean of Y	0.713	0.559	0.510	23.803	24.111	25.390
Obs.	1,765	2,379	2,554	1,765	2,383	2,563
Adj. R ²	0.047	0.065	0.102	0.085	0.070	0.093
Ever left assigned municipality	Yes	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: All panels report OLS estimates from regressions where the outcome variable is a measure of refugees' employment probability (columns 1-3) and annual earnings measured in 2015 USD (columns 4-6). We regress the outcome on the standardized quality measure of the first employer. The short-run (columns 1 and 4), medium-run (columns 2 and 5), and long-run (columns 3 and 6) effects are defined relative to the year of hiring by the first employer, normalized to 1. This sample consists of refugees subject to the 1986–1998 dispersal policy in Denmark. In all panels we employ the fixed-effects specification used throughout the analysis. Panel B restricts the sample to refugees with at least one transition between different employers, while Panel C restricts the sample to refugees who ever moved away from their municipality of assignment. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table 4: First employers and wages by group

	Ln(Hourly Wage)					
	Natives			Refugees		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
First employer quality	0.025*** (0.002)	0.001 (0.001)	0.001 (0.001)	0.098*** (0.014)	0.038*** (0.007)	0.026*** (0.009)
boot. p-val.	[0.000]	[0.234]	[0.127]	[0.000]	[0.000]	[0.005]
Obs.	536,804	439,037	367,363	4,192	3,029	2,460
Adj. R^2	0.141	0.081	0.077	0.054	0.038	0.057
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Year of Entry FE	Yes	Yes	Yes	-	-	-
Cohort FE	-	-	-	Yes	Yes	Yes
Origin FE	-	-	-	Yes	Yes	Yes

Notes: This table reports OLS estimates from regressions where the outcome variable is log of real hourly wages (2015 USD) for natives (columns 1-3) and refugees (columns 4-6). We regress the outcome on the standardized quality measure of the first employer. The short-run (columns 1 and 4), medium-run (columns 2 and 5), and long-run (columns 3 and 6) effects are defined relative to the year of hiring by the first employer, normalized to 1. The sample of refugees consists of individuals subject to the 1986–1998 dispersal policy in Denmark. The sample of natives consists of individuals entering the labor market for the first time in the same years of the dispersal policy. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table 5: Effects of co-national connections on firm hiring

	(1)	(2)	(3)	(4)
	Hired by j	Hired by j	Hired by j	Hired by j
Co-national presence	0.269*** (0.0322)	0.269*** (0.0322)	0.0556*** (0.0193)	0.108*** (0.0222)
Observations	2,370,224	2,370,224	2,370,224	2,366,900
$E[Y D=0]$	0.083	0.083	0.083	0.082
Individual controls	Yes	Yes	Yes	Yes
Education controls	No	Yes	Yes	Yes
Firm FE	No	No	Yes	-
Firm-by-Cohort FE	No	No	No	Yes

Notes: The table reports OLS estimates from a linear probability model where the outcome variable takes value 1 if refugee i is hired by establishment j and 0 otherwise, and the main regressor (“Co-national presence”) takes value 1 if one member of refugee’s i local co-national network at arrival works in establishment j and 0 otherwise. Observations are refugee-potential establishment dyads. Robust SE in parentheses are clustered at the establishment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table 6: Effects of exposure to local firms

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Geographic exposure</i>						
Avg. quality	-0.033* (0.019)	0.022 (0.027)	-0.017 (0.037)	-1.253** (0.545)	-0.787 (1.067)	-1.337 (1.628)
boot. p-val.	[0.113]	[0.466]	[0.724]	[0.032]	[0.505]	[0.470]
Co-national share	-0.002 (0.004)	-0.004 (0.005)	0.001 (0.006)	-0.053 (0.127)	-0.261 (0.225)	-0.147 (0.343)
Emp. NW immigrants	0.006 (0.006)	0.003 (0.008)	0.007 (0.011)	0.015 (0.217)	-0.046 (0.386)	0.074 (0.562)
Mean Y	0.104	0.259	0.324	2.982	8.803	13.869
Obs.	15,556	15,064	14,703	15,556	15,064	14,703
Adj. R^2	0.166	0.199	0.191	0.139	0.153	0.145
<i>Panel B: Ever hired from refugee-sending countries</i>						
Avg. quality	-0.002 (0.005)	0.002 (0.007)	0.011 (0.009)	-0.128 (0.151)	-0.168 (0.293)	0.412 (0.440)
boot. p-val.	[0.666]	[0.784]	[0.231]	[0.403]	[0.565]	[0.379]
Co-national share	-0.002 (0.004)	-0.004 (0.005)	0.001 (0.006)	-0.063 (0.128)	-0.276 (0.231)	-0.108 (0.350)
Emp. NW immigrants	0.005 (0.007)	0.001 (0.009)	0.006 (0.012)	-0.026 (0.241)	-0.286 (0.412)	-0.132 (0.623)
Mean Y	0.104	0.258	0.324	2.977	8.814	13.850
Obs.	15,118	14,644	14,287	15,118	14,644	14,287
Adj. R^2	0.165	0.198	0.190	0.139	0.154	0.145
<i>Panel C: Co-national network exposure</i>						
Avg. quality	0.009*** (0.003)	0.017*** (0.006)	0.009 (0.006)	0.328*** (0.098)	0.798*** (0.238)	0.574 (0.368)
boot. p-val.	[0.003]	[0.022]	[0.180]	[0.003]	[0.007]	[0.148]
Co-national share	-0.003 (0.004)	-0.003 (0.007)	-0.001 (0.006)	-0.005 (0.145)	-0.161 (0.290)	-0.023 (0.367)
Emp. NW immigrants	0.005 (0.014)	0.005 (0.014)	0.003 (0.018)	0.269 (0.446)	0.224 (0.597)	0.334 (0.934)
Mean Y	0.107	0.262	0.331	3.080	8.975	14.151
Obs.	10,251	9,976	9,786	10,251	9,976	9,786
Adj. R^2	0.191	0.220	0.205	0.169	0.175	0.156
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents OLS estimates of the effect of exposure to employer quality on employment and earnings of refugees subject to the 1986–1998 dispersal policy. To compute effects in the short-run, medium-run, and long-run, we set the year of admission to Denmark to 1. The quality measure used in Panel A is the average establishment effect for establishments active in the municipality of assignment at the time of arrival. The quality measure in Panel B is the average establishment effect for establishments active in the municipality of assignment at the time of arrival that are employing or have previously employing another individual from a refugee-sending country. The quality measure in Panel C is the average establishment effect for establishments active in the municipality of assignment at the time of arrival that are employing a co-national of the newly arrived refugee. Individual controls include variables observed by authorities in the dispersal process. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table 7: Network quality decomposition: Firms and members

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Average network earnings</i>						
Avg ln(earnings)	0.004 (0.005)	0.004 (0.008)	0.005 (0.010)	0.079 (0.175)	0.306 (0.355)	0.156 (0.538)
Mean Y	0.104	0.257	0.321	2.966	8.869	13.803
Obs.	10,384	10,081	9,870	10,384	10,081	9,870
Adj. R^2	0.180	0.210	0.199	0.156	0.168	0.150
<i>Panel B: Average of individual FEs as quality of members</i>						
Avg. employer quality	0.011*** (0.003)	0.017** (0.007)	0.006 (0.006)	0.364*** (0.104)	0.730*** (0.276)	0.506 (0.404)
boot. p-val.	[0.003]	[0.043]	[0.346]	[0.005]	[0.028]	[0.233]
Avg. member quality	-0.002 (0.004)	0.000 (0.007)	0.002 (0.007)	-0.035 (0.123)	-0.045 (0.279)	0.052 (0.345)
boot. p-val.	[0.575]	[0.994]	[0.774]	[0.795]	[0.887]	[0.880]
Mean Y	0.107	0.262	0.331	3.070	8.954	14.162
Obs.	9,498	9,236	9,064	9,498	9,236	9,064
Adj. R^2	0.192	0.220	0.207	0.167	0.174	0.156
<i>Panel C: Share with academic education as quality of members</i>						
Avg. employer quality	0.010*** (0.003)	0.017** (0.008)	0.008 (0.007)	0.327*** (0.121)	0.626** (0.317)	0.466 (0.478)
boot. p-val.	[0.004]	[0.095]	[0.295]	[0.015]	[0.092]	[0.364]
Avg. member quality	0.006 (0.015)	0.027 (0.023)	-0.003 (0.025)	-0.014 (0.552)	0.802 (1.089)	-0.075 (1.392)
boot. p-val.	[0.744]	[0.313]	[0.908]	[0.981]	[0.503]	[0.961]
Mean Y	0.106	0.262	0.332	3.072	8.985	14.259
Obs.	9,137	8,880	8,712	9,137	8,880	8,712
Adj. R^2	0.190	0.215	0.203	0.165	0.168	0.152
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports estimates for network measures of firm quality and member quality included together in our main estimating equation. Panel A reports estimates for the effect of aggregate network quality, using network members' average earnings in log. Panels B and C include controls for average member quality, measured using AKM worker effects in Panel B and the share of highly educated members in Panel C. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table 8: Job search outcomes: Firm quality and Job complexity

	Firm FE			Job Complexity		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Co-national network exposure - Main definition</i>						
Avg. quality	0.012** (0.006)	0.011** (0.004)	0.008** (0.004)	0.006 (0.006)	0.010* (0.006)	0.003 (0.007)
boot. p-val.	[0.037]	[0.020]	[0.061]	[0.250]	[0.110]	[0.646]
Mean Y	0.965	0.969	0.977	0.117	0.260	0.286
Obs.	1,535	2,295	2,012	10,250	9,975	9,783
Adj. R ²	0.019	0.014	0.010	0.035	0.111	0.126
<i>Panel B: Co-national network exposure - Alternative definition</i>						
Share Top-Quartile	0.026 (0.016)	0.040*** (0.014)	0.020 (0.012)	0.035** (0.017)	0.048*** (0.015)	0.035* (0.018)
boot. p-val.	[0.131]	[0.009]	[0.122]	[0.069]	[0.005]	[0.070]
Mean Y	0.965	0.969	0.977	0.117	0.260	0.286
Obs.	1,535	2,295	2,012	10,250	9,975	9,783
Adj. R ²	0.018	0.015	0.009	0.036	0.111	0.127
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	No	No	No	Yes	Yes	Yes
Municipality controls	Yes	Yes	Yes	-	-	-

Notes: This table reports OLS estimates for network measures of firm quality in our main estimating equation, using as outcomes the AKM-based establishment fixed effect of new refugees' employers (columns 1 to 3) and an index of job complexity, proxying for quality, for their jobs (columns 4 to 6). Panel A measures network exposure to employers using the baseline average quality of firms in the co-national network, while Panel B uses the share of top-quartile firms in the network. In columns 1 to 3, we use only the refugees in our sample for whom employer fixed effects are observed. Specifications in these columns do not include municipality of assignment fixed effects but control for key characteristics of the municipality of assignment (share of college-educated individuals aged 18-65, population share of the country total aged 18-65). We use our main specification in columns 4 to 6. Job quality is defined on the basis of task content, using an index equal to the ratio of communication and cognitive tasks to manual tasks. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

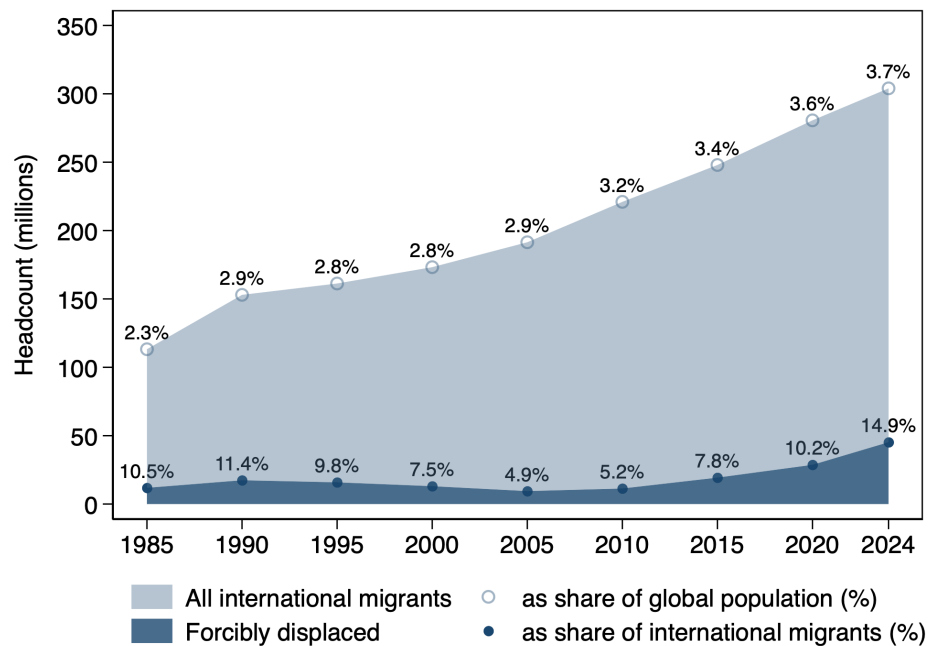
Table 9: Use of job referrals: Evidence of implications

	Wages				Turnover			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Member Present	-0.025 (0.045)	0.033 (0.030)	0.049 (0.039)	0.065* (0.037)	-0.106*** (0.040)	-0.131*** (0.049)	-0.108** (0.055)	0.006 (0.065)
Member * Tenure	0.004 (0.007)	-0.002 (0.005)	-0.006 (0.005)	-0.008* (0.005)	0.003 (0.006)	0.009 (0.007)	0.008 (0.008)	-0.003 (0.009)
Obs.	31,943	15,480	6,059	9,092	32,034	15,507	6,068	9,120
Adj. R^2	0.154	0.145	0.182	0.573	0.083	0.076	0.079	0.165
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Productivity Tercile FE	No	Yes	Yes	-	No	Yes	Yes	-
First Jobs Sample	No	No	Yes	Yes	No	No	Yes	Yes
Establishment FE	No	No	No	Yes	No	No	No	Yes

Notes: This table reports OLS estimates from equation 5. The outcome in columns 1 to 4 is log wages, while in columns 5 to 8 it is an indicator variable for leaving one's current employer. "Member present" is an indicator used as a proxy for job referral, equal to one if a newly arrived refugee starts a job in an establishment where at least one member of their initial local co-ethnic network is still employed at the time of job start. Individuals and establishment controls are listed in the text. Specifications in columns 2, 3, 6, and 7 include terciles of establishment productivity fixed effects, while columns 3, 4, 7, and 8 restrict the sample to job spells that are each refugee's first in Denmark. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Appendix A Additional Evidence and Descriptives

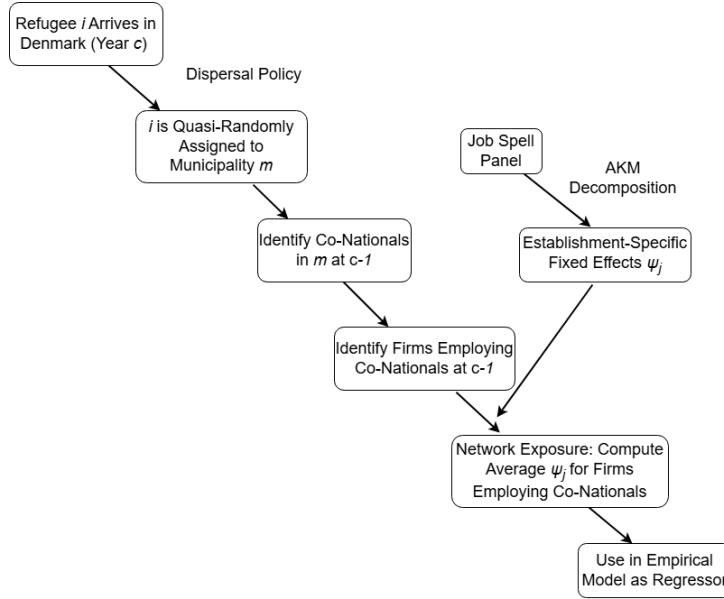
Figure A1: Global forcibly displaced population



Notes: This figure plots the worldwide number of international migrants (light blue) and the number of forcibly displaced people, including refugees, asylum seekers, and others in need of international protection but excluding internally displaced persons (dark blue), from 1985 to 2024. It also reports the share of international migrants in the global population and the share of forcibly displaced group within the total international migrant population.

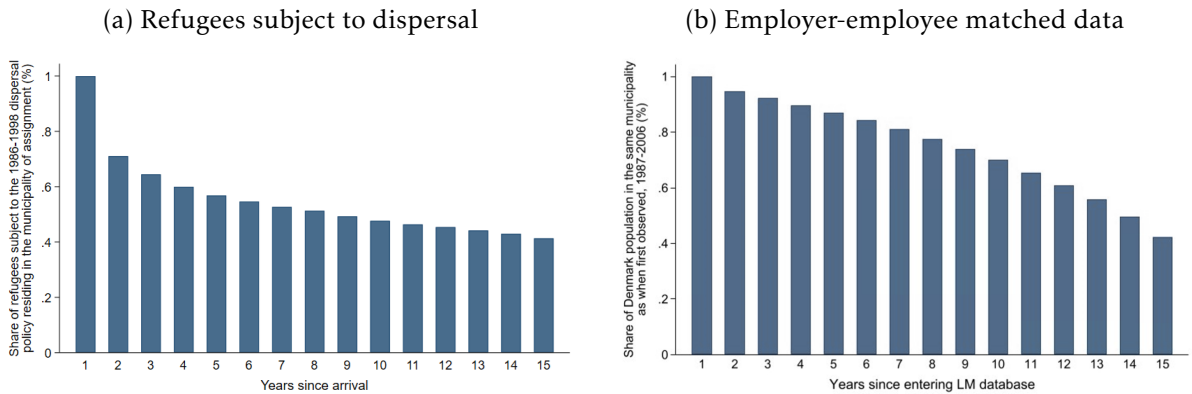
Source: United Nations High Commissioner for Refugees (UNHCR); United Nations Department of Economic and Social Affairs (UN DESA).

Figure A2: Construction of employer-based regressor



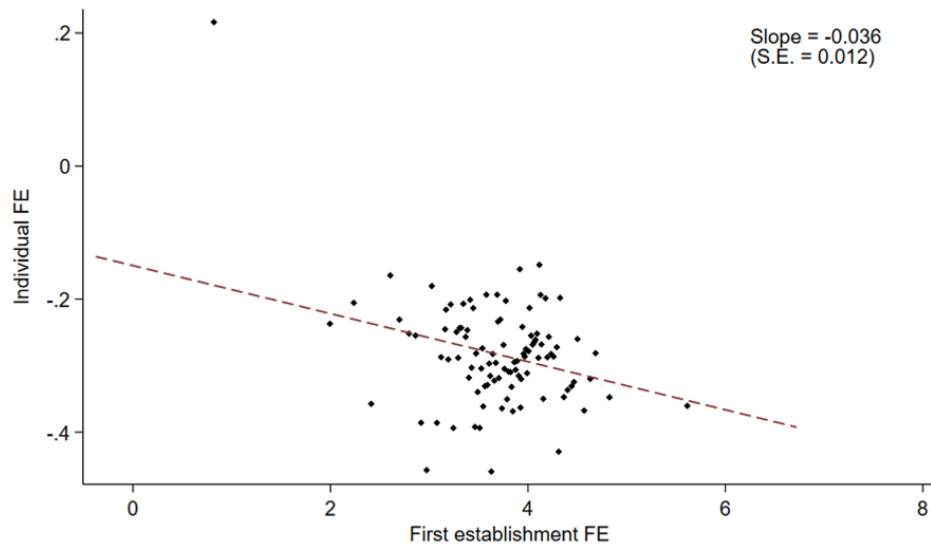
Notes: The diagram shows the methodology employed to construct the average quality of establishments employing members of the local co-ethnic network at the time of arrival, the regressor in our main empirical results.

Figure A3: Mobility from initial municipality



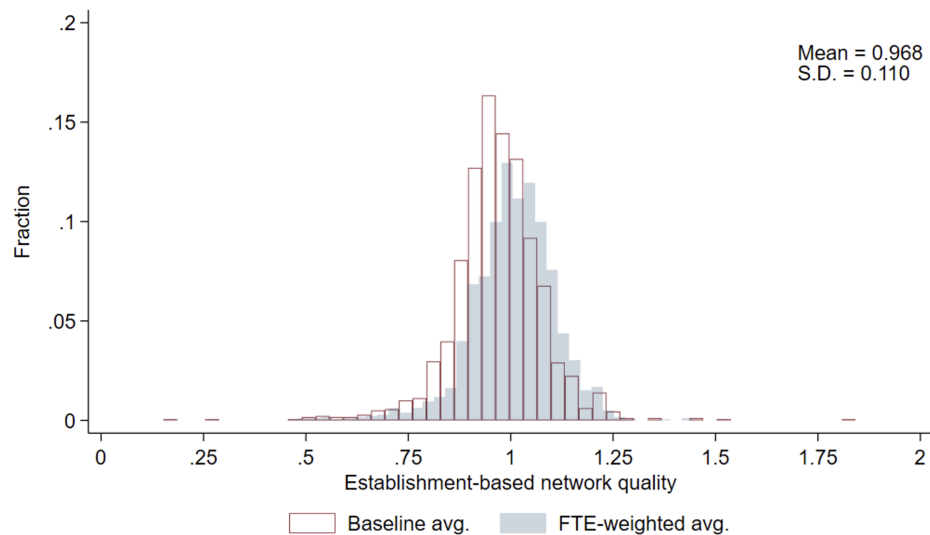
Notes: This figure plots the share of individuals who remain in the same municipality over time. Panel A considers the sample of refugees subject to the 1986–1998 dispersal policy that we use in the analysis ($N=15,578$) and plots the share who still reside in the municipality of initial assignment by year since arrival. Panel B considers the entire set of individuals included in the employer-employee matched dataset during the 20 years following the start of the dispersal policy and plots the share who still reside in the same municipality as the one they were first observed in the data.

Figure A4: Refugees' assortative matching



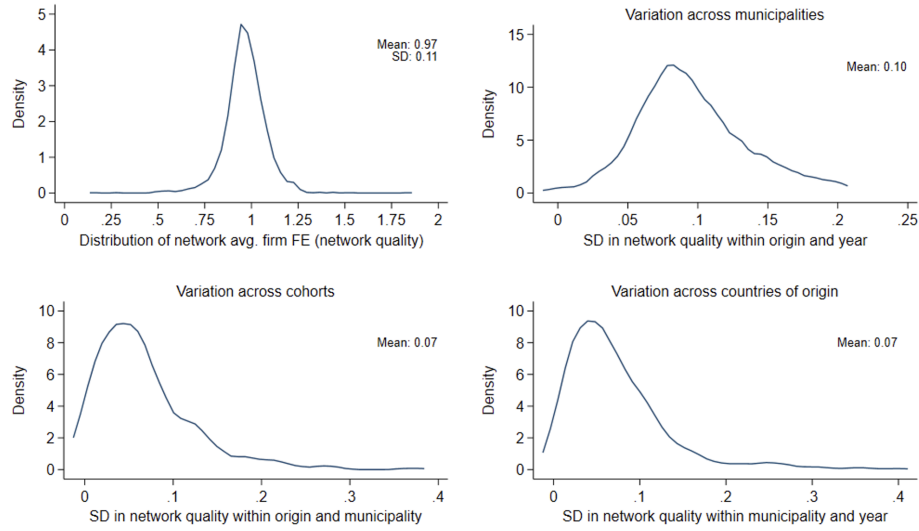
Notes: This figure plots the binned relationship between individual fixed effects for refugees, proxying for unobserved abilities, and the fixed effects of their first workplace, proxying for establishment quality. Both sets are estimated using the AKM decomposition. Each marker represents 18 individuals.

Figure A5: Variation in main treatment variables



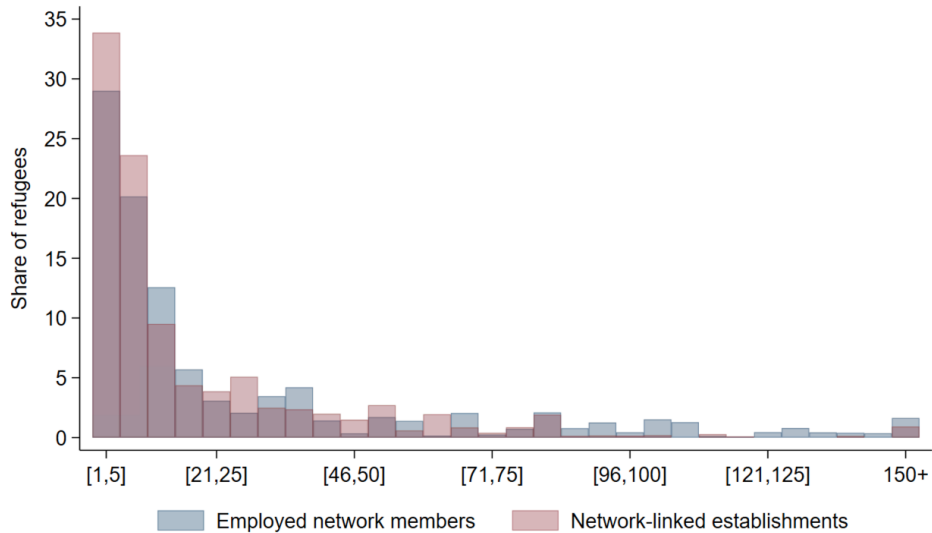
Notes: This figure displays the variation in our analysis sample for the main network quality measures (red-outlined bars for the unweighted average, light blue bars for the average weighted by full-time equivalents). The x-axis is measured in log wage points.

Figure A6: Dimensions of variation



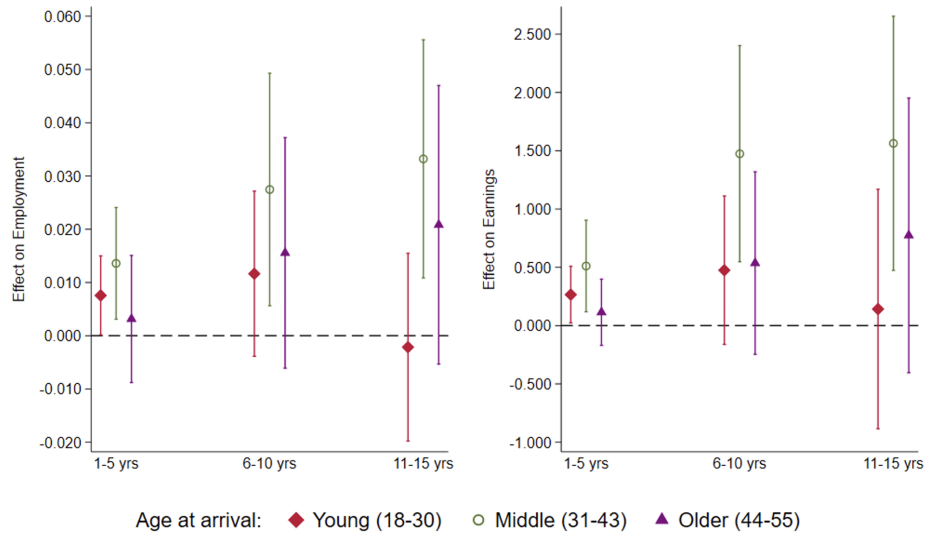
Notes: This figure shows how the main treatment variable (average firm fixed effect in the network) varies across the three dimensions of the network (municipality, cohort, origin). The top left corner plots the overall variation. The top right panel shows the distribution of standard deviations in network quality across municipalities, holding origin and cohort (year) fixed. The bottom left (right) corner plots the distribution of the standard deviations in the network quality across cohorts (countries of origin), fixing municipality and origin (year).

Figure A7: Number of connections



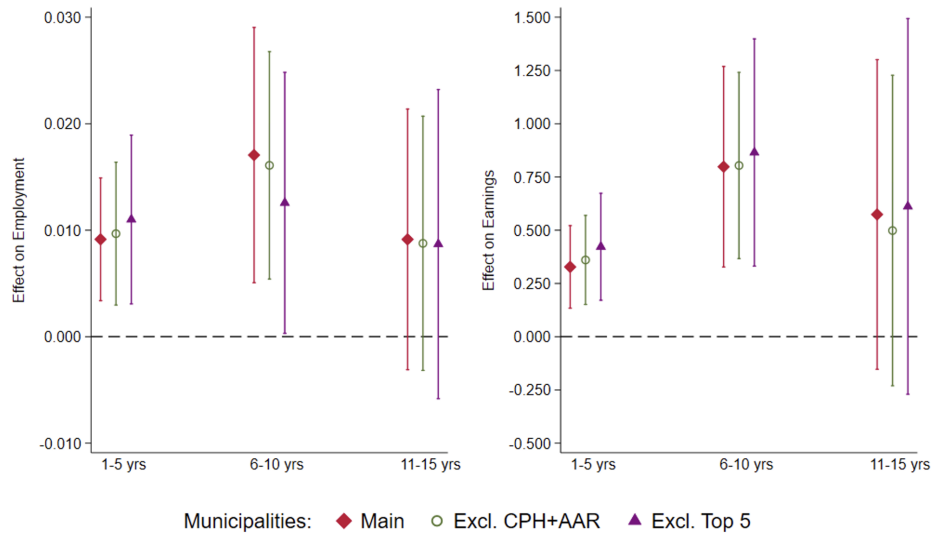
Notes: This figure shows the distribution of network connections for dispersed refugees in the analysis sample. The blue histogram shows the number of network members employed and working at establishments in the largest connected set for each refugee. The red histogram indicates the number of distinct establishments linked to each network that contribute to the average employer quality. Each histogram bar represents the share of refugees whose relevant network size or number of linked establishments falls within a given range of width five.

Figure A8: Heterogeneity analysis by age



Notes: This figure plots the estimates and 95 percent confidence intervals of the differential effect of network quality (unweighted measure) on the main outcomes (employment in the left panel and earnings in the right panel) by age at arrival. Refugees are divided into three groups spanning the same number of years: young (18 to 30 years of age), middle-aged (31 to 43 years) and older (44 to 55 years). Each outcome is examined over three time intervals: short run (1 to 5 years since arrival in Denmark), medium run (6 to 10 years), and long run (11 to 15 years).

Figure A9: Heterogeneity analysis by municipality



Notes: This figure plots the estimates and 95 percent confidence intervals of the differential effect of network quality (unweighted measure) on the main outcomes (employment in the left panel and earnings in the right panel) by municipality of assignment. Three groups of municipalities of assignment are considered: all municipalities; all excluding Copenhagen and Aarhus (top 2); and all excluding Copenhagen, Aarhus, Odense, Aalborg and Esbjerg (top 5). Each outcome is examined over three time intervals: short run (1 to 5 years since arrival in Denmark), medium run (6 to 10 years), and long run (11 to 15 years).

Table A1: Sample of refugees pre- and post-restrictions: Balance table

	Excluded			Analysis sample			Diff.
	Count (1)	Mean (2)	SD (3)	Count (4)	Mean (5)	SD (6)	
<i>Demographics</i>							
Age at Entry	4,131	28.41	7.11	15,578	29.84	8.25	1.431***
Female	4,131	0.45	0.50	15,578	0.42	0.49	-0.031**
Married/cohabiting	4,131	0.68	0.47	15,578	0.65	0.48	-0.024**
Number of Children	4,131	0.91	1.54	15,578	1.06	1.62	0.148***
Academic Education (prior to arrival)	1,293	0.26	0.44	6,566	0.26	0.44	-0.004
Latin Alphabet of Mother Tongue	4,131	0.49	0.50	15,578	0.33	0.47	-0.165***
From Predominantly Muslim Country	4,131	0.23	0.42	15,578	0.14	0.35	-0.091***
Asia	4,131	0.07	0.25	15,578	0.15	0.36	0.081***
Middle East	4,131	0.48	0.50	15,578	0.66	0.48	0.171***
Africa	4,131	0.45	0.50	15,578	0.20	0.40	-0.252***
<i>Labor market</i>							
Employment (any)	4,131	0.15	0.22	15,578	0.22	0.27	0.074***
Annual Earnings (thousand USD)	4,131	5325	10075	15,578	8245	12926	2919***
Avg. Hourly Wage Rate in Estab.	1,575	28.24	12.28	8,080	28.42	12.56	0.187
Complex Job (indicator)	4,127	0.09	0.19	15,577	0.12	0.22	0.028***

Notes: This table displays characteristics (mean and standard deviation) for individual refugees subject to the 1986–1998 dispersal policy, separated into those observed for less than 15 years in Denmark and later-arriving spouses (columns 1-3), both excluded from the final analysis sample, and those included (columns 4-6). The difference in means is presented in column (7). Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table A2: First employers and refugees' labor market outcomes: Alternative specification

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
First employer quality	0.038*** (0.007)	0.037*** (0.009)	0.036*** (0.009)	3.560*** (0.397)	2.666*** (0.442)	2.753*** (0.546)
boot. p-val.	[0.000]	[0.001]	[0.002]	[0.000]	[0.000]	[0.000]
Mean of Y	0.720	0.577	0.522	23.488	24.467	25.157
Obs.	4,217	4,111	3,885	4,217	4,115	3,897
Adj. R^2	0.052	0.075	0.100	0.068	0.053	0.065
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	No	No	No	No	No	No
Cohort FE	No	No	No	No	No	No
Origin FE	No	No	No	No	No	No

Notes: This table reports OLS estimates from regressions where the outcome variable is a measure of refugees' employment probability (columns 1-3) and annual earnings measured in 2015 USD (columns 4-6). We regress the outcome on the standardized quality measure of the first employer. The short-run (columns 1 and 4), medium-run (columns 2 and 5), and long-run (columns 3 and 6) effects are defined relative to the year of hiring by the first employer, normalized to 1. This sample consists of refugees subject to the 1986–1998 dispersal policy in Denmark. This specification does not include fixed effects. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table A3: First employers and wages on the extended sample

	Ln(Hourly Wage)			Ln(Hourly Wage)		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
First employer quality	0.102*** (0.008)	0.050*** (0.005)	0.032*** (0.005)	0.098*** (0.008)	0.046*** (0.005)	0.027*** (0.005)
boot. p-val.	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]	[0.000]
Mean of Y	3.250	3.347	3.352	3.250	3.347	3.352
Obs.	10,029	7,927	6,313	10,021	7,920	6,296
Adj. R^2	0.078	0.065	0.041	0.092	0.082	0.071
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	No	No	No	Yes	Yes	Yes
Cohort FE	No	No	No	Yes	Yes	Yes
Origin FE	No	No	No	Yes	Yes	Yes

Notes: This table reports OLS estimates from regressions where the outcome variable is log of real hourly wages (2015 USD) and main regressor is the standardized quality measure of the first employer. The sample consists of refugees who arrived in Denmark during the 1986–1998 dispersal policy. We present a specification without fixed effects (columns 1–3) and one with the main set of fixed effects (columns 4–6). The short-run (columns 1 and 4), medium-run (columns 2 and 5), and long-run (columns 3 and 6) effects are defined relative to the year of hiring by the first employer, normalized to 1. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table A4: Variation in treatment variables: Raw and residualized

	Mean	S.D.	p25	p75
Network employer quality	0.968	0.110	0.914	1.028
Residuals: net of FEs	-0.000	0.0858	-0.042	0.043
Network employer quality (FTE-weighted)	1.005	0.108	0.950	1.071
Residuals: net of FEs	0.000	0.0818	-0.034	0.040
Share top-quartile	0.2728	0.3148	0.000	0.438
Residuals: net of FEs	0.000	0.252	-0.148	0.103

Notes: This table presents summary statistics for the treatment variables used in the analysis, collapsed at the network level (municipality by year by country of origin). We report statistics for three variables: baseline average network quality (unweighted), average network quality weighted by establishment full-time equivalents, and the share of establishments in the network that fall within the top quartile of the overall quality distribution. For each variable, we report both raw variation and residualized variation after controlling for the main set of fixed effects used in the analysis (municipality fixed effects, cohort fixed effects, and origin fixed effects).

Table A5: Municipality characteristics: Correlation matrix

	Avg. Network Quality (1)	Empl. Rate (2)	Empl. Rate, Immigrants (3)	Annual Earnings (4)	Pop. Size (5)	Share of Non-Western Immigrants (6)	Share of Co-Nationals (7)	Urban Area (8)
Avg. Network Quality	1.00							
Empl. Rate	0.30***	1.00						
Empl. Rate, Non-Western Immigrants	0.29***	0.49***	1.00					
Annual Earnings	0.34***	0.72***	0.31***	1.00				
Pop. Size	0.08*	-0.18***	-0.12***	-0.11***	1.00			
Share of Non-Western Immigrants	0.23***	0.17***	0.02	0.29***	0.32***	1.00		
Share of Co-Nationals	-0.20***	-0.27***	-0.34***	-0.33***	-0.06	-0.03	1.00	
Urban Area	0.22***	0.12***	0.01	0.20***	0.44***	0.40***	-0.20***	1.00

Notes: This table presents a correlation matrix of municipality characteristics. Employment rates and earnings are measured as annual averages at the municipality level. For each refugee-sending country, we compute (i) the establishment-based average network quality and (ii) the share of co-nationals already residing in the municipality, both at the municipality–year level. We then average each of these measures across origin countries within every municipality–year cell. The analysis is restricted to municipality–years in which network quality is observed for at least one country of origin. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table A6: Geographic effects

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Geographic exposure - Unweighted average</i>						
Avg. quality	-0.014*** (0.004)	-0.003 (0.006)	0.006 (0.006)	-0.357*** (0.131)	0.020 (0.209)	0.514* (0.280)
boot. p-val.	[0.011]	[0.676]	[0.230]	[0.037]	[0.929]	[0.078]
Co-national share	0.014** (0.006)	-0.005 (0.006)	-0.007 (0.007)	0.505** (0.199)	-0.273 (0.218)	-0.275 (0.343)
Emp. NW immigrants	0.024*** (0.006)	0.016** (0.007)	0.019** (0.008)	0.773*** (0.197)	0.563** (0.264)	0.931** (0.399)
Mean Y	0.105	0.259	0.325	2.984	8.803	13.871
Obs.	15,571	15,082	14,722	15,571	15,082	14,722
Adj. R ²	0.084	0.155	0.131	0.069	0.121	0.096
<i>Panel B: Geographic exposure - Weighted average</i>						
Avg. quality	-0.014*** (0.003)	-0.004 (0.006)	0.005 (0.006)	-0.396*** (0.109)	-0.101 (0.203)	0.371 (0.289)
boot. p-val.	[0.002]	[0.578]	[0.320]	[0.004]	[0.661]	[0.206]
Co-national share	0.013** (0.006)	-0.005 (0.006)	-0.006 (0.007)	0.498** (0.198)	-0.262 (0.217)	-0.247 (0.336)
Emp. NW immigrants	0.024*** (0.006)	0.017** (0.007)	0.020** (0.008)	0.787*** (0.199)	0.595** (0.262)	0.962** (0.392)
Mean Y	0.105	0.259	0.325	2.984	8.803	13.871
Obs.	15,571	15,082	14,722	15,571	15,082	14,722
Adj. R ²	0.083	0.155	0.131	0.069	0.121	0.095
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	No	No	No	No	No	No
Origin FE	No	No	No	No	No	No
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents estimates of the effect of exposure to employer quality on employment and earnings of refugees subject to the 1986–1998 dispersal policy. To compute effects in the short-run, medium-run, and long-run, we set the year of admission to Denmark to 1. The quality measure is the average establishment effect for establishments active in the municipality of assignment at the time of arrival (unweighted in Panel A and weighted by full-time-equivalent employment in Panel B). Individual controls include variables observed by authorities in the dispersal process. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table A7: Characteristics of individual refugees matched with network

	Unmatched			Matched			
	Count	Mean	SD	Count	Mean	SD	Diff.
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>Demographics</i>							
Age at entry	5,305	29.84	7.86	10,273	29.85	8.45	0.010
Female	5,305	0.38	0.48	10,273	0.44	0.50	0.062***
Married	5,305	0.74	0.35	10,273	0.73	0.36	-0.006
Pre-arrival academic education	5,305	0.11	0.32	10,273	0.10	0.31	-0.010
Latin mother tongue	5,305	0.39	0.49	10,273	0.30	0.46	-0.092**
Predominantly muslim country	5,305	0.83	0.30	10,273	0.74	0.39	-0.097***
<i>Labor market</i>							
Mean employment dummy	5,305	0.21	0.26	10,273	0.23	0.28	0.015
First (log) labor income	3,403	8.39	1.66	6,555	8.53	1.60	0.142**
Mean complex job dummy	5,305	0.11	0.21	10,272	0.13	0.22	0.017***
<i>Municipality of assignment</i>							
(log) Conationals in initial municipality	3,997	2.58	1.56	10,124	5.23	1.63	2.649***
Conationals share, initial municipality	5,305	0.08	0.12	10,273	0.27	0.22	0.188***
Urban, initial municipality	5,301	0.28	0.45	10,273	0.72	0.45	0.437***
Initial Municipality Pop. share of country total (18-65)	5,301	0.01	0.01	10,273	0.04	0.03	0.029**
Initial Municipality Empl. rate (18-65), Any Empl.	5,301	69.42	3.67	10,273	68.90	2.78	-0.523
Initial Municipality Empl. rate (Nonwestern imm., 18-65), Any Empl.	5,301	46.56	11.87	10,273	42.98	7.74	-3.587***
(log) Avg. labor income in initial municipality (USD 2015)	5,301	10.36	0.13	10,273	10.36	0.11	-0.002
Share college educated in initial municipality (18-65)	5,301	0.15	0.05	10,273	0.18	0.05	0.031***

Notes: This table displays characteristics (mean and standard deviation) for individual refugees subject to the 1986–1998 dispersal policy and included in our sample, separated into those not matched with a network (columns 1-3) and those matched with a network (columns 4-6). The difference in means is presented in column (7). Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table A8: Effect of network employer quality in terciles relative to missing network

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Co-national network exposure - Terciles based on unweighted average</i>						
Low-quality	0.003 (0.007)	0.016 (0.012)	0.017 (0.015)	0.054 (0.218)	0.898* (0.518)	0.626 (0.771)
Mid-quality	0.010 (0.007)	0.032** (0.013)	0.021 (0.019)	0.203 (0.239)	1.197** (0.583)	0.746 (0.886)
High-quality	0.016** (0.007)	0.031*** (0.010)	0.018 (0.013)	0.549** (0.225)	1.361*** (0.497)	0.843 (0.679)
<i>Panel B: Co-national network exposure - Terciles based on weighted average</i>						
Low-quality	0.001 (0.006)	0.014 (0.012)	0.015 (0.015)	0.038 (0.184)	0.714 (0.482)	0.535 (0.765)
Mid-quality	0.018** (0.008)	0.031** (0.013)	0.017 (0.017)	0.466** (0.224)	1.369** (0.598)	0.579 (0.886)
High-quality	0.013* (0.007)	0.036*** (0.011)	0.026** (0.013)	0.441** (0.210)	1.394** (0.553)	0.925 (0.720)
Obs.	14,731	14,266	13,928	14,731	14,266	13,928
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents OLS estimates from regressions of refugees' outcomes on three dummy variables that capture different terciles of the network employer quality distribution. Missing network quality is the omitted category. Panel A uses the distribution of the unweighted network measure, while Panel B uses the weighted measure (with firms weighted by full-time-equivalent employment). Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table A9: Effects on the intensive margin of annual earnings

	Earnings			ln(Earnings)		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Conditional on reporting positive earnings</i>						
Avg. quality	0.426*	1.000**	0.057	0.094***	0.093**	-0.038
	(0.230)	(0.390)	(0.537)	(0.034)	(0.039)	(0.044)
boot. p-val.	[0.086]	[0.029]	[0.917]	[0.010]	[0.039]	[0.436]
Co-national share	0.050	0.147	-0.077	-0.027	0.020	-0.014
	(0.053)	(0.094)	(0.530)	(0.036)	(0.033)	(0.042)
Emp. NW immigrants	0.275	0.528	0.032	0.105	0.063	0.011
	(0.715)	(0.803)	(1.142)	(0.103)	(0.077)	(0.099)
Mean Y	8.403	17.566	26.892	1.134	2.020	2.478
Obs.	3,738	5,092	5,143	3,736	5,092	5,143
Adj. R ²	0.144	0.112	0.065	0.101	0.088	0.057
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents OLS estimates of the effect of exposure to network employer quality on earnings of refugees subject to the 1986–1998 dispersal policy, conditional on reporting positive earnings in the interval analyzed. To compute effects in the short-run, medium-run, and long-run, we set the year of admission to Denmark to 1. The quality measure used is the average establishment effect for establishments active in the municipality of assignment at the time of arrival. Individual controls include variables observed by authorities in the dispersal process. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Appendix B AKM Model and Evidence

B.1 Endogenous mobility

Consistent OLS estimation of equation 1 requires the assumption of conditional exogenous mobility. As discussed in [Card et al. \(2013\)](#), a sufficient condition for this assumption to hold is that, using h_j to denote establishment indicators, $E[h_j' r] = 0$ for each establishment j . This amounts to saying that the error term r in equation 1 is conditionally independent of employer transitions, implying that the probability of being assigned to an establishment for a worker depends only on worker and plant characteristics.⁷⁷ Essentially, this condition prevents forms of “endogenous mobility” driven by specific characteristics of the worker-employer match that can be interpreted as “interaction effects”.⁷⁸

A variety of tests have been proposed to ensure that data are consistent with this exogenous mobility assumption (e.g., [Card et al. \(2013\)](#) for Germany, [Card et al. \(2016\)](#) for Portugal, [Arellano-Bover and San \(2023\)](#) for Israel, [Dostie et al. \(2023\)](#) for Canada, [Song et al. \(2019\)](#) for the US, among others). Below, we replicate some of these checks in our setting to rule out problematic forms of endogenous mobility. Overall, we consider the evidence provided to support the assumptions needed for the AKM decomposition to be a reasonable approximation of labor market dynamics in Denmark during our period of study.

Following [Card et al. \(2013\)](#), in Figure B1 we begin by presenting a simple event study analysis that examines the wage effects of job transitions, where origin and destination workplaces are classified into quartiles based on the mean wages of other workers at those workplaces.⁷⁹ The figure shows that different mobility groups exhibit distinct wage levels both before and after a move, consistent with expectations based on the quartile rankings. At the same time, there is strong evidence that moving to a job with higher-paid coworkers leads to a wage increase, while transitioning to a lower quartile results in a wage reduction. Such patterns of systematic wage changes indicate that different establishments pay different average wage premia to their employees.

A crucial feature of Figure B1 is the approximate symmetry (i.e., similar magnitude, opposite sign) of the wage gains and losses for workers moving between quartile 1 and quartile 4 establishments.⁸⁰ This symmetry is inconsistent with sorting based on the idiosyncratic match component of wages, a form of endogenous mobility that would introduce bias to the AKM approach.⁸¹ Additional reassurance against this type of sorting is provided by Panel C of Table

⁷⁷Worker-employer matching is assumed to be based on a combination of the permanent component of individuals’ ability and the average pay premium offered by workplaces.

⁷⁸To examine different forms of endogeneity, we can assume, as in [Card et al. \(2013\)](#), that the composite error can be rewritten as a sum of three separate random effects:

$$r_{it} = \eta_{ij(i,t)} + \zeta_{it} + \epsilon_{it}$$

where $\eta_{ij(i,t)}$ is a match component, ζ_{it} is a unit root component, and ϵ_{it} is a transitory error. Valid OLS estimates require that firm-to-firm transitions are not related to components of r_{it} .

⁷⁹For clarity, we only report wage profiles for workers leaving quartile 1 and quartile 4 establishments.

⁸⁰A transition from quartile 1 to quartile 4 is associated with a trend-adjusted wage gain of 32.7 log points, while a transition from quartile 4 to quartile 1 with a trend-adjusted loss of 30.6 log points.

⁸¹This evidence suggests that idiosyncratic match effects are not of great importance. If variation in wages across establishments reflects differences in average wage premia rather than sorting, then movers will experience systematic wage changes. Individuals who move to establishments where other workers are highly paid (low paid)

B1, which compares the adjusted- R^2 from the AKM model to that of a fully saturated model where log wages are regressed on an indicator for each worker-employer spell (the job match effects model from [Card et al. \(2013\)](#)). While the statistical fit is slightly better for the job match effects model (adjusted- $R^2 = 0.865$), the roughly 7 percentage-point difference between the adjusted- R^2 values suggests that the additively separable AKM model of wages is fairly accurate.

More broadly, violations of the separability assumptions in the AKM model can be assessed by examining residuals for specific types of matches. Figure B2 plots the mean wage residuals across 100 cells, defined by deciles of person effects and establishment effects, as in [Card et al. \(2013\)](#). While deviations are observed among the lowest-decile establishments, even the largest deviations are less than 0.5% in magnitude, strongly supporting the conclusion that the additive structure of equation 1 provides a good approximation of the wage-setting process.

Figure B1 also shows no indication of an Ashenfelter transitory dip in movers' wages prior to a move, effectively ruling out any connection between firm-wide shocks and mobility rates.⁸² More importantly, the figure displays no evidence of systematic mobility patterns in which workers moving to higher-wage firms exhibit different wage trends prior to their move compared to those transitioning to lower-wage firms. A second form of endogenous mobility would arise if the direction of firm-to-firm mobility were correlated with transitory wage shocks, ϵ_{it} ([Card et al., 2016](#)). However, our evidence suggests that such a correlation is not present.

Finally, we consider an aspect related to the reliability of our estimates. First, since AKM decompositions require large sample sizes to ensure sufficient mobility, Panel A of Table B1 also considers the extent to which limited mobility could pose an issue in our setting. The average number of movers per employer in the sample used to estimate equation 1 is above 10, comfortably exceeding the threshold of 6 suggested by [Andrews et al. \(2012\)](#), beyond which limited mobility bias is unlikely to be a concern. Nevertheless, to deal with limited mobility concerns and measurement error, in Section 7 we will also subject our analysis to a sensitivity check introducing split-sample instrumental variable (IV) estimates to correct for possible bias.

B.2 AKM model with immigrant and native groups

A natural concern is that establishment effects estimated using the entire sample of workers active in Denmark may not reflect the actual workplace quality experienced by foreign-born individuals, and by refugees in particular. This could occur if employers engage in forms of within-firm differentiation between groups, potentially driven by differences in bargaining power, outside options, firm-specific labor supply elasticities, or reservation wages ([Adda et al., 2022](#); [Arellano-Bover and San, 2023](#); [Dustmann et al., 2024a](#)). Such a scenario would be problematic if the resulting workplace ranking of quality differ substantially between natives

will, on average, experience wage gains (losses). This is what is predicted by an additive model with exogenous mobility, and it is precisely what we observe in this exercise.

⁸²As explained in [Card et al. \(2016\)](#), workers may be more likely to leave workplaces experiencing negative shocks and move to firms undergoing positive shocks. In such cases, we would expect to observe a systematic dip in the wages of workers about to leave.

and immigrants.

To address this concern, and building on the previous AKM decomposition, we estimate a model with separate workplace effects by nativity for firms in the “dual-connected set”—i.e., the set of firms included in the connected sets for both natives and immigrants (Dostie et al., 2023; Drenik et al., 2023). In this model, y_{it} is generated by:

$$\log(y_{it}) = \alpha_i + \psi_{j(i,t)}^{B(i)} + x_{it}\beta + \epsilon_{it}, \quad (9)$$

where $\psi_{j(i,t)}^{B(i)}$ represents nativity-specific, time-invariant firm fixed effects, with birthplace $B(i)$ equal to N if worker i is native-born and to M if worker i is foreign-born—i.e., $B(i) \in N, M$. Pay premia offered by workplaces are allowed to vary by group but are assumed to be the same for all workers within each group. Clearly, since we only observe one fixed birthplace per worker, we cannot absorb potential average differences between native-born and foreign-born workers.

The magnitudes of the pay premia for native-born workers are only identified relative to those of immigrant workers by applying a normalization *across* the groups.⁸³ Therefore, we shift both the native-specific and immigrant-specific firm effect distributions by normalizing the mean of native-specific workplace effects to zero, and plot the resulting distributions in Figure B3. Workplace effects for immigrant workers are shifted downward compared to those for native workers. In the sample of “dual-connected” firms, the average pay premium for immigrants is -0.53 relative to the mean of workplace effects of native workers normalized to zero—i.e., the average pay premium is 53 log points lower for immigrants compared to natives. In other words, immigrant workers receive lower pay premia than native workers.

To examine whether firms extend their pay premia to immigrant labor, we compare the workplace pay premia earned by native workers and immigrant workers at the same workplace, following an approach similar to Arellano-Bover and San (2023) and Drenik et al. (2023). This relationship may reflect, for instance, the relative degree of rent sharing or the degree to which employers differentiate the pay of immigrant labor. We use the estimated workplace pay premia received by native workers, ψ^N , and compare those estimates to those for immigrant workers, ψ^M , at the same workplace j :

$$\psi_j^M = \alpha + \rho\psi_j^N + v_j \quad (10)$$

where ρ captures the elasticity of immigrant to native pay premia.⁸⁴ Figure B4 shows the binned relationship between native and immigrant workplace effects. The estimate of $\rho = 0.63$ implies that the pass-through of firm-level wage premium to immigrant workers is substantial, though not complete. For example, when firm A in the dual-connected set offers a 10% pay

⁸³This relies on the observation that person effects and pay premia are only identified up to a normalizing constant, such that adding a constant to all person effects and subtracting the same constant from all firm effects leaves the fitted values from the model unchanged (Abowd et al., 1999; Dostie et al., 2023). The relative pay-setting effect is identified only after normalizing the pay premia for natives and immigrants relative to each other.

⁸⁴As already noted, a normalization of workplace effects is necessary to interpret the elasticity as the proportion of workplace premia earned by native workers that immigrant workers receive at higher-paying firms (Card et al., 2016; Drenik et al., 2023). We follow these authors and normalize workplace effects to zero in the lowest vingtile, which does not affect the estimate of the slope ρ , as it would be absorbed by the constant.

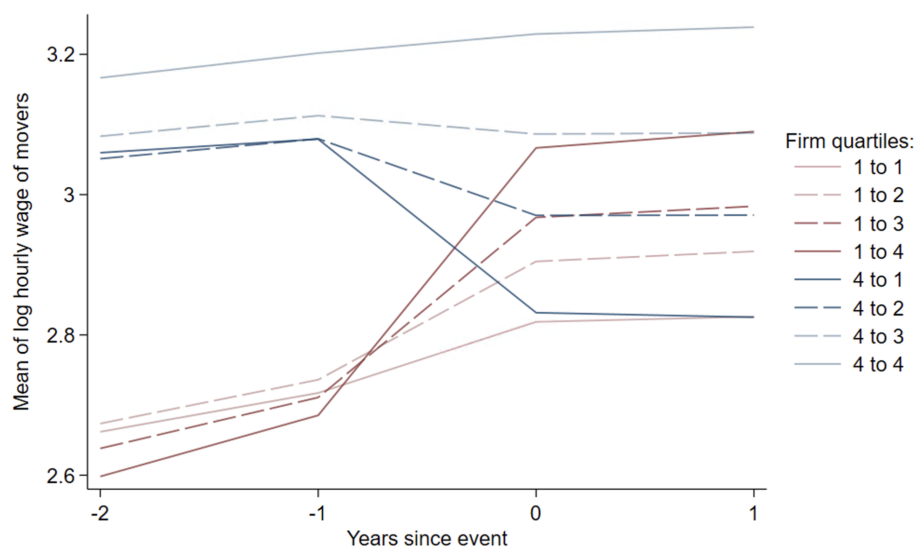
premium to its native workers compared with firm B, the corresponding pay premium for immigrant workers is 6.3% at A versus B, suggesting rather equal rent-sharing between firms and immigrant workers.⁸⁵

More importantly, despite evidence of imperfect pass-through, Figure B4 shows a strong alignment in the rankings of fixed effects across the two groups. In other words, workplaces that pay natives more also pay immigrants more. This provides additional reassurance that lower premia for immigrants relative to natives within the same firm do not threaten our AKM estimation. More broadly, [Hermansen et al. \(2025\)](#) document in a cross-country study that unequal access to higher-paying jobs is the primary driver (explaining around three fourths) of the immigrant-native earnings gap, rather than unequal pay for the same work at the same employer. Denmark, along with the U.S. and Sweden, stands out with the smallest immigrant-native differences, and in particular the smallest within-job pay gaps among the nine European and North American countries analyzed.

We complement our previous evidence with Figure B5, which plots the correlation between the establishment fixed effect estimated with the AKM and the establishment wage premium paid to immigrant workers. This correlation is virtually zero, ruling out that firms of good quality systematically pay a premium (or an under-premium) to their migrant employees. Overall, reassured by these findings, we then proceed to use the workplace effects estimated in equation 1 using the full sample of natives and non-refugee immigrants in our main analysis.

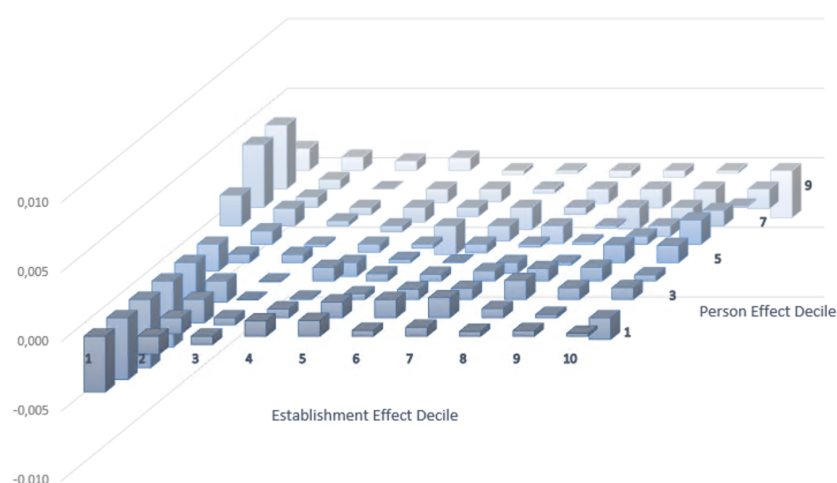
⁸⁵[Arellano-Bover and San \(2023\)](#) find a very similar correlation in Israel.

Figure B1: Wage changes for job movers by average wage quartile



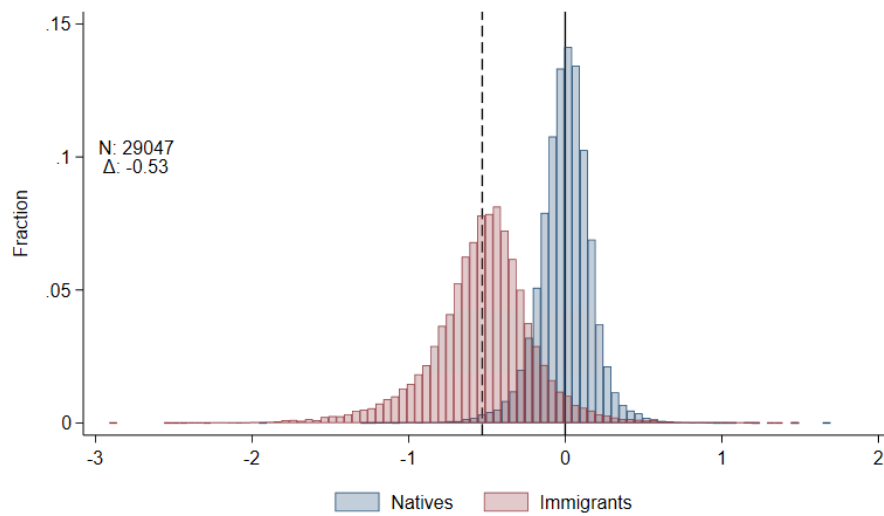
Notes: Event study analysis on the wage effects of job transitions. For any given worker, firms are categorized into quartiles based on the average wage of coworkers. Each point in the figure is the average wage by period, origin, and destination firm quartile, restricting the sample to workers who are employed for at least two years in both the origin and destination firms. The figure only displays transitions for workers leaving firms with the lowest-paid (quartile 1) and highest-paid (quartile 4) coworkers.

Figure B2: Mean residuals by person/establishment deciles



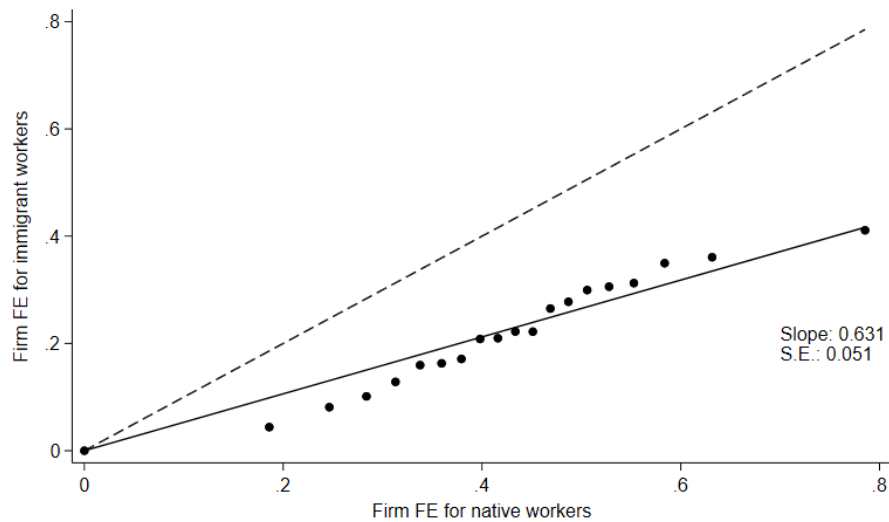
Notes: This figure plots the mean wage residuals from the AKM decomposition for specific types of matches. Mean wage residuals are displayed across 100 cells, defined by deciles of person effects (x-axis) and establishment effects (z-axis).

Figure B3: Distribution of establishment premia by birthplace



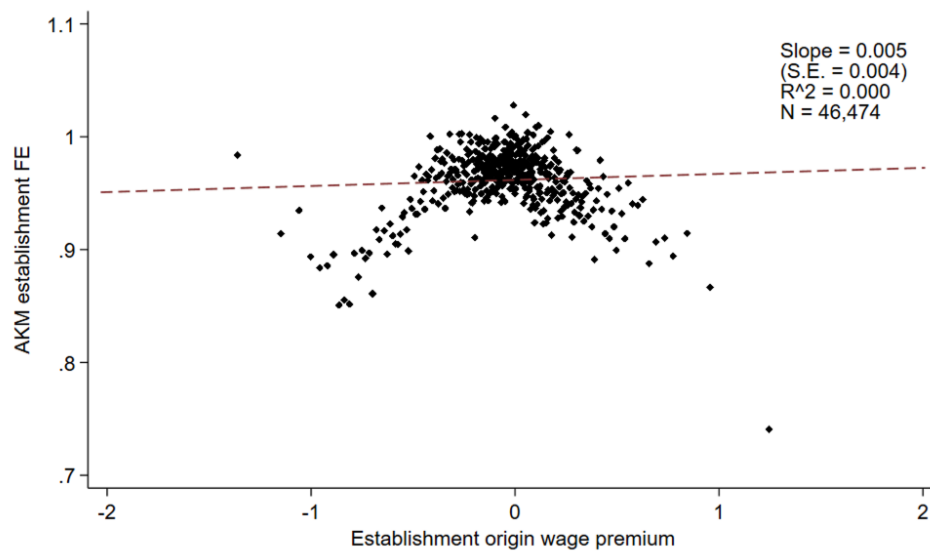
Notes: This figure plots the distribution of workplace pay premia by nativity. Native-specific (in red) and immigrant-specific (in blue) workplace effect distributions have been normalized to the mean of native-specific workplace effects. Nativity-specific workplace effects are computed for establishments in the dual-connected set ($N=29,047$).

Figure B4: Establishment pay premia sharing



Notes: This figure shows the binned relationship between native and immigrant workplace effects. We normalize workplace effects to zero in the lowest vintile (bottom 5%). The slope of the regression line captures the elasticity of immigrant to native pay premia.

Figure B5: Establishment quality and origin premium paid



Notes: This figure plots the binned relationship between the AKM-estimated establishment fixed effects and the establishment-specific average origin premia. The origin premium is computed for each establishment with at least one native and one foreign-born worker as the log of the ratio between the average real hourly wage paid to immigrants and that paid to natives, averaged over the years included in the the AKM sample (1986–1998). Each marker represents 100 establishment.

Table B1: Estimation summary for AKM model

	Full panel (pooled)	Largest connected set (pooled)
<i>Panel A: Summary statistics</i>		
No. of worker-year obs.	15,836,448	15,703,133
No. of unique workers	2,344,821	2,311,622
No. of unique employers	291,386	264,706
No. of total moves	2,804,122	2,802,315
No. of unique movers	1,312,173	1,310,531
Log hourly wage (mean)	2.93	2.93
Average no. of moves by employer		10.59
Average no. of movers by employer		10.32
<i>Panel B: Variance decomposition</i>		
Total variance		0.148
Worker FEs (% explained)		0.065 (44.0)
Employer FEs (% explained)		0.023 (15.5)
Year FEs (% explained)		0.029 (19.3)
2cov (% explained)		-0.007 (-4.5)
Residual (% explained)		0.026 (17.3)
<i>Panel C: Goodness of fit</i>		
RMSE of AKM model		0.160
Adj. R^2		0.793
RMSE of CHK match model		0.141
Adj. R^2		0.865

Notes: This table reports summary results from OLS estimation of equation 1. Panel A reports the main statistics for the 1986–1998 yearly job spell panel of natives and non-refugee migrants employed in the AKM model. Panel B shows the variance decomposition for log hourly wages (share of total variance explained in parenthesis). Panel C presents the goodness of fit from the AKM specification and compares it with a job-match effect model that includes a separate dummy for each job (person-establishment pair).

Table B2: Regressions of standardized AKM FE's on establishment characteristics

Specification:	Multivariate	Bivariate	Multivariate	Bivariate	Variable Mean / SD
Dependent Var.:	Establishment Quality				
	(1)	(2)	(3)	(4)	(5)
log(Value Added Per Worker)	0.046*** (0.005)	0.246*** (0.004)	0.036*** (0.005)	0.183*** (0.004)	12.711 [0.769]
log(Average Wage Bill Per Worker)	0.256*** (0.006)	0.408*** (0.005)	0.261*** (0.006)	0.351*** (0.005)	11.997 [0.744]
log(Employment)	0.072*** (0.002)	0.149*** (0.002)	0.077*** (0.002)	0.139*** (0.002)	2.053 [1.286]
Exporter (1 = Yes)	-0.010 (0.007)	0.304*** (0.006)	0.017** (0.007)	0.233*** (0.006)	0.408 [0.491]
Export/Sales	0.234*** (0.023)	0.642*** (0.044)	0.078** (0.019)	0.336*** (0.029)	0.075 [0.212]
High-skill Share	0.372*** (0.018)	0.721*** (0.018)	0.204*** (0.019)	0.439*** (0.019)	0.102 [0.204]
Mean of Y	0.860	0.860	0.860	0.860	
SD of Y	1.000	1.000	1.000	1.000	
Obs.	125,441	125,441	125,441	125,441	125,441
R-squared	0.108		0.170		
Industry FE	No	No	Yes	Yes	

Notes: The table reports regression results in which standardized average firm AKM fixed effects - calculated for each establishment and then averaged using worker weights - are regressed on firm characteristics for the years 1992–1998. Columns (1) and (3) display estimates from multivariate specifications, whereas columns (2) and (4) show estimates from bivariate regressions. Columns (1) and (2) are estimated without controlling for the 27-group industry codes, whereas columns (3) and (4) include these controls. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. Robust standard errors reported in parenthesis.

Table B3: Regressions of non-standardized AKM FE's on establishment characteristics

Specification:	Multivariate	Bivariate	Multivariate	Bivariate	Variable Mean / SD
Dependent Var.:	Establishment Quality				
	(1)	(2)	(3)	(4)	(5)
log(Value Added Per Worker)	0.012*** (0.001)	0.062*** (0.001)	0.00902*** (0.00121)	0.046*** (0.001)	12.711 [0.769]
log(Average Wage Bill Per Worker)	0.074*** (0.001)	0.103*** (0.001)	0.0653*** (0.00156)	0.088*** (0.001)	11.997 [0.744]
log(Employment)	0.018*** (0.001)	0.037*** (0.000)	0.0194*** (0.00057)	0.035*** (0.001)	2.053 [1.286]
Exporter (1 = Yes)	-0.002 (0.002)	0.077*** (0.001)	0.00421** (0.00166)	0.058*** (0.002)	0.408 [0.491]
Export/Sales	0.059*** (0.006)	0.162*** (0.011)	0.0196*** (0.00367)	0.084*** (0.007)	0.075 [0.212]
High-skill Share	0.094*** (0.004)	0.181*** (0.004)	0.0510*** (0.00483)	0.110*** (0.005)	0.102 [0.204]
Mean of Y	0.220	0.220	0.220	0.220	
SD of Y	0.250	0.250	0.250	0.250	
Obs.	125,441	125,441	125,441	125,441	125,441
R-squared	0.108		0.170		
Industry FE	No	No	Yes	Yes	

Notes: The table reports regression results in which non-standardized average firm AKM fixed effects - calculated for each establishment and then averaged using worker weights - are regressed on firm characteristics for the years 1992–1998. Columns (1) and (3) display estimates from multivariate specifications, whereas columns (2) and (4) show estimates from bivariate regressions. Columns (1) and (2) are estimated without controlling for the 27-group industry codes, whereas columns (3) and (4) include these controls. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. Robust standard errors reported in parenthesis.

Appendix C Other Social Outcomes

Our findings have broader policy implications for destination countries. Failures of refugee labor-market integration may increase the risk of participation in the informal economy or criminal activity (Bell et al., 2013; Pinotti, 2017). By contrast, effective labor-market integration can improve the material and psychological well-being of refugees and their families and foster broader social integration, thereby reducing social marginalization and, potentially, crime (Arendt et al., 2024). This dimension of integration is crucial for shaping natives' attitudes toward refugees, which in turn can either fuel opposition to immigration or strengthen support for additional refugee policies (Abbiati et al., 2025; Bansak et al., 2016).

In addition to examining employment and earnings, we now analyze crime outcomes for refugees and their children as measures of social integration. We study children who arrived in Denmark before age 18, no more than one year after their refugee parents (who are included in our main estimation sample), and who then lived in Denmark for at least 15 years. Our data on criminal convictions come from crime registers (KRAF, KRSI), which consist of nationwide police and court records and cover all criminal convictions and charges in Denmark. Following earlier studies, we consider all criminal convictions (excluding traffic violations) and distinguish between two categories: property crime and violent crime. Weak integration and limited ties to Danish society may leave refugees feeling marginalized and their children feeling inferior at school, experiencing learning difficulties, and becoming more prone to violent behavior. Moreover, lower disposable income may prompt efforts to supplement family resources, increasing involvement in property crime (Arendt et al., 2024; Dustmann et al., 2024c). Our crime categories follow Statistics Denmark's definitions. Violent crimes include offenses such as assault and homicide, while property crimes include theft and fraud.

Our results are reported in Tables C1 and C2. The former presents estimates of the effect of exposure to employer quality on crime outcomes for refugees covered by the 1986–1998 dispersal policy; the latter presents estimates for their children. We consider all crimes together (Panel A), property crimes (Panel B), and violent crimes (Panel C). For children, treatment is defined using the quality of firms in the network of the first-arriving parent.⁸⁶ We examine both the probability of conviction and the number of convictions, capturing the extensive and intensive margins. For adult refugees, we do not find statistically significant effects; magnitudes are small, consistent with earlier studies and with the fact that our treatment reflects exposure rather than direct intervention. By contrast, and consistent with other work on inter-generational consequences of welfare cuts and language training for refugees, we find long-run negative effects on violent crime for children, both in the probability of conviction and in the number of convictions, suggesting that labor market conditions propagate to other aspects of family integration.

⁸⁶In regressions for refugee children, we control for characteristics of the first-arriving parent (or of the father if the parents arrived together), as well as the child's gender and age at arrival.

Table C1: Effects on criminal convictions

	Probability of Conviction			Number of Convictions		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: All crime</i>						
Avg. quality	-0.009 (0.008)	-0.000 (0.005)	-0.003 (0.004)	-0.017 (0.017)	0.002 (0.012)	-0.012 (0.009)
boot. p-val.	[0.270]	[0.990]	[0.546]	[0.350]	[0.902]	[0.212]
Co-national share	-0.014** (0.006)	0.002 (0.005)	-0.008** (0.004)	-0.024* (0.013)	0.007 (0.012)	-0.027*** (0.009)
Emp. NW immigrants	-0.009 (0.013)	-0.008 (0.014)	0.010 (0.013)	0.029 (0.041)	0.055 (0.045)	0.027 (0.045)
Mean Y	0.173	0.112	0.093	0.279	0.202	0.166
Obs.	10,251	9,976	9,786	10,251	9,976	9,786
Adj. R ²	0.044	0.042	0.043	0.031	0.029	0.027
<i>Panel B: Property crime</i>						
Avg. quality	-0.011* (0.006)	0.001 (0.004)	0.002 (0.003)	-0.017 (0.012)	0.005 (0.009)	0.000 (0.007)
boot. p-val.	[0.123]	[0.851]	[0.551]	[0.211]	[0.596]	[0.968]
Co-national share	-0.010** (0.005)	0.000 (0.004)	-0.009*** (0.003)	-0.016* (0.010)	-0.002 (0.008)	-0.023*** (0.008)
Emp. NW immigrants	-0.025** (0.013)	0.003 (0.011)	0.008 (0.013)	-0.024 (0.022)	0.033 (0.032)	0.027 (0.035)
Mean Y	0.131	0.069	0.045	0.187	0.116	0.081
Obs.	10,251	9,976	9,786	10,251	9,976	9,786
Adj. R ²	0.029	0.020	0.011	0.021	0.017	0.003
<i>Panel C: Violent crime</i>						
Avg. quality	0.000 (0.003)	-0.003 (0.002)	-0.000 (0.003)	-0.001 (0.004)	-0.005* (0.003)	-0.000 (0.003)
boot. p-val.	[0.889]	[0.192]	[0.885]	[0.812]	[0.102]	[0.951]
Co-national share	-0.007*** (0.002)	-0.003 (0.002)	0.001 (0.002)	-0.008*** (0.003)	-0.004 (0.002)	0.001 (0.002)
Emp. NW immigrants	0.001 (0.005)	0.001 (0.006)	-0.003 (0.008)	0.002 (0.006)	0.005 (0.007)	-0.001 (0.009)
Mean Y	0.022	0.020	0.019	0.026	0.023	0.021
Obs.	10,251	9,976	9,786	10,251	9,976	9,786
Adj. R ²	0.017	0.013	0.016	0.011	0.010	0.013
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents OLS estimates of the effect of exposure to employer quality on crime outcomes for refugees subject to the 1986–1998 dispersal policy. Panel A shows effects on all convictions, excluding traffic, Panel B shows effects on property crimes and Panel C shows effects on violent crimes. To compute effects in the short-run, medium-run, and long-run, we set the year of admission to Denmark to 1. The quality measure used is the average establishment effect for establishments active in the municipality of assignment at the time of arrival. Individual controls include variables observed by authorities in the dispersal process. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

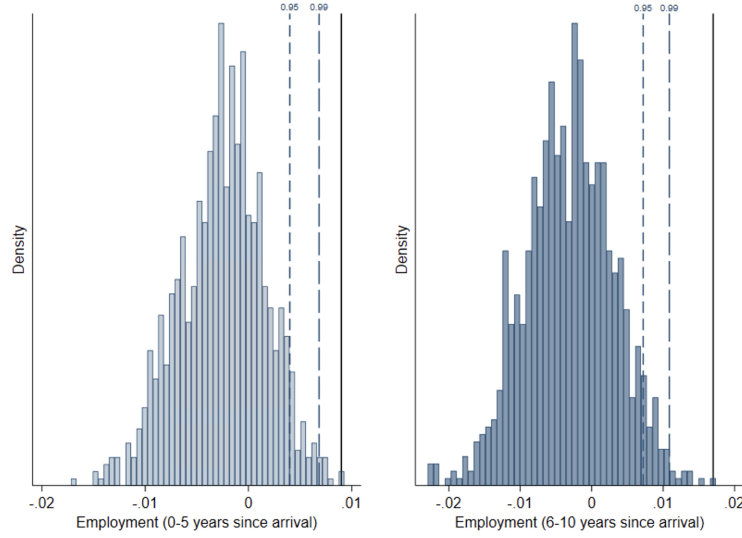
Table C2: Effects on criminal convictions for refugee children

	Probability of Conviction			Number of Convictions		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: All crime</i>						
Avg. quality	-0.000 (0.004)	-0.003 (0.009)	-0.013 (0.009)	0.006 (0.011)	0.003 (0.022)	-0.054 (0.041)
boot. p-val.	[0.921]	[0.761]	[0.186]	[0.653]	[0.914]	[0.235]
Co-national share	-0.003 (0.004)	-0.007 (0.008)	-0.001 (0.008)	-0.009 (0.013)	-0.024 (0.022)	0.022 (0.028)
Emp. NW immigrants	0.016 (0.014)	0.027 (0.020)	0.002 (0.022)	0.035 (0.027)	0.007 (0.053)	0.039 (0.076)
Mean Y	0.046	0.125	0.193	0.077	0.273	0.467
Obs.	5,412	5,330	5,264	5,412	5,330	5,264
Adj. R ²	0.120	0.138	0.134	0.078	0.107	0.101
<i>Panel B: Property crime</i>						
Avg. quality	-0.003 (0.004)	-0.001 (0.008)	-0.007 (0.007)	0.006 (0.008)	0.007 (0.017)	0.001 (0.022)
boot. p-val.	[0.535]	[0.924]	[0.378]	[0.589]	[0.739]	[0.960]
Co-national share	-0.002 (0.003)	-0.002 (0.007)	0.004 (0.007)	-0.005 (0.007)	-0.011 (0.014)	0.015 (0.013)
Emp. NW immigrants	0.020* (0.012)	0.013 (0.019)	0.006 (0.019)	0.021 (0.018)	-0.001 (0.038)	0.005 (0.040)
Mean Y	0.036	0.092	0.116	0.055	0.158	0.203
Obs.	5,412	5,330	5,264	5,412	5,330	5,264
Adj. R ²	0.095	0.083	0.068	0.061	0.072	0.054
<i>Panel C: Violent crime</i>						
Avg. quality	0.002 (0.003)	0.003 (0.004)	-0.017** (0.007)	0.002 (0.003)	0.001 (0.005)	-0.026*** (0.010)
boot. p-val.	[0.379]	[0.530]	[0.026]	[0.506]	[0.802]	[0.017]
Co-national share	-0.002 (0.003)	-0.004 (0.004)	0.002 (0.006)	-0.004 (0.005)	-0.006 (0.008)	0.002 (0.010)
Emp. NW immigrants	0.005 (0.009)	0.000 (0.011)	0.008 (0.012)	0.013 (0.011)	0.011 (0.018)	0.012 (0.017)
Mean Y	0.010	0.037	0.062	0.013	0.052	0.082
Obs.	5,412	5,330	5,264	5,412	5,330	5,264
Adj. R ²	0.030	0.066	0.067	0.026	0.072	0.054
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents OLS estimates of the effect of exposure to employer quality on crime outcomes for the children of refugees subject to the 1986–1998 dispersal policy. Panel A shows effects on all convictions, excluding traffic, Panel B shows effects on property crimes and Panel C shows effects on violent crimes. To compute effects in the short-run, medium-run, and long-run, we set the year of admission to Denmark to 1. The quality measure used is the average establishment effect for establishments active in the municipality of assignment at the time of arrival. Individual controls include parental variables observed by authorities in the dispersal process along with age at arrival and gender of the children. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

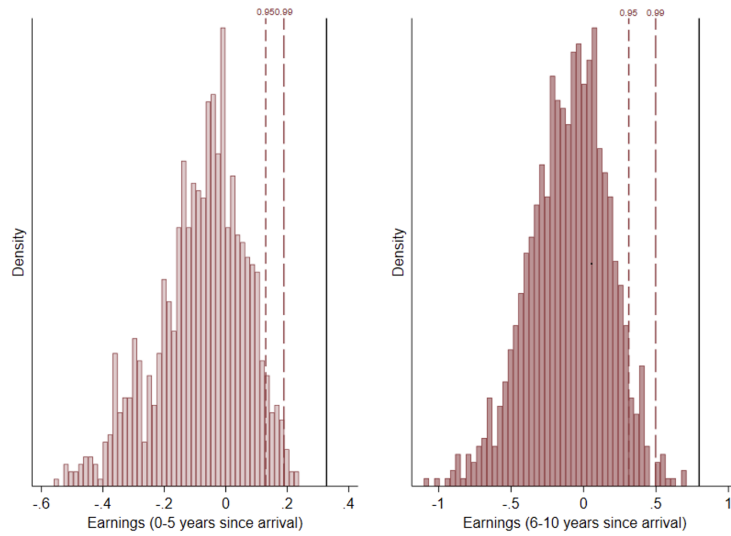
Appendix D Robustness Evidence

Figure D1: Permutation-based test: Employment



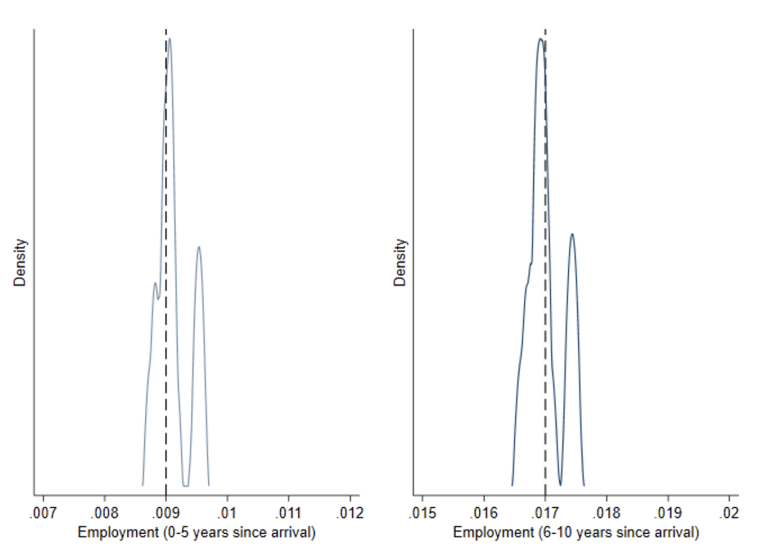
Notes: This figure plots the distribution of pseudo-placebo effects obtained by estimating coefficient β_1 in equation 3 1,000 times, using employment as outcome (short run for the left-hand side panel, medium run for the right-hand side panel). Every time we assign the network quality of one of the other eight origins to dispersed refugees from a given country of origin. The black solid line reflects the true effect estimated in our main results. The short-dotted line (long-dotted line) represents the 95th (99th) percentile of the placebo distribution.

Figure D2: Permutation-based test: Earnings



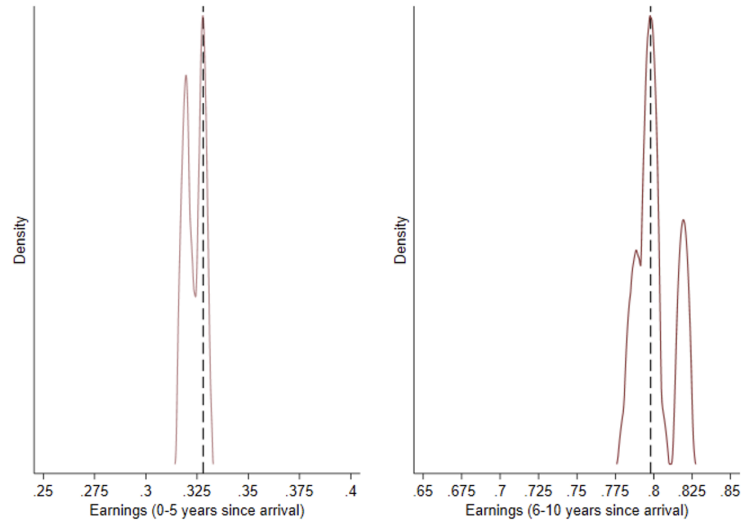
Notes: This figure plots the distribution of pseudo-placebo effects obtained by estimating coefficient β_1 in equation 3 1,000 times, using earnings as outcome (short run for the left-hand side panel, medium run for the right-hand side panel). Every time we assign the network quality of one of the other eight origins to dispersed refugees from a given country of origin. The black solid line reflects the true effect estimated in our main results. The short-dotted line (long-dotted line) represents the 95th (99th) percentile of the placebo distribution.

Figure D3: Permutations of local covariates: Employment



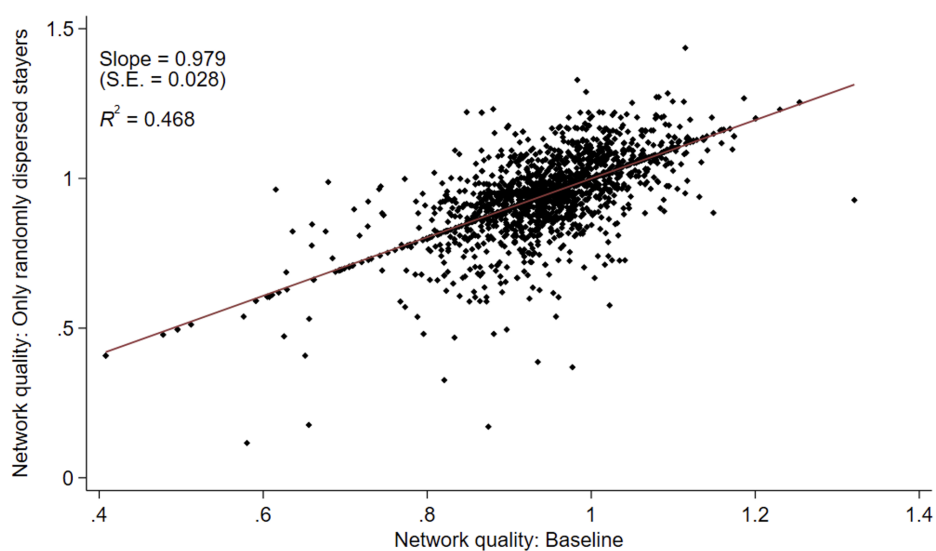
Notes: This figure plots the distribution of employment estimates (short run for the left-hand side panel, medium run for the right-hand side panel) obtained by estimating coefficient β_1 in equation 3 and iterating over possible combinations of local controls included. The pool of seven local covariates includes municipality measures for employment (refugees and co-nationals), income, education, and population density, and generates $2^7 - 1$ permutations. The black dotted line reflects the main effect estimated in our baseline specification.

Figure D4: Permutations of local covariates: Earnings



Notes: This figure plots the distribution of earnings estimates (short run for the left-hand side panel, medium run for the right-hand side panel) obtained by estimating coefficient β_1 in equation 3 and iterating over possible combinations of local controls included. The pool of seven local covariates includes municipality measures for employment (refugees and co-nationals), income, education, and population density, and generates $2^7 - 1$ permutations. The black dotted line reflects the main effect estimated in our baseline specification.

Figure D5: Construction of network measure, All co-nationals vs. randomly dispersed stayers



Notes: This figure plots the relationship between network quality based on all co-nationals in the municipality of assignment at arrival and network quality based only on co-nationals who were subject to the dispersal policy and remained in their assigned municipality. Each marker represents a local co-ethnic network, and the red solid line shows the line of best fit.

Table D1: Exposure to local firms: Alternative aggregations

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Geographic exposure</i>						
Avg. quality (weighted)	0.006 (0.009)	0.025 (0.023)	0.033 (0.038)	-0.158 (0.387)	-0.251 (1.211)	0.162 (2.011)
boot. p-val.	[0.541]	[0.487]	[0.591]	[0.695]	[0.861]	[0.956]
Co-national share	-0.002 (0.004)	-0.004 (0.005)	0.000 (0.006)	-0.059 (0.128)	-0.263 (0.230)	-0.156 (0.348)
Emp. NW immigrants	0.006 (0.006)	0.004 (0.008)	0.007 (0.011)	-0.014 (0.221)	-0.069 (0.388)	0.052 (0.561)
Mean Y	0.104	0.259	0.324	2.982	8.803	13.869
Obs.	15,556	15,064	14,703	15,556	15,064	14,703
Adj. R^2	0.165	0.199	0.191	0.139	0.153	0.145
<i>Panel B: Co-national network exposure</i>						
Avg. quality (weighted)	0.009*** (0.003)	0.018*** (0.005)	0.005 (0.007)	0.255** (0.105)	0.744*** (0.263)	0.254 (0.350)
boot. p-val.	[0.005]	[0.007]	[0.491]	[0.020]	[0.021]	[0.482]
Co-national share	-0.003 (0.004)	-0.003 (0.007)	-0.002 (0.006)	-0.012 (0.142)	-0.175 (0.284)	-0.059 (0.374)
Emp. NW immigrants	0.005 (0.014)	0.006 (0.013)	0.003 (0.018)	0.288 (0.443)	0.263 (0.583)	0.340 (0.935)
Mean Y	0.106	0.262	0.331	3.075	8.960	14.140
Obs.	10,227	9,952	9,762	10,227	9,952	9,762
Adj. R^2	0.191	0.220	0.206	0.168	0.175	0.157
<i>Panel C: Co-national network exposure</i>						
Share Top-Quartile	0.027*** (0.008)	0.048*** (0.017)	0.040** (0.018)	0.926*** (0.272)	1.991*** (0.687)	1.777 (1.086)
boot. p-val.	[0.000]	[0.031]	[0.053]	[0.002]	[0.017]	[0.132]
Co-national share	-0.004 (0.004)	-0.004 (0.007)	-0.002 (0.006)	-0.028 (0.142)	-0.220 (0.287)	-0.060 (0.375)
Emp. NW immigrants	0.005 (0.014)	0.005 (0.013)	0.004 (0.018)	0.277 (0.447)	0.249 (0.590)	0.355 (0.929)
Mean Y	0.107	0.262	0.331	3.080	8.975	14.151
Obs.	10,251	9,976	9,786	10,251	9,976	9,786
Adj. R^2	0.191	0.220	0.206	0.169	0.175	0.156
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Panel A and Panel B of this table replicate, respectively, Panel A and Panel C of Table 6, replacing the original measures with weighted versions. Average quality is calculated as the weighted average of workplace effects, using full-time equivalents as weights. Panel C considers the share of top employers in the network, defined by assigning a value of one to establishments in the top quartile of the Danish workplace pay premium distribution and then averaging this indicator within each network. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table D2: Exposure to local firms: Alternative specifications and restrictions

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Exposure measures jointly</i>						
Avg. geographic quality	-0.047 (0.029)	-0.013 (0.045)	-0.032 (0.058)	-1.691** (0.829)	-2.306 (1.774)	-2.061 (2.558)
boot. p-val.	[0.162]	[0.848]	[0.788]	[0.083]	[0.338]	[0.616]
Avg. network quality	0.010*** (0.003)	0.017*** (0.006)	0.010 (0.006)	0.365*** (0.095)	0.848*** (0.233)	0.620* (0.367)
boot. p-val.	[0.001]	[0.019]	[0.149]	[0.001]	[0.004]	[0.116]
Co-national share	-0.003 (0.004)	-0.003 (0.007)	-0.001 (0.006)	0.003 (0.144)	-0.151 (0.283)	-0.013 (0.362)
Emp. NW immigrants	0.005 (0.013)	0.005 (0.014)	0.004 (0.018)	0.280 (0.444)	0.240 (0.593)	0.351 (0.938)
Mean Y	0.107	0.262	0.331	3.080	8.975	14.151
Obs.	10,251	9,976	9,786	10,251	9,976	9,786
Adj. R ²	0.192	0.220	0.205	0.169	0.175	0.156
<i>Panel B: Network exposure - Including industry controls</i>						
Avg. quality	0.009*** (0.003)	0.017*** (0.006)	0.008 (0.006)	0.299*** (0.104)	0.755*** (0.241)	0.522 (0.370)
boot. p-val.	[0.008]	[0.026]	[0.232]	[0.012]	[0.011]	[0.189]
Co-national share	-0.003 (0.004)	-0.002 (0.007)	-0.000 (0.006)	0.018 (0.146)	-0.119 (0.292)	0.022 (0.361)
Emp. NW immigrants	0.007 (0.013)	0.004 (0.014)	0.002 (0.018)	0.255 (0.428)	0.138 (0.610)	0.234 (0.961)
Mean Y	0.107	0.262	0.331	3.080	8.975	14.151
Obs.	10,251	9,976	9,786	10,251	9,976	9,786
Adj. R ²	0.191	0.220	0.205	0.169	0.175	0.156
<i>Panel C: Network exposure - Early movers sample</i>						
Avg. quality	0.000 (0.005)	0.000 (0.008)	-0.005 (0.009)	0.150 (0.198)	0.119 (0.338)	-0.095 (0.461)
boot. p-val.	[0.929]	[0.960]	[0.595]	[0.474]	[0.724]	[0.834]
Co-national share	0.007 (0.008)	-0.002 (0.011)	0.023* (0.013)	0.500 (0.359)	0.381 (0.617)	1.514 (0.935)
Emp. NW immigrants	-0.015 (0.015)	0.003 (0.016)	0.003 (0.026)	-0.643 (0.588)	-0.567 (0.820)	0.065 (1.338)
Mean Y	0.113	0.266	0.331	3.382	9.188	14.749
Obs.	3,549	3,472	3,392	3,549	3,472	3,392
Adj. R ²	0.251	0.236	0.219	0.227	0.192	0.164
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Panel A of this table reports estimates from regressions that include both main measures of exposure to firm quality as regressors (geographic exposure and network-based exposure to employers hiring at least one co-national at arrival). Panel B examines network-level quality controlling for the industry shares of the network. Panel C repeats the estimation for network-level quality restricting the sample to refugees who relocate to a municipality different from their assigned one within their first five years after arrival ("early movers"). All quality measures are unweighted (calculated as the unweighted average of workplace effects). Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table D3: Effects of exposure to local firms: Alternative fixed effects

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Co-national network exposure</i>						
Avg. quality	0.008** (0.003)	0.013** (0.006)	0.005 (0.007)	0.231** (0.107)	0.558** (0.253)	0.342 (0.379)
boot. p-val.	[0.017]	[0.085]	[0.511]	[0.037]	[0.052]	[0.400]
Co-national share	-0.002 (0.004)	-0.005 (0.007)	-0.004 (0.008)	0.075 (0.140)	-0.139 (0.319)	-0.001 (0.435)
Emp. NW immigrants	-0.000 (0.012)	-0.001 (0.014)	0.002 (0.019)	0.094 (0.408)	0.211 (0.603)	0.399 (0.974)
Mean of Y	0.107	0.262	0.331	3.080	8.979	14.156
Obs.	10,246	9,971	9,781	10,246	9,971	9,781
Adj. R^2	0.205	0.226	0.210	0.185	0.180	0.159
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort-by-Origin FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Panel B: Co-national network exposure</i>						
Avg. quality	0.012** (0.005)	0.023*** (0.008)	0.010 (0.009)	0.370** (0.166)	1.112*** (0.354)	1.080** (0.529)
boot. p-val.	[0.035]	[0.025]	[0.279]	[0.053]	[0.013]	[0.065]
Co-national share	-0.004 (0.005)	-0.004 (0.008)	-0.007 (0.007)	-0.094 (0.184)	-0.253 (0.359)	-0.322 (0.427)
Mean Y	0.105	0.260	0.327	3.034	8.888	13.998
Obs.	10,030	9,758	9,569	10,030	9,758	9,569
Adj. R^2	0.205	0.226	0.213	0.182	0.187	0.159
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality-by-Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table replicates Panel C of Table 6, replacing the main set of fixed effects with alternative, more demanding ones. Panel A includes origin-by-cohort fixed effects along with municipality fixed effects. Panel B uses municipality-by-cohort fixed effects along with origin fixed effects. The specifications are otherwise left unchanged. Individual controls include variables observed by authorities in the dispersal process. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table D4: Sample restrictions and sensitivity

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Network size</i>						
Avg. quality (w/o bottom 10%)	0.008*** (0.003)	0.016*** (0.006)	0.008 (0.006)	0.307*** (0.101)	0.769*** (0.241)	0.543 (0.371)
boot. p-val.	[0.008]	[0.029]	[0.280]	[0.008]	[0.011]	[0.193]
Obs.	10,101	9,831	9,641	10,101	9,831	9,641
Avg. quality (w/o top 10%)	0.008*** (0.003)	0.015*** (0.006)	0.007 (0.006)	0.297*** (0.102)	0.736*** (0.246)	0.487 (0.367)
boot. p-val.	[0.018]	[0.025]	[0.254]	[0.006]	[0.012]	[0.226]
Obs.	8,734	8,517	8,355	8,734	8,517	8,355
<i>Panel B: Municipality of assignment's population</i>						
Avg. quality (w/o bottom 10%)	0.010*** (0.003)	0.017*** (0.006)	0.010 (0.006)	0.379*** (0.094)	0.823*** (0.241)	0.574 (0.372)
boot. p-val.	[0.002]	[0.024]	[0.151]	[0.001]	[0.005]	[0.145]
Obs.	10,054	9,782	9,595	10,054	9,782	9,595
Avg. quality (w/o top 10%)	0.009*** (0.003)	0.018*** (0.006)	0.010 (0.006)	0.346*** (0.095)	0.837*** (0.232)	0.575 (0.383)
boot. p-val.	[0.006]	[0.013]	[0.163]	[0.002]	[0.004]	[0.183]
Obs.	8,801	8,590	8,450	8,801	8,590	8,450
<i>Panel C: Municipality of assignment's share of national FTE</i>						
Avg. quality (w/o bottom 10%)	0.010*** (0.003)	0.017*** (0.006)	0.009 (0.006)	0.368*** (0.096)	0.826*** (0.240)	0.547 (0.375)
boot. p-val.	[0.001]	[0.035]	[0.192]	[0.001]	[0.008]	[0.201]
Obs.	10,064	9,793	9,605	10,064	9,793	9,605
Avg. quality (w/o top 10%)	0.010*** (0.003)	0.020*** (0.006)	0.013** (0.006)	0.359*** (0.095)	0.942*** (0.219)	0.727* (0.372)
boot. p-val.	[0.005]	[0.006]	[0.077]	[0.000]	[0.001]	[0.085]
Obs.	8,853	8,640	8,497	8,853	8,640	8,497
Main controls	Yes	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports estimates for the network-level measure of firm quality after trimming the top or bottom 10% of observations in our refugee sample based on size of co-ethnic network at arrival (Panel A), population in the municipality of assignment (Panel B), and municipality of assignment's share of national full-time equivalents (Panel C). Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table D5: Split-sample IV estimates

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Co-national network exposure - Unweighted average</i>						
Avg. quality	0.015*** (0.005)	0.022** (0.011)	0.023* (0.014)	0.351** (0.165)	0.879** (0.453)	1.090 (0.723)
Mean Y	0.106	0.262	0.331	3.070	8.956	14.176
Obs.	10,119	9,853	9,666	10,119	9,853	9,666
Adj. R^2	0.068	0.107	0.100	0.056	0.081	0.077
Kleib.-Paap F	63.01	62.57	62.50	63.01	62.57	62.50
<i>Panel B: Co-national network exposure - Weighted average</i>						
Avg. quality	0.010** (0.005)	0.021** (0.009)	0.008 (0.011)	0.210 (0.153)	0.766** (0.368)	0.341 (0.533)
Mean Y	0.106	0.262	0.331	3.070	8.947	14.168
Obs.	10,098	9,832	9,645	10,098	9,832	9,645
Adj. R^2	0.070	0.108	0.100	0.057	0.082	0.078
Kleib.-Paap F	87.70	85.48	83.38	87.70	85.48	83.38
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Share of co-nationals	Yes	Yes	Yes	Yes	Yes	Yes
Empl. rate NW	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports IV estimates from the following split-sample procedure. We randomly split the sample of workers used in the AKM estimation into two equal sized samples, calculate two sets of AKM estimates separately in both subsamples, aggregate establishment effects at the network level, and then use the network-level quality from one set of establishment effects as instrumental variable for the other in our main equation. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table D6: Additional controls

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Co-national employment</i>						
Avg. quality	0.009*** (0.003)	0.017*** (0.006)	0.009 (0.006)	0.321*** (0.097)	0.787*** (0.235)	0.566 (0.367)
boot. p-val.	[0.003]	[0.022]	[0.178]	[0.004]	[0.007]	[0.152]
Co-national empl.	0.013 (0.006)	0.012 (0.008)	0.001 (0.009)	0.306 (0.188)	0.505 (0.407)	0.328 (0.528)
Mean Y	0.107	0.262	0.331	3.080	8.975	14.151
Obs.	10,251	9,976	9,786	10,251	9,976	9,786
Adj. R ²	0.192	0.220	0.205	0.169	0.175	0.156
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Share of co-nationals	Yes	Yes	Yes	Yes	Yes	Yes
Empl. rate NW	No	No	No	No	No	No
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes
<i>Panel B: Native attitudes</i>						
Avg. quality	0.014*** (0.003)	0.021*** (0.007)	0.013* (0.007)	0.493*** (0.122)	1.043*** (0.278)	0.876** (0.432)
boot. p-val.	[0.000]	[0.007]	[0.108]	[0.001]	[0.002]	[0.063]
Anti-migration votes	-0.006 (0.005)	-0.030** (0.012)	-0.026* (0.014)	-0.111 (0.162)	-0.983* (0.506)	-0.772 (0.723)
Mean Y	0.108	0.265	0.329	3.111	9.363	14.241
Obs.	8,470	8,233	8,066	8,470	8,233	8,066
Adj. R ²	0.200	0.225	0.205	0.174	0.181	0.153
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Share of co-nationals	Yes	Yes	Yes	Yes	Yes	Yes
Empl. rate NW	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports robustness checks that incorporate additional controls into the main specification. Panel A replaces the employment rate of non-Western immigrants with the co-national employment rate (in the municipality of assignment, at the time of arrival). Panel B adds the municipality-level vote share for parties with anti-migration platforms across six election cycles (1989, 1990, 1993, 1994, 1997, 1998). Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table D7: Robustness with municipality-by-origin fixed effects

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
<i>Panel A: Co-national network exposure - Main definition</i>						
Avg. quality	0.008** (0.003)	-0.002 (0.008)	-0.015 (0.010)	0.274** (0.111)	0.024 (0.311)	-0.687 (0.466)
boot. p-val.	[0.019]	[0.844]	[0.162]	[0.023]	[0.952]	[0.164]
Mean Y	0.106	0.262	0.330	3.074	8.975	14.140
Obs.	10,220	9,944	9,753	10,220	9,944	9,753
Adj. R^2	0.191	0.204	0.171	0.168	0.169	0.132
<i>Panel B: Co-national network exposure - Alternative definition</i>						
Share Top-Quartile	0.029*** (0.010)	0.013 (0.027)	0.002 (0.030)	0.824*** (0.319)	0.501 (0.964)	-0.505 (1.373)
boot. p-val.	[0.006]	[0.686]	[0.949]	[0.015]	[0.666]	[0.763]
Mean Y	0.106	0.262	0.330	3.074	8.975	14.140
Obs.	10,220	9,944	9,753	10,220	9,944	9,753
Adj. R^2	0.192	0.204	0.171	0.168	0.169	0.132
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality-by-Continent FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort bin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table replicates our main estimates using a different, more demanding set of fixed effects that include region-by-municipality and arrival cohort fixed effects. Specifically, we grouped origin countries by broad geographic region (Middle East, Africa, and Southwest Asia) and grouped cohorts into three 4-year cohort bins. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table D8: Split-sample IV estimates with firm effects as outcome

	Firm FE		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)
<i>Panel A: Co-national network exposure - Unweighted average</i>			
Avg. quality (IV)	0.009 (0.010)	0.014 (0.009)	0.016** (0.008)
boot. p-val.	[0.373]	[0.139]	[0.042]
Mean Y	0.964	0.969	0.978
Obs.	1520	2268	1990
Adj. R^2	-0.008	-0.006	-0.009
Kleib.-Paap F	65.06	103.32	80.34
<i>Panel B: Co-national network exposure - Weighted average</i>			
Avg. quality (IV)	0.006 (0.009)	0.014* (0.008)	0.014** (0.006)
boot. p-val.	[0.571]	[0.114]	[0.026]
Mean Y	0.964	0.969	0.978
Obs.	1518	2265	1989
Adj. R^2	-0.007	-0.002	-0.010
Kleib.-Paap F	95.35	154.29	94.71
Controls	Yes	Yes	Yes
Municipality FE	No	No	No
Cohort FE	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes

Notes: This table reports IV estimates from the following split-sample procedure. We randomly split the sample of workers used in the AKM estimation into two equal sized samples, calculate two sets of AKM estimates separately in both subsamples, aggregate establishment effects at the network level, and then use the network-level quality from one set of establishment effects as instrumental variable for the other in our main equation. The remainder of the specification is as in Table 8. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Appendix E Additional Results on Mechanisms

E.1 Residential clustering

The sharing of information among network members might also pertain to other aspects beyond the labor market. One of the most prominent theories explaining ethnic segregation is related to networks, as social connections may lead to a concentration of members of the same origin in the same residential areas (or the same workplaces) through the exchange of information about residential (and job) opportunities (Glitz, 2014).⁸⁷ We then start by assessing whether the quality of networks, according to our employer-based definition, is linked to refugees' outcomes outside the labor market, such as residential mobility decisions, which would suggest an even more extensive flow of information among network members. While proximity to contacts is itself valuable, social connections provide hard-to-find local information that is useful when choosing among alternative locations (Büchel et al., 2020).

Hypothesizing that network members employed in better firms might be individuals who are more aware of good job opportunities and better integrated in the host country search market, and hence more active in sharing relevant information, we test whether higher network employer quality predicts more residential clustering. We construct an individual measure of residential clustering, or co-location rate, for each refugee by computing the ratio between co-nationals living in their parish (minus self) and co-nationals living in the whole municipality (minus self). We run the following specification:

$$y_{ioc,t} = \alpha + \beta_1 f[\hat{\psi}_j]_{om,c-1} + \sum_{\tau=2}^{15} \phi_{\tau} \mathbf{1}\{YSM_{it} = \tau\} + \sum_{\tau=2}^{15} \eta_{\tau} \mathbf{1}\{YSM_{it} = \tau\} * f[\hat{\psi}_j]_{om,c-1} \quad (11) \\ + \beta_2 Sh.Conat_{om,c} + \beta_3 Emp.NW_{m,c} + X_{i,t} + \gamma_m + \delta_o + \mu_c + \epsilon_{ioc,t}$$

where $y_{ioc,t}$ represents this individual co-location rate capturing how locally embedded members become over time. Our coefficients of interest η come from the interaction between indicators for years since migration (τ) and initial network employer quality. The model also includes the set of controls and fixed effects used in the main analysis.

Figure E2 plots the total effect of network quality at each year since migration, showing a clear and significant pattern over time. Refugees assigned to co-national networks with high-quality employers at arrival are more likely to live near other co-nationals, and this effect strengthens the longer they have been in the country. The increasing effect in later years may arise for two reasons. First, since refugees receive housing assistance and most of them use it, relocation is relatively costly in the initial years. Second, as more individuals move closer together, others are likely to follow, giving rise to a cumulative mechanism consistent with increasing information flow within the network over time.

⁸⁷Various studies have documented the link between labor market networks and residential proximity (e.g., Hellerstein et al. (2011)).

E.2 Co-ethnic managers

Co-ethnic connections to higher-quality firms may be particularly valuable when they involve individuals employed as managers, who often have greater influence over hiring decisions ([Åslund et al., 2025](#)). We therefore examine whether the positive effects of these connections are stronger when the employed co-ethnic contacts hold managerial positions.⁸⁸

We begin by identifying managers using occupation codes and classifying workers by task content. For 1987–1990, we adopt Statistics Denmark’s definition of managers; for 1991–1998, we follow [Bernard et al. \(2024\)](#), who classify workers into five broad categories, one of which is managers.⁸⁹ These definitions are fairly broad, encompassing salaried managers across small and large businesses, from small-store managers to managers of large corporations. We identify 743 distinct individuals from refugee-sending countries who held managerial positions and were part of the networks of the newly arrived refugees during the years of the dispersal policy we study.

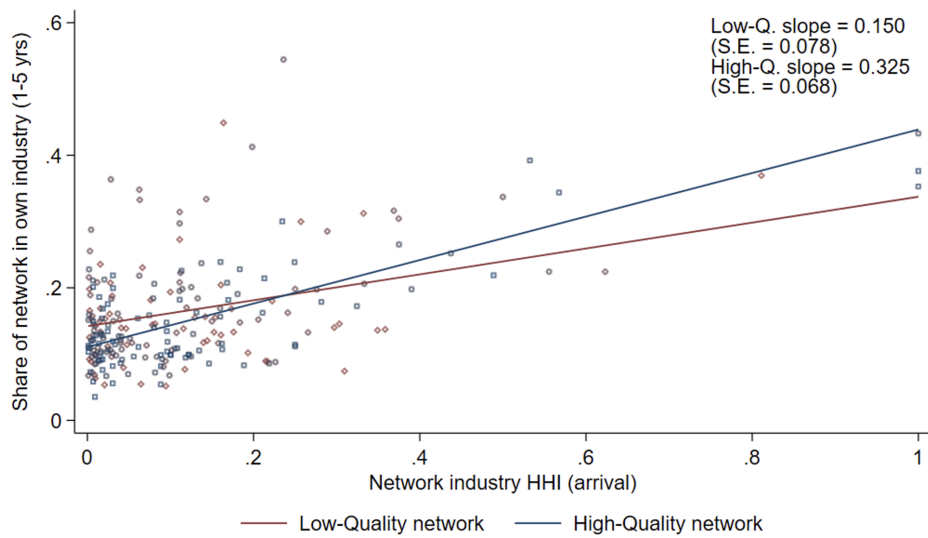
Next, having classified managers in our data, we define an indicator equal to one if a refugee in our sample had at least one co-ethnic employed as a manager at an establishment in the municipality to which the refugee was initially assigned. We then interact this indicator with both our baseline measure of network employer quality, constructed from all co-ethnics employed in any position at local establishments, and indicators for years since arrival. In doing so, our goal is to estimate whether the effect of establishment quality varies with the presence of co-ethnic managers in the network.

Figure E3 shows the estimated annual differential effects of network establishment quality when co-ethnic managers are part of the network, relative to having no co-ethnic managers in the network at arrival. The effect of employer quality is stronger in networks with managers, as such connections lead to higher employment (left panel) and annual earnings (right panel). The coefficients, however, are statistically significant only for some of the first five years after migration for earnings. Perhaps not surprisingly, the positive patterns then fade over time. Given the limited number of managers in the sample, these estimates are imprecise and should be interpreted with caution. Nonetheless, managerial connections may represent an additional mechanism contributing to the broader effects documented in our main results.

⁸⁸A related perspective is offered by [Dagnelie et al. \(2019\)](#), who study whether co-ethnic entrepreneurs, rather than managers, help refugees by hiring them.

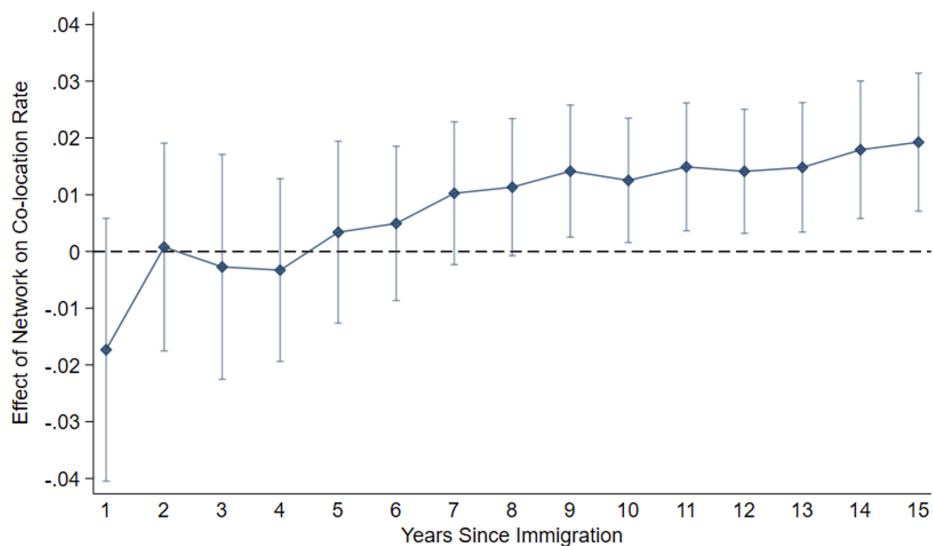
⁸⁹Since DISCO-88 codes, which are occupation codes nearly identical to ISCO-88, are available in the registers only from 1991 onward, we use the NYSTGR variable to identify managers in earlier years. Using NYSTGR, we apply Statistics Denmark’s definition; using DISCO codes, we follow [Bernard et al. \(2024\)](#). In both cases, occupation codes come from the Employment Classification Module (AKM) register.

Figure E1: Industry concentration: Unconditional correlations



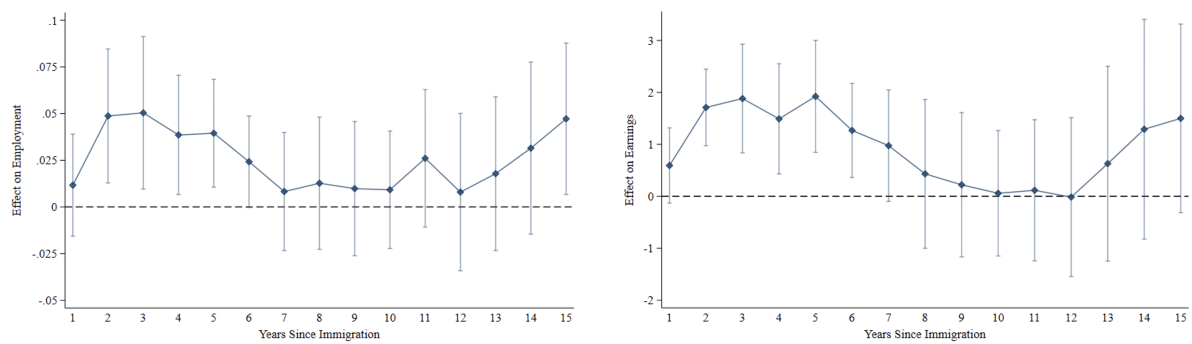
Notes: This figure plots the binned relationship between the share of an individual refugee's network in their own industry over the first five years after arrival and the network's industry Herfindahl-Hirschman Index (HHI) at arrival. Variable definitions are provided in the main text. Observations are split into two groups based on network quality (navy indicates networks with above-median quality, while maroon represents those below the median). Each marker represents 5 refugees.

Figure E2: Residential clustering



Notes: This figure plots the effect of network quality (unweighted measure) at each year since migration on refugees' individual co-location rate. The co-location rate is computed as the ratio between co-ethnics living in their parish (minus self) and co-ethnics living in the whole municipality (minus self). The coefficients are displayed with the 95% confidence intervals.

Figure E3: Role of co-ethnic managers



Notes: The figures display the yearly effects of co-ethnic links to firms employing co-ethnic managers. Variable definitions are provided in the main text. The left panel shows the effects on employment, while right panel presents the effects on earnings. Coefficients are shown with 95% confidence intervals.

Table E1: Job search outcomes: Geographic mobility

	Out-mobility			Timing	
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Hazard (4)	Hazard (5)
<i>Panel A: Co-national network exposure - Main definition</i>					
Avg. quality	0.008 (0.013)	0.014 (0.012)	0.023** (0.011)	0.061 (0.048)	0.060 (0.047)
boot. p-val.	[0.571]	[0.276]	[0.058]		
Mean Y	0.347	0.460	0.534	–	–
Obs.	10,251	9,976	9,786	21,932	21,932
Adj. R ²	0.177	0.203	0.230	–	–
<i>Panel B: Co-national network exposure: Alternative definition</i>					
Share Top-Quartile	0.001 (0.039)	0.009 (0.033)	0.036 (0.033)	0.084 (0.136)	0.086 (0.131)
boot. p-val.	[0.982]	[0.808]	[0.323]		
Mean Y	0.347	0.460	0.534	–	–
Obs.	10,251	9,976	9,786	21,932	21,932
Adj. R ²	0.176	0.203	0.230	–	–
Individual controls	Yes	Yes	Yes	Yes	Yes
Share of co-nationals	Yes	Yes	Yes	Yes	Yes
Empl. rate NW	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes
Interval FE	–	–	–	Yes	Yes

Notes: This table reports estimates of the effect of network measures of firm quality on refugees' out-mobility from the municipality of initial assignment. In columns 1 to 3, we use the same specification as in the main equation 3 and define the outcome as an indicator for geographic mobility equal to one if the refugee ever moved within 15 years after initial assignment. In columns 4 and 5, we estimate a discrete-time hazard model in which out-mobility for each refugee is grouped into three five-year intervals and the indicator outcome takes the value one in the first interval in which the move occurs. The reported coefficients capture the effect on the probability of moving out in a given interval, conditional on having remained in the municipality up to that point. Panel A measures network exposure to employers using the baseline average quality of firms in the co-national network, while Panel B uses the share of top-quartile firms in the network. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table E2: Industry concentration and network employer quality

	Network share in own industry						
	Yr. 1-5 (1)	Yr. 1-5 (2)	Yr. 6-10 (3)	Yr. 6-10 (4)	Yr. 1-3 (5)	Yr. 4-6 (6)	Yr. 7-10 (7)
Network HHI	0.276*** (0.051)		0.110** (0.048)				
High network HHI		0.075*** (0.020)		0.033* (0.018)	0.035* (0.019)	0.017 (0.017)	0.011 (0.028)
High network quality					-0.004 (0.019)	-0.006 (0.017)	-0.022 (0.028)
Interaction					0.051** (0.019)	0.047*** (0.017)	0.023 (0.028)
Mean Y	0.169	0.169	0.136	0.136	0.184	0.149	0.132
Obs.	1,292	1,292	1,117	1,117	610	1,205	823
Adj. R ²	0.232	0.142	0.091	0.076	0.175	0.108	0.072
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Municipality quality	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports OLS estimates on industry sorting, where the outcome variable is the share of a refugee's co-national network working in the same industry as the individual during their first years after arrival. A Herfindahl–Hirschman Index (HHI) based on the distribution of network members across industries measures the industry concentration of the co-national network at arrival. Columns 1 and 3 use a continuous HHI as the main regressor ("Network HHI"), while columns 2 and 4 use a binary HHI indicator ("High network HHI", implying that networks with below-median HHI are the excluded category). Columns 5 to 7 also include employer quality in the network as a regressor to study the interaction between employer quality and industry concentration in the network. Relative to the main specification, these regressions replace municipality fixed effects with a measure municipality quality, given the smaller sample size available, and include an additional control for individuals' academic attainment to account for sorting. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Table E3: Effect of employer quality in the network by network size

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
Avg. quality	0.016*** (0.004)	0.015* (0.008)	0.003 (0.008)	0.540*** (0.148)	1.003*** (0.310)	0.533 (0.485)
boot. p-val.	[0.001]	[0.060]	[0.681]	[0.002]	[0.004]	[0.329]
Small network	0.158* (0.084)	-0.075 (0.105)	-0.108 (0.116)	4.115 (3.067)	-0.444 (4.505)	-0.454 (5.785)
Large network	0.132** (0.056)	0.027 (0.129)	-0.067 (0.113)	5.444*** (2.082)	9.415* (5.511)	1.448 (6.063)
Avg. quality*Small netw.	-0.016* (0.009)	0.011 (0.011)	0.014 (0.012)	-0.369 (0.334)	0.172 (0.479)	0.158 (0.598)
boot. p-val.	[0.088]	[0.313]	[0.267]	[0.284]	[0.727]	[0.800]
Avg. quality*Large netw.	-0.015** (0.006)	-0.002 (0.014)	0.009 (0.011)	-0.596** (0.236)	-0.990 (0.614)	-0.092 (0.624)
boot. p-val.	[0.077]	[0.903]	[0.456]	[0.036]	[0.182]	[0.877]
Obs.	10,251	9,976	9,786	10,251	9,976	9,786
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Empl. rate NW	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table presents OLS estimates from regressions of refugees' outcomes on average employer quality in the network interacted with dummy variables that capture different terciles of the network size distribution. Mid-sized network is the omitted category. Controls used in main equation 3 are included. Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$. P-values from wild cluster bootstrap are reported in square brackets.

Table E4: Co-national network exposure: Employer quality and willingness to hire

	Employment			Earnings		
	Yr. 1-5 (1)	Yr. 6-10 (2)	Yr. 11-15 (3)	Yr. 1-5 (4)	Yr. 6-10 (5)	Yr. 11-15 (6)
Avg. quality	0.004 (0.006)	0.010 (0.012)	0.003 (0.012)	0.176 (0.186)	0.295 (0.515)	0.329 (0.621)
Medium will.	-0.132** (0.061)	-0.064 (0.103)	0.014 (0.112)	-4.363* (2.275)	-5.550 (4.924)	-0.797 (5.267)
High will.	-0.041 (0.086)	-0.126 (0.142)	-0.183 (0.145)	-0.588 (2.260)	-7.591 (6.093)	-10.118 (6.774)
Avg. quality*Medium will.	0.014** (0.007)	0.006 (0.012)	0.001 (0.013)	0.482* (0.262)	0.596 (0.559)	0.117 (0.595)
Avg. quality*High will.	0.004 (0.010)	0.015 (0.016)	0.024 (0.017)	0.054 (0.257)	0.885 (0.680)	1.127 (0.800)
Obs.	8,897	8,693	8,522	8,897	8,693	8,522
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Municipality FE	Yes	Yes	Yes	Yes	Yes	Yes
Cohort FE	Yes	Yes	Yes	Yes	Yes	Yes
Origin FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports OLS estimates of the effect of the interaction between terciles of network average willingness to hire refugees (“Low will.”, “Medium will.”, and “High will.”) and network employer quality on refugees’ labor market outcomes. The omitted category is the bottom tercile of openness to hire a refugee (“Low”). Robust SE in parentheses are clustered at the municipality of assignment level. Significance levels: *** for $p < 0.01$, ** for $p < 0.05$, * for $p < 0.1$.

Appendix F Additional Results on Optimal Assignment

We train and test two supervised machine learning models: a Least Absolute Shrinkage and Selection Operator (LASSO) constraint to a logit model, and a gradient boosted regression tree (GBRT). We also compare these models with two second-best alternatives: a naive constant estimator and a standard logit model that includes individual characteristics but does not account for municipality-specific effects.

The LASSO estimates a high-dimensional regression model by minimizing the sum of squared residuals subject to an ℓ_1 penalty on the coefficients, which shrinks some estimates toward zero and sets others exactly to zero, thereby performing variable selection. The resultant reduction in variability may be sufficiently large to offset the bias introduced by shrinkage, leading to a lower mean squared error. Our initial high-dimensional specification included individual refugee characteristics (e.g., age, gender, family composition, origin) and their interactions with municipalities. In our refined version, we further incorporate network measures to account for the potential influence of co-ethnic connections on employment outcomes. When applied to a logit specification for a binary outcome, the same penalization principle applies to the log-likelihood, resulting in a model that retains only the most relevant predictors. Importantly, the LASSO selects network measures among the predictors most useful for explaining the outcome.

GBRT builds an ensemble of classification trees sequentially, where each new tree is fit to the residuals from the previous stage to reduce prediction errors. In the binary outcome setting, the algorithm uses a logistic link so that predictions are probabilities. We apply it on the same specification as in the LASSO.

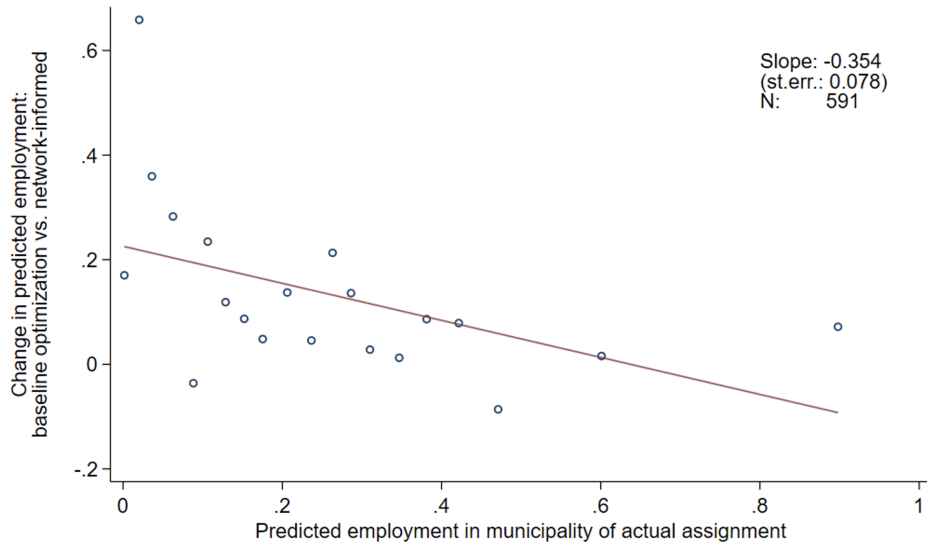
Table F1 reports the performance of all models on the holdout sample of refugees from the 1987–1997 cohorts across several metrics. Both the LASSO and GBRT outperform the second-best benchmarks on all measures, with the LASSO exceeding GBRT on every metric except precision. Incorporating network measures into the LASSO further improves prediction accuracy in the test set relative to the baseline LASSO on all metrics except deviance. Since we use the predicted probabilities from this refined LASSO as our preferred predictions, we finally check that our optimization problem is meaningful by ensuring sufficient variation in probabilities across municipalities for each refugee. We find an average within-individual standard deviation in predicted employment probability across municipalities of 0.195 (average individual predicted probability = 0.283), making the exercise policy relevant.

Table F1: Performance of machine learning models

Model	Test sample					
	Misc. error (1)	Recall (2)	Precision (3)	AUC-ROC (4)	Brier (5)	Deviance (6)
Constant	0.258	0.000	0.000	0.500	0.191	1.141
Logit	0.256	0.010	0.778	0.662	0.179	1.077
LASSO	0.240	0.246	0.578	0.730	0.166	0.959
GBRT	0.242	0.078	0.826	0.721	0.174	1.054
LASSO with netw.	0.227	0.284	0.592	0.739	0.161	0.970

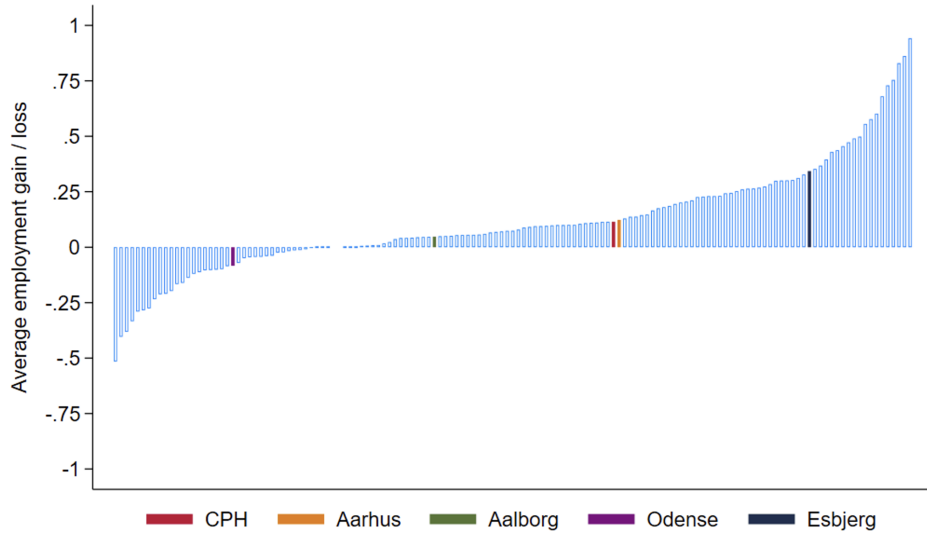
Notes: *Misclassification (Misc.) error* is the proportion of observations incorrectly classified. *Recall* measures the proportion of correctly predicted employed refugees among refugees actually employed (true positives over true positives plus false negatives). *Precision* measures the proportion of correctly predicted employment cases among all predicted employment cases (true positives over true positives plus false positives). All of these measures refer to a binary classification with a threshold set at the standard value of 0.5. Because our measure of quality scores uses predicted probabilities of employment, this specific threshold does not affect optimal allocations. Area under the receiver operating characteristic curve (*AUC-ROC*) measures the area under the receiver operating characteristic curve for each model. The *Brier* score measures the accuracy of probabilistic predictions for binary outcomes, computing the mean squared difference between the predicted probability and the actual outcome (coded as 0 or 1). *Deviance* also measures the quality of probabilistic predictions but penalizes errors logarithmically (i.e., it punishes high-confidence wrong predictions much more harshly than the Brier score).

Figure F1: Gains along the distribution of individual predicted probabilities



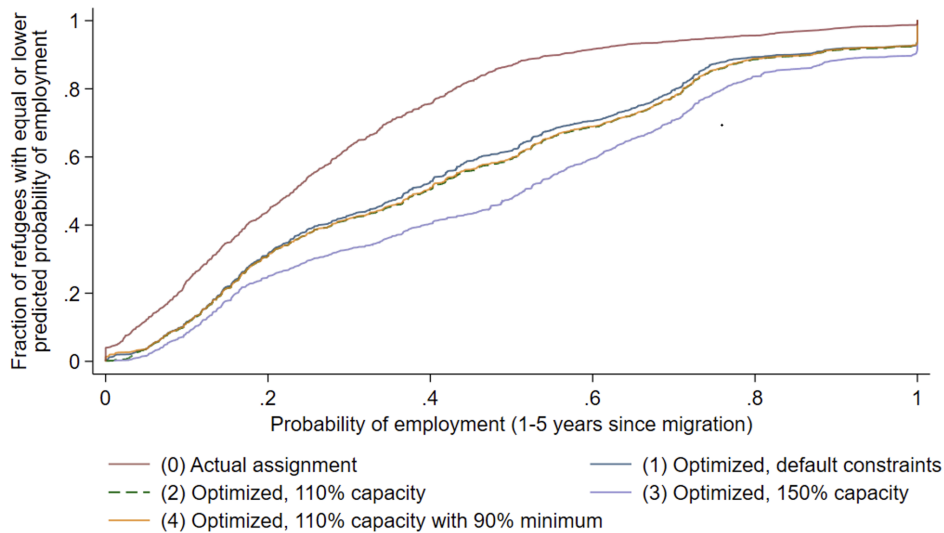
Notes: This figure shows the binned relationship between gains (or losses) in predicted employment probability for refugees who would be reassigned from municipalities optimally chosen without accounting for networks to those selected optimally with networks (y-axis) and their predicted probability in the municipality of actual assignment under the dispersal policy (x-axis). Each marker represents one-twentieth of the refugees subject to these hypothetical relocations with nonzero gains. The red solid line represents the line of best fit (slope and standard error are reported).

Figure F2: Average gains by municipality



Notes: This figure displays the average gains and losses in predicted refugee employment by municipality, comparing the status quo assignment under the dispersal policy with the optimized assignment generated using predictions that include network measures. The five largest municipalities (Copenhagen, Aarhus, Aalborg, Odense, and Esbjerg) are highlighted.

Figure F3: Counterfactual policy scenarios



Notes: This figure displays the empirical cumulative distribution functions (ECDFs) of the refugees' predicted employment probabilities within the first five years since migration under different assignment scenarios. In all scenarios, probabilities are computed with the refined LASSO that includes network measures. The solid red line refers to predicted probabilities in municipality of assignment under the dispersal policy. The solid blue line (counterfactual 1) refers to the optimized assignment using default constraints (municipality-specific family structure restrictions and inferred quotas). The dashed green line (counterfactual 2) refers to the optimized assignment subject to family structure restrictions and allowing for 110% of inferred municipality quotas (with no lower bound). The solid purple line (counterfactual 3) refers to the optimized assignment subject to family structure restrictions and allowing for 150% of inferred municipality quotas (with no lower bound). The solid yellow line (counterfactual 4) refers to the optimized assignment subject to family structure restrictions and allowing for 110% of inferred municipality quotas, with a lower bound of 90%.